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MOHSIN IBNA HOSSAIN

AIUB - MATH - 4 - NOTE

MATH - 4

MATRICES, VECTORS,

FOURIER ANALYSIS

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#Matrix:—

⇒ Rectangular array of numbers arranged systematically this called an matrix.

Q:- How to Define a matrix:-

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

here,

m = Column.

n = Row.

$m \times n$ = element = size

↓

Called m by n

#Types of Matrices:-

⇒ The various types of matrices are row matrix, column, null matrix, square, diagonal, upper triangular, Lower triangular, symmetric, Antisymmetric matrix.

Note:- The scalar matrix is similar to a square matrix.

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Row Matrix:-

=> Its a type of matrix that has a single row.

Ex:-

$B = [1, 2, 3]$

Q:- What is the size of the matrix?

=> The size is 1 by 3.

Column Matrix:-

=> It is a type of matrix that has only one column.

Ex:- $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ size = 3 by 1

Equal matrix:-

=> Two matrix are said to be equal if: Both the matrix are the same order. That mean's they have the same number of rows, and columns. Also If the elements of the first matrix are equal to the the corresponding elements in the second matrix.

Ex:-

$$A=\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}, \quad B=\begin{bmatrix} 1 & x \\ 2 & 3 \end{bmatrix}$$

if $x=-1$ then those two matrix are equal.

Unit/Identity Matrix:-

⇒ Main diagonal = 1 or equals to 1, and zeroes every where

Ex:-

$$I_2=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_3=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Transpose Matrix:-

⇒ A Matrix which is formed by turning all the rows of given matrix into columns and vice-versa.

Ex:-

$$A=\begin{bmatrix} 2 & 1 & -4 \\ 3 & 0 & 2 \end{bmatrix} \rightarrow 2\times3$$

Sol: $$A^T=\begin{bmatrix} 2 & 3 \\ 1 & 0 \\ -4 & 2 \end{bmatrix} \rightarrow 3\times2$$

#Symmetricical Matrix:-

$\Rightarrow$ A Square matrix that is equal to its transpose is called a symmetric matrix.

Ex:- if $A = A^T$

$$\therefore A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & -1 & 0 \\ 4 & 0 & 3 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & 3 & 4 \\ 3 & -1 & 0 \\ 4 & 0 & 3 \end{bmatrix}$$

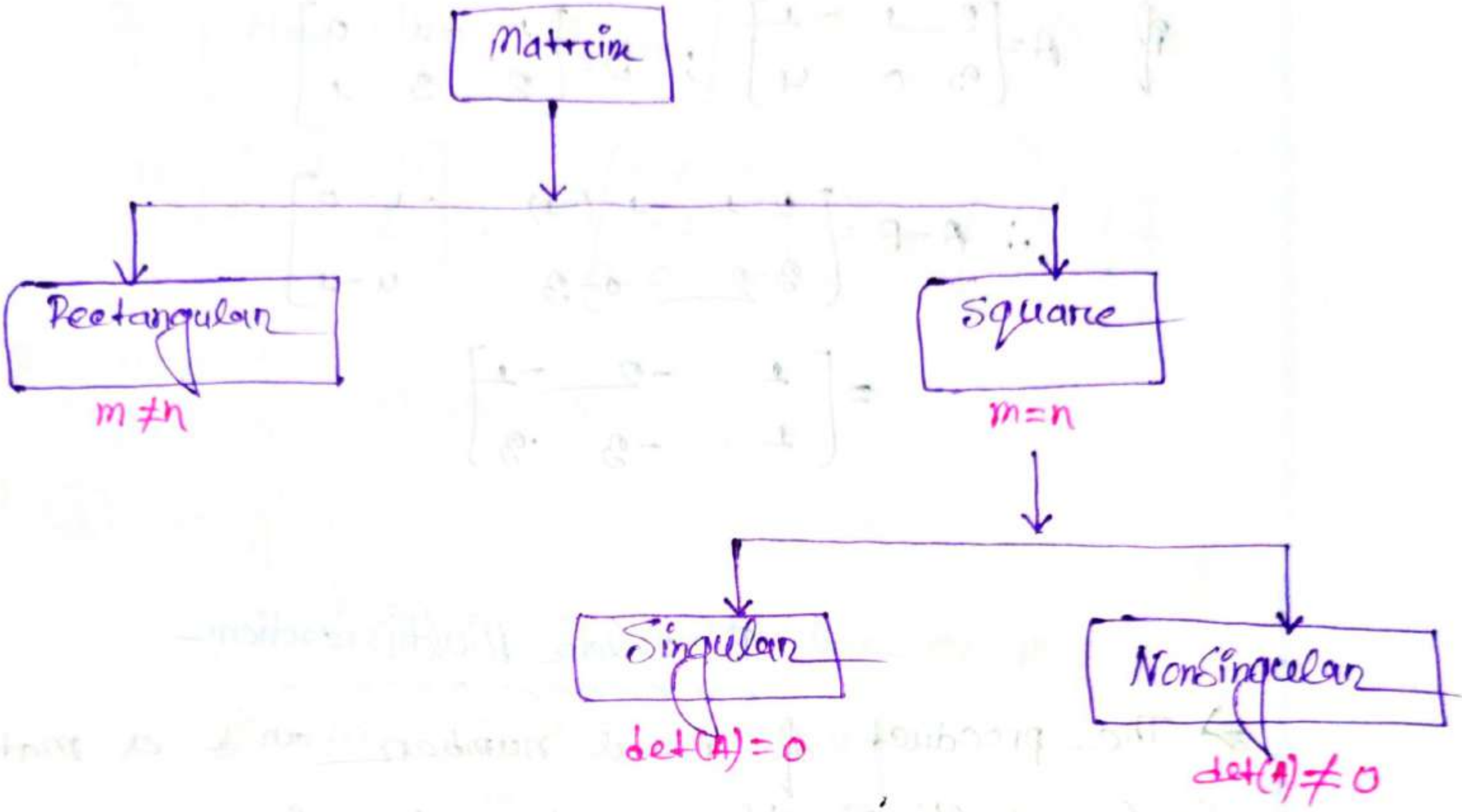
#Skew Symmetric matrix:-

$\Rightarrow$ A square matrix is skew symmetric if and only if $A = -A^T$

Ex:-

$$A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \Rightarrow -A^T = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = A$$

Classification of matrix:—



Matrix adding:—

if $A = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 0 & 4 \end{bmatrix}$; $B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \end{bmatrix}$

$$\therefore A+B = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2+1 & 1+(-1) & -1+0 \\ 3+2 & 0+3 & 4+1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & -1 \\ 5 & 3 & 5 \end{bmatrix}$$

Matrix Subtraction:-

if $A = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 0 & 4 \end{bmatrix}$; $B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 1 \end{bmatrix}$

$$\therefore A-B = \begin{bmatrix} 2-1 & 1-(-1) & -1-0 \\ 3-2 & 0-3 & 4-1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & -1 \\ 1 & -3 & 3 \end{bmatrix}$$

Scalar Multiplication:-

⇒ The product of real number and a matrix. In Scalar multiplication, each entry in the matrix is multiplied by the given scalar.

if, $A = \begin{bmatrix} 2 & 3 & 1 \\ -1 & 0 & 4 \end{bmatrix}$

$$\therefore 2A = \begin{bmatrix} 4 & 6 & 2 \\ -2 & 0 & 8 \end{bmatrix}$$

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Matrix Multiplication:-

If $\boxed{A_{m\times p} \times B_{p\times n} = AB_{m\times n}}$

$A=\begin{bmatrix} 2 & 1 & -4 \\ 3 & 0 & 2 \end{bmatrix}$, $B=\begin{bmatrix} -4 & 2 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$, $C=\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix}$

Q. ① $A\times B = ?$
② $B\times A = ?$ ③ $A\times C = ?$

⇒ ① In given:-

$A = 2\times 3$
$B = 3\times 2$ so its possible cz $p=p$

So, $A\times B = \begin{bmatrix} -8-1-0 & 4+2-4 \\ -12-0-0 & 6+0+2 \end{bmatrix}$

$= \begin{bmatrix} -9 & 2 \\ -12 & 8 \end{bmatrix} = 2\times 2 = AB$

② In given:-

$A = 2\times 3$
$B = 3\times 2$

$\therefore BA = 3\times 3 = m\times n \Rightarrow p=p$ so, its possible.

So, $B\times A = \begin{bmatrix} -8+6 & -4+0 & 16+4 \\ -2+6 & -1+0 & 4+4 \\ 0+3 & 0+0 & 0+2 \end{bmatrix}$

$= \begin{bmatrix} -2 & -4 & 20 \\ 4 & -1 & 8 \\ 3 & 0 & 2 \end{bmatrix}$

(3) Is given:—

$A = 2\times 3$

$C = 2\times 2$

So, $A\times C = m\times n \Rightarrow p\neq p$

thats means:— AC = is not possible.

Q:— If $IA = AI = A$

and $A \rightarrow m\times n$ then what's the size of I?

$\Rightarrow I_m A = A I_n = I$

Some properties:—

1. $A = (A^T)^T$
2. $(A+B)^T = A^T + B^T$
3. $(AB)^T = B^T A^T$
4. $(\alpha A)^T = \alpha A^T$ [α is a scaler]

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Determinant:—

if $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

So $\det(A) = |A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$

$$\boxed{|A| = a_{11}a_{12} - a_{12}a_{21}}$$

or, if $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

So, $\det(A) = |A| = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$

$$\Rightarrow \boxed{A = a_{11}(a_{22}\cdot a_{33} - a_{32}\cdot a_{23}) - a_{12}(a_{21}\cdot a_{33} - a_{31}\cdot a_{23}) + a_{13}(a_{21}\cdot a_{32} - a_{31}\cdot a_{22})}$$

H.W:—

① $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 0 & 5 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 7 & -2 & 8 \\ 0 & 6 & -2 \\ -1 & 0 & 9 \end{bmatrix}$

prove that $(AB)^T = B^T A^T$

$\Rightarrow$ In given:—

$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 0 & 5 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 7 & -2 & 8 \\ 0 & 6 & -2 \\ -1 & 0 & 9 \end{bmatrix}$

$$\therefore A \times B = \begin{bmatrix} 14+0-3 & -4-6+0 & 16+2+27 \\ 28+0+1 & -8+0+0 & 32+0-9 \\ 0+0-2 & 0+30+0 & 0-10+18 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & -10 & 45 \\ 29 & -8 & 23 \\ -2 & 30 & 8 \end{bmatrix}$$

So, $$[AB]^T = \begin{bmatrix} 11 & 29 & -2 \\ -10 & -8 & 30 \\ 45 & 23 & 8 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & 4 & 0 \\ -1 & 0 & 5 \\ 3 & -1 & 2 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 7 & 0 & -1 \\ -2 & 6 & 0 \\ 8 & -2 & 9 \end{bmatrix}$$

So, $$B^T \times A^T = \begin{bmatrix} 14-0-3 & 28+0+1 & 0+0-2 \\ -4-6+0 & -8+0+0 & 0+30+0 \\ 16+2+27 & 32+0-9 & 0-10-18 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 29 & -2 \\ -10 & -8 & 30 \\ 45 & 23 & 8 \end{bmatrix}$$

proved

② Find the values of α if the matrix is singular.

$$A = \begin{bmatrix} -10 & 4 & 4\alpha \\ -1 & 2 & 0 \\ 1 & \alpha & 2 \end{bmatrix}$$

$\Rightarrow$

$$\det(A) = |A| = -10(-4+0) - 4(2+0) + 4\alpha(-\alpha-2)$$

$$= +40 - 8 + (-4\alpha^2 - 8\alpha)$$

Now,

$$4\alpha^2 + 8 - 32 = 0$$

$$(\alpha-2)(\alpha+4) = 0$$

$$\therefore \alpha = -4, 2$$

Ans.

Multiplication properties:-

① $(AB)C = A(BC)$ [Commutative law]

② $A(B+C) = AB+AC$ [distributive law]

③ $(A+B)C = AC+BC$. [distributive law]

④ $\alpha(AB) = (\alpha A)B$ [α is a scalar]

⑤ $AB = BA$ [Not true always]

⑥ $AA^{-1} = A^{-1}A = I$

⑦ if $A \rightarrow n \times n$

then $A^{-1} \rightarrow n \times n$

so, $I \rightarrow I_n$

Inverse of a matrix:-

⇒ Two commonly used methods for finding the inverse are:-

1. Using adjoint matrix/cofactor method.
2. Using elementary row operation.

Cofactor of an element of a matrix:-

⇒ If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

so, cofactor of the element a_{ij} is represented by A_{ij}

$$\therefore \boxed{A_{ij} = (-1)^{i+j} \times M_{ij}}$$

OR, $A_{ij} = (-1)^{i+j} \times [m_{ij}]$

OR, $A_{11} = (-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

Q:-/Ex:- If $A = \begin{bmatrix} 3 & 4 & 7 \\ -2 & 5 & 6 \\ 7 & 3 & -9 \end{bmatrix}$ Find the cofactor of 6.

⇒ The cofactor of 6 is:-

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 3 & 4 \\ 7 & 3 \end{vmatrix}$$
$$= -(9-28) = 19$$

Inverse using adjoint Matrix:—

Formula:—

(1) $A^{-1} = \frac{1}{\det(A)}(\operatorname{adj} A)$ [A is a nonsingular Matrix]

(2) $\operatorname{adj}(A) = (\text{Cofactor } A)^T$

Ex:- Find the inverse of the given matrix using cofactor (adjoint) matrix.

$$A = \begin{bmatrix} 1 & 0 & -4 \\ -2 & 2 & 5 \\ 3 & -1 & 2 \end{bmatrix}$$

⇒ We know:—

$$A^{-1} = \frac{1}{\det(A)}(\operatorname{adj} A)$$

So, $\det(A) = \begin{vmatrix} 1 & 0 & -4 \\ -2 & 2 & 5 \\ 3 & -1 & 2 \end{vmatrix}$

$= 1(4+5) - 0(-4-15) + (-4)(2-6)$

$= 25 \neq 0$ ∴ A is not-singular, and A^{-1} exists.

Now Cofactor of A:—

$A_{11} = (-1)^{1+1}\begin{vmatrix} 2 & 5 \\ -1 & 2 \end{vmatrix} = 9$

$A_{12} = (-1)^{1+2}\begin{vmatrix} -2 & 5 \\ 3 & 2 \end{vmatrix} = 19$

$A_{13} = (-1)^{1+3}\begin{vmatrix} -2 & 2 \\ 3 & -1 \end{vmatrix} = -4$

$A_{21} = (-1)^{2+1}\begin{vmatrix} 0 & -4 \\ -1 & 2 \end{vmatrix} = 4$

$A_{22} = (-1)^{2+2}\begin{vmatrix} 1 & -4 \\ 3 & 2 \end{vmatrix} = 14$

$A_{23} = (-1)^{2+3}\begin{vmatrix} -2 & 0 \\ 3 & -1 \end{vmatrix} = 2$

$A_{31} = (-1)^{3+1} \begin{vmatrix} 0 & -4 \\ 2 & 5 \end{vmatrix} = 8$

$A_{32} = (-1)^{3+2} \begin{vmatrix} 1 & -4 \\ -2 & 5 \end{vmatrix} = 3$

$A_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 0 \\ -2 & 2 \end{vmatrix} = 2$

So, $\text{adj } A = (\text{Cofactor } A)^T$

$$= \begin{bmatrix} 9 & 19 & -4 \\ 4 & 14 & 1 \\ 8 & 3 & 2 \end{bmatrix}^T$$

$$= \begin{bmatrix} 9 & 4 & 8 \\ 19 & 14 & 3 \\ -4 & 1 & 2 \end{bmatrix}$$

Now, $A^{-1} = \frac{1}{\det(A)} \times (\text{adj } A)$

$$= \frac{1}{25} \begin{bmatrix} 9 & 4 & 8 \\ 19 & 14 & 3 \\ -4 & 1 & 2 \end{bmatrix}$$

Q. Now Justify your Answer!—

⇒ Justification!—

We know!—

if $A^{-1}A = AA^{-1} = I$

Then it's a Inverse Matrix!—

$$\Rightarrow A^{-1}A = \frac{1}{25} \begin{bmatrix} 9 & 4 & 8 \\ 19 & 14 & 3 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -4 \\ -2 & 2 & 5 \\ 3 & -1 & 2 \end{bmatrix}$$

$$= \frac{1}{25} \begin{bmatrix} 9-8+24 & 0+8-8 & -36+20+16 \\ 19-28+9 & 0+28-3 & -76+70+6 \\ -4-2+6 & 0+2-2 & 16+5+4 \end{bmatrix}$$

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$$= \frac{1}{25}\begin{bmatrix} 25 & 0 & 0 \\ 0 & 25 & 0 \\ 0 & 0 & 25 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= I$$

(Justified)

Elementary row operations:—

Rules:—

1. Swap (Interchanges) Any two rows.
2. Multiply any row by a nonzero constant.
3. Replace any row by adding a multiple of another row.

Row Echelon Form (REF)

Rules:—

1. All nonzero rows are above the rows with all zeros.

Ex:- If $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \\ 2 & 0 & 1 \end{bmatrix}$ then for REF $R_1 \longleftrightarrow R_3$ $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

2. Leading co-efficient of a non-zero row is always. Strictly to the right right of the leading co-efficient of the row above it.

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Reduced - Row - Echelon Form (RREF)

Rules:

(1) Leading co-efficient is equal in equal to 1 & this is the only nonzero element in the column.

Ex:- $A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

Example:- Find row echelon form (REF) of the following matrix and then convert in to reduced row echelon form (RREF).

$A = \begin{bmatrix} 2 & -3 & 1 & 2 \\ 4 & -6 & 2 & 4 \\ -4 & 6 & -1 & 0 \end{bmatrix}$

⇒ Is given:-

$$A = \begin{bmatrix} 2 & -3 & 1 & 2 \\ 4 & -6 & 2 & 4 \\ -4 & 6 & -1 & 0 \end{bmatrix}$$

Now, REF:-

$$= \begin{bmatrix} 2 & -3 & 1 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{bmatrix} \quad \begin{matrix} \tilde{r_2} = r_2 + r_3 \\ \tilde{r_3} = r_3 + 2r_1 \end{matrix}$$

$$= \begin{bmatrix} 2 & -3 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 \end{bmatrix} \quad \tilde{r_2} = r_2 - r_3$$

$$= \begin{bmatrix} 2 & -3 & 1 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{r_2} = r_2 \leftrightarrow r_3$$

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Now, RREF:-

$$= \begin{bmatrix} 2 & -3 & 0 & -2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{r}_1 = r_1 - r_2$$

$$= \begin{bmatrix} 1 & -\frac{3}{2} & 0 & -1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{r}_1 = \frac{r_1}{2}$$

Inverse of a Matrix Using elementary row operations:-

Rules:-

$[A|I]$

$\downarrow$

Elementary row operation

$\downarrow$

$[I|A^{-1}]$

Example:- Find the inverse of $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

$\Rightarrow$ Now,

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1 & 0 & 0 \\ 1 & 4 & 3 & 0 & 1 & 0 \\ 1 & 3 & 4 & 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right] \quad \begin{array}{l} \tilde{r}_2 = r_2 - r_1 \\ \tilde{r}_3 = r_3 - r_1 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 4 & 0 & -3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right] \quad \tilde{r}_1 = r_1 - 3r_3$$

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$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & -3 & -3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

$$= [I \mid A^{-1}]$$

$$\therefore A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Example:-2:- Find A^{-1} Using both the methods:-

$$A = \begin{bmatrix} -1 & -8 & -3 \\ 2 & 0 & 5 \\ 3 & 6 & 7 \end{bmatrix}$$

⇒ **Using Adjoint matrix:-**

We know:-

$$A^{-1} = \frac{1}{\det(A)} \times (\text{adj } A).$$

and:- $\text{adj} A = (\text{Cofactor})^T$

$$\det(A) = \begin{vmatrix} -1 & -8 & -3 \\ 2 & 0 & 5 \\ 3 & 6 & 7 \end{vmatrix}$$

$$= -1(0-30) + 8(14-15) - 3(12-0)$$

$$= 30 - 8 - 36$$

$$= -14 \neq 0$$ ∴ A is not singular

so, A^{-1} is exist.

Now co-factor of A:-

$$A_{11} = (-1)^{1+1}\begin{vmatrix} 0 & 5 \\ 6 & 7 \end{vmatrix} = -30$$

$$A_{12} = (-1)^{1+2}\begin{vmatrix} 2 & 5 \\ 3 & 7 \end{vmatrix} = 1$$

$$A_{13} = (-1)^{1+3}\begin{vmatrix} 2 & 0 \\ 3 & 6 \end{vmatrix} = 12$$

$$A_{21} = (-1)^{2+1}\begin{vmatrix} -8 & -3 \\ 6 & 7 \end{vmatrix} = 38$$

$$A_{22} = (-1)^{2+2}\begin{vmatrix} -1 & -3 \\ 3 & 7 \end{vmatrix} = 2$$

$$A_{23} = (-1)^{2+3}\begin{vmatrix} -1 & -8 \\ 3 & 6 \end{vmatrix} = -18$$

$$A_{31} = (-1)^{3+1}\begin{vmatrix} -8 & -3 \\ 0 & 5 \end{vmatrix} = -40$$

$$A_{32} = (-1)^{3+2}\begin{vmatrix} -1 & -3 \\ 2 & 5 \end{vmatrix} = -1$$

$$A_{33} = (-1)^{3+3}\begin{vmatrix} -1 & -8 \\ 2 & 0 \end{vmatrix} = 16$$

So, $\text{adj } A = (\text{cofactor } A)^T$

$$= \begin{bmatrix} -30 & 1 & 12 \\ 38 & 2 & -18 \\ -40 & -1 & 16 \end{bmatrix}^T$$

$$= \begin{bmatrix} -30 & 38 & -40 \\ 1 & 2 & -1 \\ 12 & -18 & 16 \end{bmatrix}$$

$\therefore A^{-1} = \frac{1}{\det(A)} \times \text{adj}(A)$

$$= \frac{1}{-14}\begin{bmatrix} -30 & 38 & -40 \\ 1 & 2 & -1 \\ 12 & -18 & 16 \end{bmatrix}$$

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#Rank of a Matrix:-

⇒ The rank of a matrix is the maximum number of non-zero rows in the matrix is called rank of a matrix.

Procedure/Formula/Rules:-

① Reduce the matrix to REF

② Count the number of nonzero rows.

Ex:- Find the Rank of the matrix.

$$A = \begin{bmatrix} 2 & 4 & 1 & 3 \\ -1 & -2 & 1 & 0 \\ 0 & 0 & 2 & 2 \\ 3 & 6 & 2 & 5 \end{bmatrix}$$

⇒ 1st we need to Reduce the matrix to REF

$$\therefore A = \begin{bmatrix} 2 & 4 & 1 & 3 \\ -1 & -2 & 1 & 0 \\ 0 & 0 & 2 & 2 \\ 3 & 6 & 2 & 5 \end{bmatrix} \begin{matrix} \widetilde{r_1} \to r_1 + r_2 \\ \widetilde{r_3} \to \frac{r_3}{2} \end{matrix}$$

$$= \begin{bmatrix} 1 & 2 & 2 & 3 \\ -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 3 & 6 & 2 & 5 \end{bmatrix} \begin{matrix} \widetilde{r_2} \to r_2 + r_1 \\ \widetilde{r_4} \to r_4 - 3r_1 \end{matrix}$$

$$= \begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -4 & -4 \end{bmatrix} \quad \widetilde{r_2} \to r_2 - 3r_3$$

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$$=\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -4 & -4 \end{bmatrix} \quad r_2 \leftrightarrow r_4$$

$$=\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & -4 & -4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{r}_2 \rightarrow \frac{r_2}{-4}$$

$$=\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{r}_3 \rightarrow r_3 - r_2$$

$$=\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ So, the rank(A) = 2

Ans

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Exercise 1.1:–

① If $A=\begin{bmatrix}-5 & 6 & 1\\ 2 & -2 & 9\end{bmatrix}$; $B=\begin{bmatrix}5 & 8 & -4\\ 1 & 2 & -9\end{bmatrix}$ then find the

(i) $7A$,

(ii) $3A+2B$

(iii) $2A-3B$.

(i)

$$\Rightarrow \quad 7A = 7\begin{bmatrix}-5 & 6 & 1\\ 2 & -2 & 9\end{bmatrix}$$

$$\therefore = \begin{bmatrix}-35 & 42 & 7\\ 14 & -14 & 63\end{bmatrix}$$

Ans:–

(ii)

$$\Rightarrow \quad 3A+2B = 3\begin{bmatrix}-5 & 6 & 1\\ 2 & -2 & 9\end{bmatrix} + 2\begin{bmatrix}5 & 8 & -4\\ 1 & 2 & -9\end{bmatrix}$$

$$= \begin{bmatrix}-15 & 18 & 3\\ 6 & -6 & 27\end{bmatrix} + \begin{bmatrix}10 & 16 & -8\\ 2 & 4 & -18\end{bmatrix}$$

$$= \begin{bmatrix}-15+10 & 18+16 & 3-8\\ 6+2 & -6+4 & 27-18\end{bmatrix}$$

$$= \begin{bmatrix}-5 & 34 & -5\\ 8 & -2 & 9\end{bmatrix}$$

Ans:–

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(iii)

$\Rightarrow 2A - 3B = 2\begin{bmatrix} -5 & 6 & 1 \\ 2 & -2 & 9 \end{bmatrix} - 3\begin{bmatrix} 5 & 8 & -4 \\ 1 & 2 & 9 \end{bmatrix}$

$= \begin{bmatrix} -10-15 & 12-24 & 2+12 \\ 4-3 & -4-6 & 18-27 \end{bmatrix}$

$= \begin{bmatrix} -25 & -12 & 14 \\ 1 & -10 & -9 \end{bmatrix}$

Ans:-

(4) If $A = \begin{bmatrix} 6 & 0 \\ -4 & -2 \\ 9 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 5 \\ -2 & -3 \\ 0 & 3 \end{bmatrix}$ then find AB and BA (if possible);

$\Rightarrow$ In given information:

$A = 3 \times 2$

$B = 3 \times 2$

So, $AB = 2 \times 3$ so, it's not possible. Also

$BA = 2 \times 3$ that means it's not possible.

Ans:-

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(5) If $A=\begin{bmatrix}-1 & 4\\ 0 & 3\end{bmatrix}$, $B=\begin{bmatrix}7 & 0\\ 6 & 5\end{bmatrix}$. then prove that

$(AB)^T = B^T A^T$

$\Rightarrow \quad AB = \begin{bmatrix}-1 & 4\\ 0 & 3\end{bmatrix}\begin{bmatrix}7 & 0\\ 6 & 5\end{bmatrix}$

$= \begin{bmatrix}-7+24 & 0+20\\ 0+18 & 0+15\end{bmatrix}$

$= \begin{bmatrix}17 & 20\\ 18 & 15\end{bmatrix}$

$\therefore (AB)^T = \begin{bmatrix}17 & 18\\ 20 & 15\end{bmatrix}$

Now, $A^T = \begin{bmatrix}-1 & 0\\ 4 & 3\end{bmatrix}$, $B^T = \begin{bmatrix}7 & 6\\ 0 & 5\end{bmatrix}$

$\therefore A^T \times B^T = \begin{bmatrix}-1 & 0\\ 4 & 3\end{bmatrix}\begin{bmatrix}7 & 6\\ 0 & 5\end{bmatrix}$

$= \begin{bmatrix}-7+0 & -6+0\\ 28+0 & 24+15\end{bmatrix}$

$= \begin{bmatrix}-7 & -6\\ 28 & 39\end{bmatrix}$

Now, $B^T \times A^T = \begin{bmatrix}7 & 6\\ 0 & 5\end{bmatrix}\begin{bmatrix}-1 & 0\\ 4 & 3\end{bmatrix}$

$= \begin{bmatrix}-7+24 & 0+18\\ 0+20 & 0+15\end{bmatrix}$

$= \begin{bmatrix}17 & 18\\ 20 & 15\end{bmatrix}$

(Proved)

Exercise 1.2

1. Find the value of determinant

$$A=\begin{bmatrix} 7 & -3 & 0 \\ 3 & 4 & 3 \\ -5 & 0 & -5 \end{bmatrix}$$

⇒ we know—

$$\det(A)=\begin{vmatrix} 7 & -3 & 0 \\ 3 & 4 & 3 \\ -5 & 0 & -5 \end{vmatrix}$$

$$=7(-20+0)+3(-15+15)+0(0+20)$$

$$=-140$$

(2) if $A=\begin{bmatrix} 3 & -1 & 5 \\ -3 & 4 & -3 \\ 5 & -1 & 0 \end{bmatrix}$ then find the $\det(A)$?

⇒ we know—

$$\det(A)=\begin{vmatrix} 3 & -1 & 5 \\ -3 & 4 & -3 \\ 5 & -1 & 0 \end{vmatrix}$$

$$=3(0-3)+1(0+15)+5(3-20)$$

$$=-79$$

Ans—

Exercise: 1.3

① Find the inverse of the following matrices (if possible) using co-factor method (for odd number) and elementary row operation (for Even Number) (ERO), also justify your Answers:

ⓘ $A = \begin{bmatrix} 3 & 2 \\ 1 & -4 \end{bmatrix}$

⇒ we know:-

$$A^{-1} = \frac{1}{\det(A)} (\text{cofacto } A)^T$$

∴ Now, $\det(A) = \begin{vmatrix} 3 & 2 \\ 1 & -4 \end{vmatrix}$

$$= -12 - 2 = -14 \neq 0$$

Now, cofactor A:-

$$\therefore \text{adj} A = (\text{cofactor})^T = \begin{bmatrix} -4 & -1 \\ -2 & 3 \end{bmatrix}^T$$

$$= \begin{bmatrix} -4 & -2 \\ -1 & 3 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{-1}{14} \begin{bmatrix} -4 & -2 \\ -1 & 3 \end{bmatrix}$$

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Now— Justify the Answer—

We know!—

$$AA^{-1} = A^{-1}A = I$$

$$\therefore A^{-1}A = -\frac{1}{14}\begin{bmatrix}-4 & -2\\ -1 & 3\end{bmatrix}\begin{bmatrix}3 & 2\\ 1 & -4\end{bmatrix}$$

$$= -\frac{1}{14}\begin{bmatrix}-12-2 & -8+8\\ -3+3 & -2-12\end{bmatrix}$$

$$= \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$= I$$

Justified!—

(ii) $A = \begin{bmatrix}4 & -2\\ 7 & 6\end{bmatrix}$

Using ERO!—

We know!—

$$[A|I] = \left[\begin{array}{cc|cc}4 & -2 & 1 & 0\\ 7 & 6 & 0 & 1\end{array}\right] \quad \tilde{r}_1 = r_1\times 3 + r_2$$

$$= \left[\begin{array}{cc|cc}19 & 0 & 3 & 1\\ 7 & 6 & 0 & 1\end{array}\right] \quad \tilde{r}_2 = r_2 + 12 - r_1$$

$$= \left[\begin{array}{cc|cc}19 & 0 & 3 & 1\\ 0 & 18 & 0 & 12\end{array}\right] \quad \tilde{r}_1 = \frac{r_1}{19},\ \tilde{r}_2 = \frac{r_2}{18}$$

$$= \left[\begin{array}{cc|cc}1 & 0 & \frac{3}{19} & \frac{1}{19}\\ 0 & 1 & \frac{1}{2} & \frac{2}{3}\end{array}\right] = [A^{-1}|I]$$

$$\therefore A^{-1} = \begin{bmatrix}\frac{3}{19} & \frac{1}{19}\\ \frac{1}{2} & \frac{2}{3}\end{bmatrix}$$

(ii) $A = \begin{bmatrix} 4 & -2 \\ 7 & 6 \end{bmatrix}$

Using ERO:-

We know:-

$$= [A|I]$$

$$= \left[\begin{array}{cc|cc} 4 & -2 & 1 & 0 \\ 7 & 6 & 0 & 1 \end{array}\right] \quad \tilde{R}_2 = R_2 \times 4 - R_1 \times 7$$

$$= \left[\begin{array}{cc|cc} 4 & -2 & 1 & 0 \\ 0 & 38 & -7 & 4 \end{array}\right] \quad \tilde{R}_1 = R_1 \times 19 + R_2$$

$$= \left[\begin{array}{cc|cc} 76 & 0 & 12 & 4 \\ 0 & 38 & -7 & 4 \end{array}\right] \quad \tilde{R}_1 = \frac{R_1}{76}, \quad \tilde{R}_2 = \frac{R_2}{38}$$

$$= \left[\begin{array}{cc|cc} 1 & 0 & \frac{3}{19} & \frac{1}{19} \\ 0 & 1 & -\frac{7}{38} & \frac{2}{19} \end{array}\right]$$

$$\therefore A^{-1} = \begin{bmatrix} \frac{3}{19} & \frac{1}{19} \\ -\frac{7}{38} & \frac{2}{19} \end{bmatrix}$$

Now, justify the Answer:-

We know:-

$$AA^{-1} = A^{-1}A = I$$

$$\therefore A^{-1}A = \begin{bmatrix} \frac{3}{19} & \frac{1}{19} \\ -\frac{7}{38} & \frac{2}{19} \end{bmatrix} \begin{bmatrix} 4 & -2 \\ 7 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{12}{19} + \frac{7}{19} & -\frac{6}{19} + \frac{6}{19} \\ -\frac{14}{19} + \frac{14}{19} & \frac{7}{19} + \frac{12}{19} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$

So, justified!

(iii) $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & -1 & 0 \end{bmatrix}$

using Co-factor method:-

we know:-

$$A^{-1} = \frac{1}{\det(A)} * (\operatorname{adj} A)$$

$$\operatorname{adj} A = (\text{cofact } A)^T$$

Now, $\det(A) = \begin{vmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & -1 & 0 \end{vmatrix}$

$$= 1(0-0) + 1(0-0) + 1(-2+1)$$

$$= -1 \neq 0$$

Now cofactor of A:-

$A_{11} = \begin{vmatrix} -1 & 0 \\ -1 & 0 \end{vmatrix} = 0$

$A_{12} = -\begin{vmatrix} 2 & 0 \\ 1 & 0 \end{vmatrix} = 0$

$A_{13} = \begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix} = -1$

$A_{21} = -\begin{vmatrix} -1 & 1 \\ -1 & 0 \end{vmatrix} = -1$

$A_{22} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1$

$A_{23} = -\begin{vmatrix} 1 & -1 \\ 1 & -1 \end{vmatrix} = 0$

$A_{31} = \begin{vmatrix} -1 & 1 \\ -1 & 0 \end{vmatrix} = 1$

$A_{32} = -\begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = +2$

$A_{33} = \begin{vmatrix} 1 & -1 \\ 2 & -1 \end{vmatrix} = 1$

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$\therefore$ Cofactor $A = \begin{bmatrix} 0 & 0 & -1 \\ -1 & -1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

$\therefore$ Adj A = (Cofactor A)^T

$= \begin{bmatrix} 0 & -1 & 1 \\ 0 & -1 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

$\therefore \; A^{-1} = \frac{1}{det(A)}$ adj A

$= -1 \times \begin{bmatrix} 0 & -1 & 1 \\ 0 & -1 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 0 & 1 & -1 \\ 0 & 1 & -2 \\ 1 & 0 & -1 \end{bmatrix}$

Ans:

Now, justify the Answer:

we know:

$A^{-1}A = AA^{-1} = I$

$\therefore AA^{-1} = \begin{bmatrix} 0 & 1 & -1 \\ 0 & 1 & -2 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & -1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 0+2-1 & 0-1+1 & 0+0+0 \\ 0+2-2 & 0-1+2 & 0+0+0 \\ 1+0-1 & -1+0+1 & 1+0+0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

(proved)

(d) $B = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -3 \\ 1 & -1 & 1 \end{bmatrix}$

$\Rightarrow$ Using ERO:—

We know:—

$[A|I] = [I|A^{-1}]$

$$\therefore [A|I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 & 0 & 1 \end{array}\right] \quad \begin{array}{l} \tilde{R}_3 \to R_3 - R_1 \\ \tilde{R}_2 \to R_2 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -5 & -1 & 1 & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \end{array}\right] \quad \tilde{R}_2 \to R_2 - 5R_3$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 6 & 0 & 4 & 1 & -5 \\ 0 & -1 & -1 & -1 & 0 & 1 \end{array}\right] \quad \tilde{R}_3 \to 6R_3 + R_2$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 6 & 0 & 4 & 1 & -5 \\ 0 & 0 & -6 & -2 & 1 & 1 \end{array}\right] \quad \tilde{R}_1 \to 3R_1 + R_3$$

$$= \left[\begin{array}{ccc|ccc} 3 & 0 & 0 & 1 & 1 & 1 \\ 0 & 6 & 0 & 4 & 1 & -5 \\ 0 & 0 & -6 & -2 & 1 & 1 \end{array}\right] \quad \begin{array}{l} \tilde{R}_1 \to \frac{R_1}{3} \\ \tilde{R}_2 \to \frac{R_2}{6} \\ \tilde{R}_3 \to \frac{R_3}{-6} \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 1 & 0 & \frac{2}{3} & \frac{1}{6} & \frac{-5}{2\times 3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{1}{6} & -\frac{1}{6} \end{array}\right] = [A^{-1}|I]$$

$$\therefore A^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{6} & \frac{-5}{2\times 3} \\ \frac{1}{3} & -\frac{1}{6} & -\frac{1}{6} \end{bmatrix}$$

Ans:—

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Justify the Answer:-

we know:-

$$A^{-1}A = AA^{-1} = I$$

$$\therefore A^{-1}A = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{6} & -\frac{5}{6} \\ \frac{1}{3} & -\frac{1}{6} & -\frac{1}{6} \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -3 \\ 1 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{3}+\frac{1}{3}+\frac{1}{3} & 0+\frac{1}{3}-\frac{1}{3} & \frac{2}{3}-1+\frac{1}{3} \\ \frac{2}{3}+\frac{1}{6}-\frac{5}{6} & 0+\frac{1}{6}+\frac{5}{6} & \frac{4}{3}-\frac{3}{6}-\frac{5}{6} \\ \frac{1}{3}-\frac{1}{6}-\frac{1}{6} & 0-\frac{1}{6}+\frac{1}{6} & \frac{2}{3}+\frac{3}{6}-\frac{1}{6} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= I$$

$\therefore$ Justified

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(2) If $A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$

Find:—

(i) $A^{-1}B$

(ii) $B^{-1}A$

(iii) $(AB)^{-1}$ Also justified your Answer.

using co-factor method:— (i)

we know—

$$A^{-1} = \frac{1}{\det A} \, adj(A)$$

$$adj(A) = (cofactor\ A)^T$$

$$\therefore \rightarrow \det(A) = \begin{vmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{vmatrix}$$

$$= -1(5-0) - 2(10-0) - 3(-4-4)$$

$$= -1 \neq 0$$

∴ Now co-factor of A:—

$$A_{11} = \begin{vmatrix} 1 & 0 \\ -2 & 5 \end{vmatrix} = 5 \qquad A_{21} = -\begin{vmatrix} 2 & -3 \\ -2 & 5 \end{vmatrix} = -4 \qquad A_{31} = \begin{vmatrix} 2 & -3 \\ 1 & 0 \end{vmatrix} = 3$$

$$A_{12} = -\begin{vmatrix} 2 & 0 \\ 4 & 5 \end{vmatrix} = -10 \qquad A_{22} = \begin{vmatrix} -1 & -3 \\ 4 & 5 \end{vmatrix} = 7 \qquad A_{32} = -\begin{vmatrix} -1 & -3 \\ 2 & 0 \end{vmatrix} = -6$$

$$A_{13} = \begin{vmatrix} 2 & 1 \\ 4 & -2 \end{vmatrix} = -8 \qquad A_{23} = -\begin{vmatrix} -1 & 2 \\ 4 & -2 \end{vmatrix} = 6 \qquad A_{33} = \begin{vmatrix} -1 & 2 \\ 2 & 1 \end{vmatrix} = -5$$

$$\therefore \ adj(A) = \begin{bmatrix} 5 & -10 & -8 \\ -4 & 7 & -6 \\ 3 & -6 & -5 \end{bmatrix}^T$$

$$= \begin{bmatrix} 5 & -4 & 3 \\ -10 & 7 & -6 \\ -8 & 6 & -5 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{-1} \begin{bmatrix} 5 & -4 & 3 \\ -10 & 7 & -6 \\ -8 & 6 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 4 & -3 \\ 10 & -7 & +6 \\ 8 & -6 & 5 \end{bmatrix}$$

Now justify the Inverse of A

we know:—

$$AA^{-1} = A^{-1}A = I$$

$$\therefore A^{-1}A = \begin{bmatrix} -5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{bmatrix} \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 5+8-12 & -10+4+6 & 15+0-15 \\ -10-14+24 & 20-7-12 & -30+0+30 \\ -8-12+20 & 16-6-10 & -24+0+25 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= I$$

verified

Now:- $A^{-1}B = \begin{bmatrix} 5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$

$$= \begin{bmatrix} 10+0-15 & 5+8-6 & -5+4+9 \\ 20+0+30 & 10-14+12 & -10-7-18 \\ 16+0+25 & 8-12+10 & -8-6-15 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 7 & 8 \\ 50 & 8 & -35 \\ 41 & 6 & -29 \end{bmatrix}$$

Ans:-

(ii)

Using ERO method:-

We know:-

$$[A|I] = [I|A^{-1}]$$

$$\therefore [B|I] = \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 5 & 2 & -3 & 0 & 0 & 1 \end{array}\right] \quad \tilde{R}_3 = 2R_3 - 5R_1$$

$$= \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & -1 & -1 & -5 & 0 & 2 \end{array}\right] \quad \begin{array}{l} \tilde{R}_1 = R_3 + R_1 \\ \tilde{R}_2 = R_3 + R_2 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 2 & 1 & -2 & -4 & 0 & 2 \\ 0 & 1 & 0 & -5 & 1 & 2 \\ 0 & -1 & -1 & -5 & 0 & 2 \end{array}\right] \quad \tilde{R}_3 = R_2 + R_3$$

$$= \left[\begin{array}{ccc|ccc} 2 & 0 & -2 & -4 & 0 & 2 \\ 0 & 1 & 0 & -5 & 1 & 2 \\ 0 & 0 & -1 & -10 & 1 & 4 \end{array}\right] \quad \tilde{R}_1 = R_1 - 2R_3$$

$$= \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 16 & -2 & -6 \\ 0 & 1 & 0 & -5 & 1 & 2 \\ 0 & 0 & -1 & -10 & 1 & 4 \end{array}\right] \quad \begin{array}{l} \tilde{R}_1 = \frac{R_1}{2} \\ \tilde{R}_3 = \frac{R_3}{-1} \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & -1 & -3 \\ 0 & 1 & 0 & -5 & 1 & 2 \\ 0 & 0 & 1 & 10 & -1 & -4 \end{array}\right] = [B^{-1}|I]$$

$$\therefore B^{-1} = \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}$$

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Now Justify the Inverse of A:—

We know:—

$$AA^{-1} = A^{-1}A = I$$

$$\therefore A^{-1}A = \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 16+0-15 & 8-2-6 & -8-1+9 \\ -10+0+10 & -5+2+4 & 5+1-6 \\ 20+0-20 & 10-2-8 & -10-1+12 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= I$$

Justified

$$\therefore B^{-1}A = \begin{array}{c} \begin{matrix} -3 & \quad 0 & \quad 5 \end{matrix} \\ \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix} \end{array}\begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} -8-2-12 & 16-1+6 & -24+0-15 \\ 40+2+8 & -10+1-4 & 15+0+10 \\ -10-2-16 & 20-1+8 & -30+0-20 \end{bmatrix}$$

$$= \begin{bmatrix} -22 & 21 & -39 \\ 50 & -13 & 25 \\ -28 & 27 & -50 \end{bmatrix}$$

Ans:—

(iii)

In given:-

$$AB = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix} \times \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -2+0-15 & -1+4-6 & 1+2+9 \\ 4+0+0 & 2+2+0 & -2+1+0 \\ 8+0+25 & 4-4+10 & -4-2-15 \end{bmatrix}$$

$$= \begin{bmatrix} -17 & -3 & 12 \\ 4 & 4 & -1 \\ 33 & 10 & -21 \end{bmatrix}$$

For $(AB)^{-1}$ using ERO:-

we know:-

$$[A|I] = [I|A^{-1}]$$

$$\therefore [AB|I] = \left[\begin{array}{ccc|ccc} -17 & -3 & 12 & 1 & 0 & 0 \\ 4 & 4 & -1 & 0 & 1 & 0 \\ 33 & 10 & -21 & 0 & 0 & 1 \end{array}\right] \quad \begin{array}{l} \tilde{R}_1 = 7R_1 + 4R_3 \\ \tilde{R}_2 = 21R_2 - R_3 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 13 & 19 & 0 & 7 & 0 & 4 \\ 51 & 74 & 0 & 0 & 21 & -1 \\ 33 & 10 & -21 & 0 & 0 & 1 \end{array}\right] \quad \begin{array}{l} \tilde{R}_1 = 74R_1 - 19R_2 \\ \tilde{R}_3 = 51R_3 - 33R_2 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} -7 & 0 & 0 & 518 & -399 & 315 \\ 51 & 74 & 0 & 0 & 21 & -1 \\ 0 & -1932 & -1071 & 0 & -693 & 84 \end{array}\right] \quad \tilde{R}_2 = 51R_1 + 7R_2$$

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$$= \left[\begin{array}{ccc|ccc} -7 & 0 & 0 & 518 & -399 & 315 \\ 0 & 518 & 0 & 26418 & -20202 & 16058 \\ 0 & -1932 & -1071 & 0 & -693 & 84 \end{array}\right] \quad \tilde{R_3} = 1932R_2 + 518R_3$$

$$= \left[\begin{array}{ccc|ccc} -7 & 0 & 0 & 518 & -399 & 315 \\ 0 & 518 & 0 & 26418 & -20202 & 16058 \\ 0 & 0 & -554778 & 51039576 & -39389238 & 31067568 \end{array}\right] \quad \begin{array}{l} \tilde{R_1} = \frac{R_1}{-7} \\ \tilde{R_2} = \frac{R_2}{518} \\ \tilde{R_3} = \frac{R_3}{-554778} \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -74 & 57 & -45 \\ 0 & 1 & 0 & 51 & -39 & 31 \\ 0 & 0 & 1 & -92 & 71 & -56 \end{array}\right]$$

$$= \left[I \mid (AB)^{-1} \right]$$

$$\therefore (AB)^{-1} = \begin{bmatrix} -74 & 57 & -45 \\ 51 & -39 & 31 \\ -92 & 71 & -56 \end{bmatrix}$$

Now Justify the Answer:—

We know:—

$$AA^{-1} = A^{-1}A = I$$

$$\therefore A^{-1}A = \begin{bmatrix} -74 & 57 & -45 \\ 51 & -39 & 31 \\ -92 & 71 & -56 \end{bmatrix} \begin{bmatrix} -17 & -3 & 12 \\ 4 & 4 & -1 \\ 33 & 10 & -21 \end{bmatrix}$$

$$= \begin{bmatrix} 1258+228-1485 & 222+228-450 & -888-57+945 \\ -867-156+1023 & -153-156+310 & 612+39-651 \\ 1564+284-1848 & 276+284-560 & -1104-71+1176 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Justified

(3) If $A=\begin{bmatrix}-1 & -8 & -3\\ 2 & 0 & 5\\ 3 & 6 & 7\end{bmatrix}$ and $B=\begin{bmatrix}3\\ -1\\ 5\end{bmatrix}$ find $A^{-1}B$.

$\Rightarrow$ using cofactor of A method:—

We know:—

$$A^{-1}=\frac{1}{\det A}\times \operatorname{adj}(A)$$

$$\operatorname{adj}(A)=(\text{cofactor } A)^{T}$$

$$\therefore \det(A)=\begin{vmatrix}-1 & -8 & -3\\ 2 & 0 & 5\\ 3 & 6 & 7\end{vmatrix}$$

$$=-1(-30)+8(14-15)-3(12-0)$$

$$=-14\neq 0$$

Now, Cofactor of A:—

$A_{11}=\begin{vmatrix}0 & 5\\ 6 & 7\end{vmatrix}=-30$

$A_{12}=-\begin{vmatrix}2 & 5\\ 3 & 7\end{vmatrix}=1$

$A_{13}=\begin{vmatrix}2 & 0\\ 3 & 6\end{vmatrix}=12$

$A_{21}=-\begin{vmatrix}-8 & -3\\ 6 & 7\end{vmatrix}=38$

$A_{22}=\begin{vmatrix}-1 & -3\\ 3 & 7\end{vmatrix}=2$

$A_{23}=-\begin{vmatrix}-1 & -8\\ 3 & 6\end{vmatrix}=-18$

$A_{31}=\begin{vmatrix}-8 & -3\\ 0 & 5\end{vmatrix}=-40$

$A_{32}=-\begin{vmatrix}-1 & -3\\ 2 & 5\end{vmatrix}=-1$

$A_{33}=\begin{vmatrix}-1 & -8\\ 2 & 0\end{vmatrix}=16$

$$\therefore \operatorname{adj}(A)=\begin{bmatrix}-30 & 1 & 12\\ 38 & 2 & -18\\ -40 & -1 & 16\end{bmatrix}^{T}=\begin{bmatrix}-30 & 38 & -40\\ 1 & 2 & -1\\ 12 & -18 & 16\end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{-14}\begin{bmatrix} -30 & 38 & -40 \\ 1 & 2 & -1 \\ 12 & -18 & 16 \end{bmatrix}$$

$$= \frac{1}{14}\begin{bmatrix} 30 & -38 & 40 \\ -1 & -2 & 1 \\ -12 & 18 & -16 \end{bmatrix}$$

$$\therefore A^{-1}B = \frac{1}{14}\begin{bmatrix} 30 & -38 & 40 \\ -1 & -2 & 1 \\ -12 & 18 & -16 \end{bmatrix}\begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix}$$

$$= \frac{1}{14}\begin{bmatrix} 90 + 38 + 200 \\ -3 + 2 + 5 \\ -36 - 18 - 80 \end{bmatrix}$$

$$= \frac{1}{14}\begin{bmatrix} 328 \\ 4 \\ -134 \end{bmatrix}$$

Exercise : 1.4

(1.) Reduce the following matricees to row echelon form and find the rank of the following matricees.

(i) $M = \begin{bmatrix} 6 & 2 & 0 & 4 \\ -2 & -1 & 3 & 4 \\ -1 & -1 & 6 & 10 \end{bmatrix}$

⇒ Is given:—

$$M = \begin{bmatrix} 6 & 2 & 0 & 4 \\ -2 & -1 & 3 & 4 \\ -1 & -1 & 6 & 10 \end{bmatrix} \quad \begin{matrix} \tilde{R}_2 \rightarrow R_2 \times 3 + R_1 \\ \tilde{R}_3 \rightarrow R_3 \times 6 + R_1 \end{matrix}$$

$$= \begin{bmatrix} 6 & 2 & 0 & 4 \\ 0 & -1 & 9 & 16 \\ 0 & -4 & 36 & 64 \end{bmatrix} \quad \tilde{R}_3 = R_3 - 4R_2$$

$$= \begin{bmatrix} 6 & 2 & 0 & 4 \\ 0 & -1 & 9 & 16 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

∴ So the rank of $(M) = 2$

(ii) $B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 3 & 2 \\ 5 & 9 & 12 & 14 \end{bmatrix}$

⇒ Is given:—

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 3 & 2 \\ 5 & 9 & 12 & 14 \end{bmatrix} \quad \begin{matrix} \tilde{R}_2 = R_2 - 2R_1 \\ \tilde{R}_3 = R_3 - 5R_1 \end{matrix}$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -3 & -6 \\ 0 & -1 & -3 & -6 \end{bmatrix} \quad \tilde{R}_3 = R_3 - R_2$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -3 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{R_1} = R_1 + R_2$$

$$= \begin{bmatrix} 1 & 1 & 0 & -2 \\ 0 & -1 & -3 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ the rank of $(A) = 2$

(iii) $$P = \begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix} \quad \begin{aligned} \tilde{R_2} &= R_2 + 3R_1 \\ \tilde{R_3} &= R_3 + 2R_1 \\ \tilde{R_4} &= R_4 + 4R_1 \end{aligned}$$

$$= \begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 0 & -1 & 2 & 12 & 16 & -5 \\ 0 & -1 & 2 & 12 & 16 & -5 \\ 0 & -1 & 2 & 12 & 16 & -5 \end{bmatrix} \quad \begin{aligned} \tilde{R_4} &= R_4 - R_2 \\ \tilde{R_3} &= R_3 - R_2 \end{aligned}$$

$$= \begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 0 & -1 & 2 & 12 & 16 & -5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ the rank of $(P) = 2$

Ans:-

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(2) Find the rank of the following matrices:-

(i) $A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 1 \\ 0 & 3 & 0 \end{bmatrix}$ $\tilde{R}_2 = R_2 + R_1$

$\Rightarrow = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 0 \\ 0 & 3 & 0 \end{bmatrix}$ $\tilde{R}_3 = R_2 - R_3$

$= \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\therefore$ So, the rank of $(A) = 2$

Ans:-

(ii) $A = \begin{bmatrix} 1 & -2 & -1 \\ -2 & 4 & 2 \\ -1 & 2 & 6 \end{bmatrix}$ $\tilde{R}_2 = R_2 + 2R_1$, $\tilde{R}_3 = R_3 + R_1$

$= \begin{bmatrix} 1 & -2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ $R_2 \leftrightarrow R_3$

$= \begin{bmatrix} 1 & -2 & -1 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{bmatrix}$

$\therefore$ the rank of $(A) = 2$

Ans:-

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(iii) $A = \begin{bmatrix} 1 & -2 & -1 \\ -200 & 3 & 4 \\ -1 & 2 & 6 \end{bmatrix}$ $\tilde{R}_3 = R_3 + R_1$

$= \begin{bmatrix} 1 & -2 & -1 \\ -200 & 3 & 4 \\ 0 & 0 & 5 \end{bmatrix}$ $\tilde{R}_2 = R_2 + 200R_1$

$= \begin{bmatrix} 1 & -2 & -1 \\ 0 & -397 & -196 \\ 0 & 0 & 5 \end{bmatrix}$

So, the rank of $(A) = 3$

Ans.

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#Cryptographically Problem:-

Ch: 02

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Cryptography

⇒ the process to write (encode) and read (decoded) any secret messages by using matrices is known as Cryptography.

Rules!—

① Message $\xrightarrow{A}$ encode

② Encoded $\xrightarrow{A^{-1}}$ Message

Algorithm to Encode a Message:—

① Assign the numbers 1-26 to the letters in the alphabet given below.

② Assign the number 0 to a blank to provide for space between words.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	2	3	4	5	6	7	8	9	10	11	12	13	14

O	P	Q	R	S	T	U	V	W	X	Y	Z	Blank
15	16	17	18	19	20	21	22	23	24	25	26	0

Example:- Encode the message "SECRET CODE" using the matrix $A = \begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}$

⇒ The message corresponds to the sequence.

19 5 3 18 5 20 0 3 15 4 5

$$\therefore B = \begin{bmatrix} 19 & 3 & 5 & 0 & 15 & 5 \\ 5 & 18 & 20 & 3 & 4 & 0 \end{bmatrix}$$

Now:- Encode process:-

$$= AB$$

$$= \begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 19 & 3 & 5 & 0 & 15 & 5 \\ 5 & 18 & 20 & 3 & 4 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 76+15 & 12+54 & 20+60 & 0+9 & 60+12 & 20+0 \\ 5+19 & 3+18 & 5+20 & 0+3 & 15+4 & 5+0 \end{bmatrix}$$

$$= \begin{bmatrix} 91 & 66 & 80 & 9 & 72 & 20 \\ 24 & 21 & 25 & 3 & 19 & 5 \end{bmatrix}$$

So the coded message is:

91 24 66 21 80 25 9 3 72 19 20 5.

Example:-2 The encoded message is 21 24 66 21 80 25 9 3 72 19 20 5. Decode the message by using $A = \begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}$

⇒ We know—

For decode any message.

(i) Encoded $\xrightarrow{A^{-1}}$ Message.

(ii) $A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$

(iii) $\operatorname{adj}(A) = (\text{cofact } A)^T$

$\therefore \det(A) = \begin{vmatrix} 4 & 3 \\ 1 & 1 \end{vmatrix}$

$= 4 - 3 = 1 \neq 0$

Now:- cofactor of A:-

$A_{11} = 1$
$A_{12} = -1$
$A_{21} = -3$
$A_{22} = 4$

$\therefore \operatorname{adj}(A) = (\text{cofactor } A)^T$

$= \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}$

$\therefore A^{-1} = \frac{1}{1} \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}$

In given:–

$$B=\begin{bmatrix} 91 & 66 & 80 & 9 & 72 & 20 \\ 24 & 21 & 25 & 3 & 19 & 5 \end{bmatrix}$$

Now:– Decode process:–

$$= A^{-1}B$$

$$=\begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}\begin{bmatrix} 91 & 66 & 80 & 9 & 72 & 20 \\ 24 & 21 & 25 & 3 & 19 & 5 \end{bmatrix}$$

$$=\begin{bmatrix} 91+(-72) & 66-63 & 80-75 & 9-9 & 72-57 & 20-15 \\ -91+96 & -66+84 & -80+100 & -9+12 & -72+76 & -20+20 \end{bmatrix}$$

$$=\begin{bmatrix} 19 & 3 & 5 & 0 & 15 & 5 \\ 5 & 18 & 20 & 3 & 4 & 0 \end{bmatrix}$$

so, the Sequence =

19 5 3 18 5 20 0 3 15 4 5 0

Now the message is:–

"SECRET CODE"

Ans:–

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Example: 3 The Encode message is "28 13 28 20 23 5" Now Decode the message by using matrix $A=\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

$\Rightarrow$ We know:—

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

$$\therefore \det(A) = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0$$

$$\therefore \operatorname{adj}(A) = (\text{cofactor})^T = \begin{bmatrix} 0 & -1 \\ -1 & 1 \end{bmatrix}^T = \begin{bmatrix} 0 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{-1}\begin{bmatrix} 0 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$$

In given:—

$$B = \begin{bmatrix} 28 & 28 & 23 \\ 13 & 20 & 5 \end{bmatrix}$$

$$\therefore A^{-1}B = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 28 & 28 & 23 \\ 13 & 20 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0+13 & 0+20 & 0+5 \\ 28-13 & 28-20 & -5+23 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & 20 & 5 \\ 15 & 8 & 18 \end{bmatrix}$$

Now the sequence:— 13 15 20 8 5 18

So, the message:— "MOTHER"

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#Exercise 2

(13.) Encode the message "TRY YOUR BEST" by using matrix, $A=\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}$

$\Rightarrow$ The message corresponds to the sequence.

20, 18, 25, 0, 25, 15, 21, 18, 0, 2, 5, 19, 20.

$\therefore B=\begin{bmatrix}20 & 25 & 25 & 21 & 0 & 5 & 20\\ 18 & 0 & 15 & 18 & 2 & 19 & 0\end{bmatrix}$

Now Encoded process:-

$=AB$

$=\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}20 & 25 & 25 & 21 & 0 & 5 & 20\\ 18 & 0 & 15 & 18 & 2 & 19 & 0\end{bmatrix}$

$=\begin{bmatrix}20+18 & 25+0 & 25+15 & 21+18 & 0+2 & 5+19 & 20+0\\ 20+0 & 25+0 & 25+0 & 21+0 & 0+0 & 5+0 & 20+0\end{bmatrix}$

$=\begin{bmatrix}38 & 25 & 40 & 39 & 2 & 24 & 20\\ 20 & 25 & 25 & 21 & 0 & 5 & 20\end{bmatrix}$

So, the coded message is:-

38 20 25 25 40 25 39 21 2 0 24 5 20 20

Ans:-

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(15) Encrypt the message "EFFORT NEVER DIES" by the provided matrix $B=\begin{bmatrix}1 & 2\\ 1 & 1\end{bmatrix}$

$\Rightarrow$ The message corresponds to the sequence:-

5, 6, 6, 15, 18, 20, 0, 14, 5, 22, 5, 18, 0, 4, 9, 5, 19

$$\therefore C=\begin{bmatrix}5 & 6 & 18 & 0 & 5 & 5 & 0 & 9 & 19\\ 6 & 15 & 20 & 14 & 22 & 18 & 4 & 5 & 0\end{bmatrix}$$

Now Encoded process:-

$$=BC$$

$$=\begin{bmatrix}1 & 2\\ 1 & 1\end{bmatrix}\begin{bmatrix}5 & 6 & 18 & 0 & 5 & 5 & 0 & 9 & 19\\ 6 & 15 & 20 & 14 & 22 & 18 & 4 & 5 & 0\end{bmatrix}$$

$$=\begin{bmatrix}5+12 & 6+30 & 18+40 & 0+28 & 5+44 & 5+36 & 0+8 & 9+10 & 19+0\\ 5+6 & 6+15 & 18+20 & 0+14 & 5+22 & 5+18 & 0+4 & 9+5 & 19+0\end{bmatrix}$$

$$=\begin{bmatrix}17 & 36 & 58 & 28 & 49 & 41 & 8 & 19 & 19\\ 11 & 21 & 38 & 14 & 27 & 23 & 4 & 14 & 19\end{bmatrix}$$

So, the coded Message is:-

17 11 36 21 58 38 28 14 49 27 41 23 8 4 19 14 19 19

Ans:-

GOOD LUCK

(16.) The Encoded message is "→ 6 28 20 23 5"
Decode the message by using matrix $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

⇒ we know:—

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

$$\therefore \det(A) = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix}$$

$$= -1$$

$$\operatorname{adj}(A) = (\text{Cofactor } A)^T$$

$$= \begin{bmatrix} 0 & -1 \\ -1 & 1 \end{bmatrix}^T$$

$$= \begin{bmatrix} 0 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{-1}\begin{bmatrix} 0 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$$

Is given:—

$$B = \begin{bmatrix} 7 & 28 & 23 \\ 6 & 20 & 5 \end{bmatrix}$$

Now:— Decode process:—

$$= A^{-1}B$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 7 & 28 & 23 \\ 6 & 20 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0+6 & 0+20 & 0+5 \\ 7-6 & 28-20 & 23-5 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 20 & 5 \\ 1 & 8 & 18 \end{bmatrix}$$

So, the sequence:— 6 1 20 8 5 18

Now, Message is:— "FATHFR"

System of Linear Equations (SLE)

Ch: 2

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Linear equation:

⇒ A linear equation an equation in which the highest power of the variable is always 1.

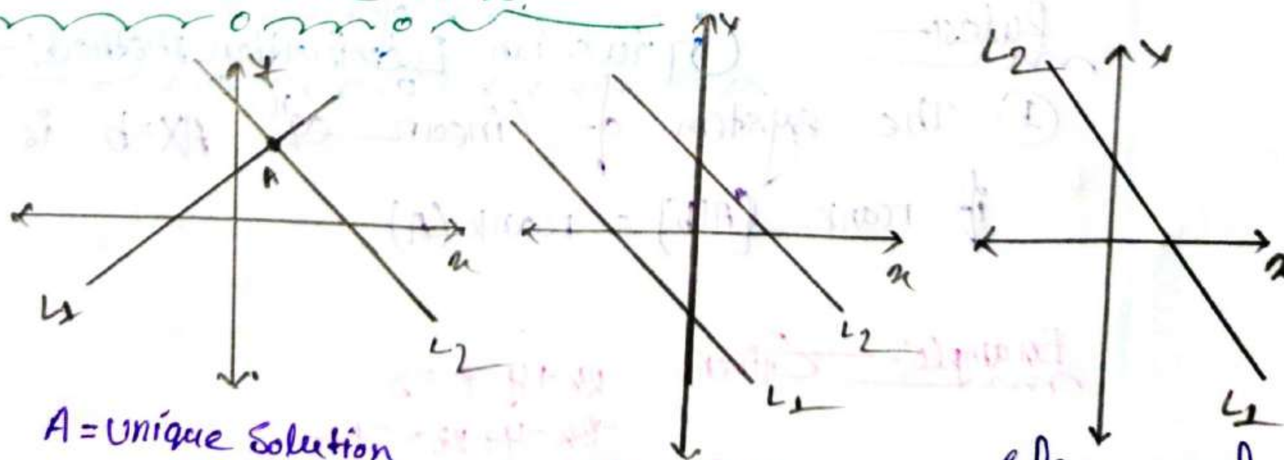
Example:

1. $2x+3y=8$ (linear)
2. $x_1+x_2+\cdots+x_n=1$ (linear)
3. $x^2+4x=8$ ....→ (Non-linear)

System of linear equations:

⇒ A sequence of numbers $\alpha_1, \alpha_2, \ldots, \alpha_n$ is called solution of the system of linear equation.

There are 3 case arise:



A = Unique Solution
It's called Consistent & Determinate

No solution
It's called Inconsistent

infinite No. of sol^n
It's called Consistent & Indeterminate.

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Matrices and system of LEq^n:

Suppose you have:—

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$
$$\vdots$$
$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$

In matrix form:—

$$AX = b$$

So, Size of $A \to n \times n$

$b \to n \times 1$

$\therefore X \to n \times 1$

Consistency Theorems: & Gaussian Elimination method:—

Rules:—

① The system of Linear eq^n $AX=b$ is Consistency if rank $[A|b]$ = rank(A)

Example:— Given,

$$2x+y-z=8$$
$$-3x-y+2z=-11$$
$$-2x+y+2z=-3$$

Test Consistency. If consistence find the sl^n:—

⇒ we know:—

In matrix $AX = b$

here, $A = \begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$

$b = \begin{bmatrix} 8 \\ -11 \\ -3 \end{bmatrix}$

For Consistency rank(A) = rank of [A|b]

Now:— $[A|b] = \left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array}\right]$ $\tilde{R}_3 = R_3 + R_2$

$= \left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ 0 & 2 & 1 & 5 \end{array}\right]$ $\tilde{R}_2 = 2R_2 + 3R_1$

$= \left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1 & 1 & 2 \\ 0 & 2 & 1 & 5 \end{array}\right]$ $\tilde{R}_3 = R_3 - 2R_2$

$= \left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & 1 \end{array}\right]$ $\tilde{R}_1 = R_1 - R_2$

$= \left[\begin{array}{ccc|c} 2 & 0 & -2 & 6 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & 1 \end{array}\right]$ $\tilde{R}_1 = \frac{R_1}{2}$, $\tilde{R}_3 = -R_3$

$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right]$

So, this in REF

rank of (A) = 3

rank of [A|b] = 3

So, this consistency.

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Now the Solution in:—

For Soln we need to Convert It RREF:—

$$=\left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right] \quad \begin{array}{l} R_1 \to R_1 + R_3 \\ R_2 \to R_2 - R_3 \end{array}$$

$\therefore x = 4$

$y = 3$

$z = -1$

So, It's Consistency and determinate.

Ans:—

Example: 2 Given:—

$3x + y - 6z = -10$

$2x + y - 5z = -8$

$6x - 3y + 3z = 0$

Test the Consistency if Consistence find the soln.

We know:—

In Matrix $AX = B$

here, $A = \begin{bmatrix} 3 & 1 & -6 \\ 2 & 1 & -5 \\ 6 & -3 & 3 \end{bmatrix}$

$B = \begin{bmatrix} -10 \\ -8 \\ 6 \end{bmatrix}$

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Now, For Consistency $[A|B]$ rank = A rank

$$\therefore [A|B] = \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 2 & 1 & -5 & -8 \\ 6 & -3 & 3 & 0 \end{array}\right] \quad \begin{array}{l} \tilde{R}_2 \to 3R_2 - R_3 \\ \tilde{R}_3 \to R_3 - 2R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 0 & 6 & -18 & +24 \\ 0 & -5 & 15 & +20 \end{array}\right] \quad \begin{array}{l} \tilde{R}_3 \to \frac{R_3}{5} \\ \tilde{R}_2 \to \frac{R_2}{6} \end{array}$$

$$= \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 0 & 1 & -3 & -4 \\ 0 & -1 & 3 & 4 \end{array}\right] \quad \tilde{R}_3 \to \tilde{R}_3 + R_2$$

$$= \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\therefore$ rank$(A) = 2$

rank$[A|B] = 2$

So, the system is consistent.

Now the soln:

$$= \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \tilde{R}_1 \to R_1 - R_2$$

$$= \left[\begin{array}{ccc|c} 3 & 0 & -3 & -6 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \tilde{R}_1 = \frac{R_1}{3}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

here z is a free variable

Let, $\boxed{z = a}$

$\therefore y - 3z = -4$

$\Rightarrow \boxed{y = -4 + 3a}$

$x - z = -2$

$\therefore \boxed{x = -2 + a}$

Cramer's Rule:-

Let, a system of linear equations is given $AX=B$ and $|A|=D$. This system is.

1. Inconsistent if $B\neq 0$ but $D=0$;
2. Consistent if $B=0$ and $D=0$.
3. Consistent if $D\neq 0$ & in this case the sol^n is given by $x_i=\frac{D_i}{D}$ $[i=1,2,\dots,n]$

Example:1:- Given:-

$$-3x+2y-3z=-8$$
$$2x-y+z=4$$
$$x+2y-4z=-2$$

Solve using Cremer's rule:-

$\Rightarrow$ We know:- In matrix form $AX=B$

where, $A=\begin{bmatrix}-3 & 2 & -3\\ 2 & -1 & 1\\ 1 & 2 & -4\end{bmatrix}$

$B=\begin{bmatrix}-8\\ 4\\ -2\end{bmatrix}$

$X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$

Now, $|A| = \begin{vmatrix} -3 & +2 & -3 \\ 2 & -1 & +1 \\ 1 & +2 & -4 \end{vmatrix}$

$= -3(4-2) - 2(-8-1) - 3(4+1)$

$= -3\times2 - 2\times-9 - 3\times5$

$= -3$

$\therefore |A_x| = \begin{vmatrix} -8 & 2 & -3 \\ 4 & -1 & 1 \\ -2 & 2 & -4 \end{vmatrix}$

$= -8(4-2) - 2(-16+2) - 3(8-2)$

$= -6$

$|A_y| = \begin{vmatrix} -3 & -8 & -3 \\ 2 & 4 & 1 \\ 1 & -2 & -4 \end{vmatrix}$

$= -3(-16+2) + 8(-8-1) - 3(-4-4)$

$= -6$

$|A_z| = \begin{vmatrix} -3 & 2 & -8 \\ 2 & -1 & 4 \\ 1 & 2 & -2 \end{vmatrix}$

$= -3(2-8) - 2(-4-4) - 8(4+1)$

$= -6.$

$\therefore x = \dfrac{|A_x|}{|A|} = \dfrac{-6}{-3} = 2$

$y = \dfrac{|A_y|}{|A|} = \dfrac{-6}{-3} = 2$

$z = \dfrac{|A_z|}{|A|} = \dfrac{-6}{-3} = 2$

Ans.

Matrix Inversion Method:-

In Matrix form:-

$$AX = B$$

$$\Rightarrow A^{-1}AX = A^{-1}B$$

$$\Rightarrow IX = A^{-1}B$$

$$\Rightarrow X = A^{-1}B$$

Example 1:- Given,

$$x_1 + x_2 + 2x_3 = 8$$

$$-x_1 - 2x_2 + 3x_3 = 1$$

$$3x_1 - 7x_2 + 4x_3 = 10$$

Solve using matrix inversion method:-

⇒ We know:-

$$X = A^{-1}B$$

where, $A = \begin{bmatrix} 1 & 1 & 2 \\ -1 & -2 & 3 \\ 3 & -7 & 4 \end{bmatrix}$

$B = \begin{bmatrix} 8 \\ 1 \\ 10 \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Now for A^{-1}:-

We know,

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

$$\operatorname{adj}(A) = (\text{Cofactor } A)^T$$

∴ Cofactor of A:-

$$A_{11} = \begin{vmatrix} -2 & 3 \\ -7 & 4 \end{vmatrix} = 13$$

$$A_{12} = -\begin{vmatrix} -1 & 3 \\ 3 & 4 \end{vmatrix} = 13$$

$$A_{13} = \begin{vmatrix} -1 & -2 \\ 3 & -7 \end{vmatrix} = 13$$

$A_{21} = -\begin{vmatrix} 1 & 2 \\ -7 & 4 \end{vmatrix} = -18$

$A_{22} = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2$

$A_{23} = -\begin{vmatrix} 1 & 1 \\ 3 & -7 \end{vmatrix} = 10$

$A_{31} = \begin{vmatrix} 1 & 2 \\ -2 & 3 \end{vmatrix} = 7$

$A_{32} = -\begin{vmatrix} 1 & 2 \\ -1 & 3 \end{vmatrix} = -5$

$A_{33} = \begin{vmatrix} 1 & 1 \\ -1 & -2 \end{vmatrix} = -1$

$$\therefore \ adj(A) = \begin{bmatrix} 13 & 13 & 13 \\ -18 & -2 & 10 \\ 7 & -5 & -1 \end{bmatrix}^T$$

$$= \begin{bmatrix} 13 & -18 & 7 \\ 13 & -2 & -5 \\ 13 & 10 & -1 \end{bmatrix}$$

Now, $\det(A) = \begin{vmatrix} 1 & 1 & 2 \\ -1 & -2 & 3 \\ 3 & -7 & 4 \end{vmatrix}$

$$= 1(-8+21) - 1(-4-9) + 2(7+6)$$

$$= 52$$

$$\therefore \ X = A^{-1}B$$

$$= \frac{1}{52}\begin{bmatrix} 13 & -18 & 7 \\ 13 & -2 & -5 \\ 13 & 10 & -1 \end{bmatrix}\begin{bmatrix} 8 \\ 1 \\ 10 \end{bmatrix}$$

$$= \frac{1}{52}\begin{bmatrix} 104-18+70 \\ 104-2-50 \\ 104+10-10 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

$\therefore \ x_1 = 3, \ x_2 = 1, \ x_3 = 2$

Ans.

System of LE:—

Formula:—

If $[A|b] \sim \sim \sim \left[\begin{array}{ccc|c} \checkmark & \checkmark & \checkmark & \checkmark \\ 0 & \checkmark & \checkmark & \checkmark \\ 0 & 0 & p & q \end{array}\right]$

1. Consistent with unique solⁿ if $p \neq 0$
2. Consistent with infinite solⁿ if $p = 0, q = 0$
3. Consistent with No solⁿ if $p = 0, q \neq 0$

Example:- 1:-

⇒ Determine the value(s) of λ such that the following linear equation

(i) no solution

(ii) more than solution

(iii) a unique solution.

$$x - 3z = -3$$
$$2x + \lambda y - z = -2$$
$$x + 2y + \lambda z = 1$$

⇒ here,

$$A = \begin{bmatrix} 1 & 0 & -3 \\ 2 & \lambda & -1 \\ 1 & 2 & \lambda \end{bmatrix}$$

$$b = \begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix}$$

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Now, 1st convert REF:—

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 0 & -3 & -3 \\ 2 & \lambda & -1 & -2 \\ 1 & 2 & \lambda & 1 \end{array}\right] \quad \begin{array}{l} \tilde{R}_2 = R_2 - 2R_1 \\ \tilde{R}_3 = R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -3 & -3 \\ 0 & \lambda & 5 & 4 \\ 0 & 2 & (\lambda+3) & 4 \end{array}\right] \quad \tilde{R}_3 = R_3 \times \lambda - 2R_2$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -3 & -3 \\ 0 & \lambda & 5 & 4 \\ 0 & 0 & (\lambda^2+3\lambda-10) & 4\lambda-8 \end{array}\right]$$

Now,

(i) For No Soln:—

$\lambda^2+3\lambda-10=0$

$\Rightarrow (\lambda+5)(\lambda-2)=0$

$\Rightarrow \lambda = -5$, only soln

$\therefore$

$4\lambda-8 \neq 0$

$\lambda \neq \frac{8}{4}$

$\lambda \neq \frac{8}{4}$

$\lambda \neq 2$

(ii) For more than 1 soln:—

$\lambda^2+3\lambda-10=0$

$(\lambda+5)(\lambda-2)=0$

$4\lambda-8=0$

$\lambda-2=0$

$\therefore \lambda = 2$, only

(iii) For unique soln:—

$\lambda^2+3\lambda-10 \neq 0$

$\therefore (\lambda+5)(\lambda-2) \neq 0$

$\therefore \lambda \neq -5, 2$

Example: 2

⇒ Determine the values of λ and μ such that the following system of linear equation has

(i) no solution.

(ii) more than one soln.

(iii) unique soln

$$\begin{cases} x+y+z=6 \\ x+2y+3z=10 \\ x+2y+\lambda z=\mu \end{cases}$$

here,

$$[A|b]=\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right] \quad \begin{array}{l} \tilde{R}_2 = R_2 - R_1 \\ \tilde{R}_3 = R_3 - R_1 \end{array}$$

$$=\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & (\lambda-1) & (\mu-6) \end{array}\right] \quad \tilde{R}_3 = R_3 - R_2$$

$$=\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & (\lambda-3) & (\mu-10) \end{array}\right]$$

Now,

(i) For No soln:

$\lambda - 3 = 0$

$\therefore \lambda = 3$

$\mu - 10 \neq 0$

$\mu \neq 10$

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(ii) For more than one sol^n:—

$\lambda - 3 = 0$ | $\mu - 10 = 0$

$\therefore \lambda = 3$ | $\mu = 10$

(iii) For unique Sol^n:—

$\lambda - 3 \neq 0$

$\therefore \lambda \neq 3$

Examples:—

⇒ Determine the value of λ and μ such that the following system of linear eq has (i)

(i) No Sol^n.

(ii) more than one sol^n.

(iii) a unique $soln$.

$$\begin{cases} 2x + 3y + z = 5 \\ 3x - y + \lambda z = 2 \\ x + 7y - 6z = \mu \end{cases}$$

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$\Rightarrow$ here, $[A|b] = \left[\begin{array}{ccc|c} 2 & 3 & 1 & 5 \\ 3 & -1 & \lambda & 2 \\ 1 & 7 & -6 & \mu \end{array}\right]$ $\tilde{R}_2 = R_2 \times 2 - R_1 \times 3$
$\tilde{R}_3 = R_3 \times 2 - R_1$

$= \left[\begin{array}{ccc|c} 2 & 3 & 1 & 5 \\ 0 & -11 & (2\lambda-3) & -11 \\ 0 & 11 & -13 & 2\mu-5 \end{array}\right]$ $\tilde{R}_3 = R_3 + R_2$

$= \left[\begin{array}{ccc|c} 2 & 3 & 1 & 5 \\ 0 & -11 & (2\lambda-3) & -11 \\ 0 & 0 & 2\lambda-16 & 2\mu-16 \end{array}\right]$

(i) For No soln:—

$2\lambda-16=0$ | $2\mu-16\neq 0$

$\lambda=8$ | $\mu\neq 8$

(ii) For more than One soln:—

$2\lambda-16=0$ | $2\mu-16=0$

$\therefore \lambda=8$ | $\mu=8$

(iii) For Unique soln:—

$2\lambda-16\neq 0$

$\therefore \lambda\neq 8$

Ans:—

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Example of Gaussian Elimination:-

Example 1:-

⇒ Solve the following system of equations using Gaussian elimination method.

$$2x+y+3z=1$$
$$2x+6y+8z=3$$
$$6x+8y+18z=5$$

⇒ Here:-

$$(A|b)=\left[\begin{array}{ccc|c} 2 & 1 & 3 & 1 \\ 2 & 6 & 8 & 3 \\ 6 & 8 & 18 & 5 \end{array}\right] \quad \begin{array}{l} \tilde{R}_2 = R_2 - R_1 \\ \tilde{R}_3 = R_2 - 3R_1 \end{array}$$

$$=\left[\begin{array}{ccc|c} 2 & 1 & 3 & 1 \\ 0 & 5 & 5 & 2 \\ 0 & 5 & 9 & 2 \end{array}\right] \quad \tilde{R}_3 = \tilde{R}_3 - R_2$$

$$=\left[\begin{array}{ccc|c} 2 & 1 & 3 & 1 \\ 0 & 5 & 5 & 2 \\ 0 & 0 & 4 & 0 \end{array}\right]$$

$\therefore$

$$z=0$$
$$5z+5y=2$$
$$y=\frac{2}{5}$$
$$2x+y+3z=1$$
$$x=\left(1-\frac{2}{5}\right)/2$$
$$=\frac{3}{10}$$

$$\therefore \quad (x,y,z)=\left(\frac{3}{10},\frac{2}{5},0\right)$$

Example: 2:-

⇒ Solve the following system of equations using elementary row operations.

$$3x+y-6z=-10$$
$$2x+y-5z=-8$$
$$6x-3y+3z=0$$

⇒ Here,

$$[A|b] = \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 2 & 1 & -5 & -8 \\ 6 & -3 & 3 & 0 \end{array}\right] \quad \begin{array}{l} \tilde{r}_2 = r_2\times 3 - r_1\times 2 \\ \tilde{r}_3 = r_3 - r_1\times 2 \end{array}$$

$$= \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 0 & 1 & -3 & -4 \\ 0 & -5 & 15 & 20 \end{array}\right] \quad \tilde{r}_3 = r_3 + 5r_2$$

$$= \left[\begin{array}{ccc|c} 3 & 1 & -6 & -10 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

here, z = Free variable.

$\therefore$ let, $z = a$

$\therefore 3x+y-6z = -10$

$\therefore 3x = -10+6z-y$

$= -10+6a-(-4+3a)$

$= -10+6a+4-3a$

$= -6+3a$

$\therefore x = -2+a$

$y-3z=-4$

$\therefore y = -4+3a$

$\therefore (x,y,z) = \{(-2+a), (-4+3a), a\}$

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Example-3:

⇒ Solve the following system of equations using Gaussian elemination method.

$$x + z = 1$$
$$x + y + z = 2$$
$$x - y + z = 1$$

here:-

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 1 \end{array}\right] \begin{array}{l} \tilde{r}_2 = r_2 - r_1 \\ \tilde{r}_3 = r_3 - r_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & 1 \end{array}\right] \tilde{r}_3 = r_3 + r_2$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

here, 3rd row:-

$0 = 1$ it does not Exit.

So, the system in inconsistent

Thath means no solution

Ans:

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Example of Cramer's Rule

Example:1:- Given

$2x+y+z=3$

$x-y-z=0$

$x+2y+z=0$

Solve the following SLE Using Cramer's Rule:-

⇒ We know:-

In Matrix From $AX=B$

here $A=\begin{bmatrix} 2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 1 \end{bmatrix}$

$x=\begin{bmatrix} x \\ y \\ z \end{bmatrix}$; $B=\begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$

$\therefore |A| = \begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 2 & 1 \end{vmatrix}$

$= 2(-1+2) - 1(1+1) + 1(2+1)$

$= 3$

$\therefore |Ax| = \begin{vmatrix} 3 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 2 & 1 \end{vmatrix}$

$= 3(-1+2) - 1(0-0) + 1(0+0)$

$= 3$

$$|A_y| = \begin{vmatrix} 2 & 3 & 1 \\ 1 & 0 & -1 \\ 1 & 0 & 1 \end{vmatrix}$$

$$= 2(0+0) - 3(1+1) + 1(0+0)$$

$$= -6$$

$$|A_z| = \begin{vmatrix} 2 & 1 & 3 \\ 1 & -1 & 0 \\ 1 & 2 & 0 \end{vmatrix}$$

$$= 2(0+0) - 1(0+0) + 3(2+1)$$

$$= 9$$

$$x = \frac{|A_x|}{|A|} = \frac{3}{3} = 1$$

$$y = \frac{|A_y|}{|A|} = \frac{-6}{3} = -2$$

$$z = \frac{|A_z|}{|A|} = \frac{6+3}{3} = 3$$

Ans:-

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Example 2:-

⇒ Solve the equations of SLE using Cramer's Rule:-

$2x+3y+z=10$

$x-y+z=4$

$4x-y-5z=-8$.

⇒ We know,

In matrix from:-

$AX=B$

$$A=\begin{bmatrix} 2 & 3 & 1 \\ 1 & -1 & 1 \\ 4 & -1 & -5 \end{bmatrix}$$

$$X=\begin{bmatrix} x \\ y \\ z \end{bmatrix} ; \quad B=\begin{bmatrix} 10 \\ 4 \\ -8 \end{bmatrix}$$

Now,

$$|A|=\begin{vmatrix} 2 & 3 & 1 \\ 1 & -1 & 1 \\ 4 & -1 & -5 \end{vmatrix}$$

$$=2(5+1)-3(-5-4)+1(-1+4)$$

$$=42$$

$$|Ax|=\begin{vmatrix} 10 & 3 & 1 \\ 4 & -1 & 1 \\ -8 & -1 & -5 \end{vmatrix}$$

$$=10(5+1)-3(-20+8)+1(-4-8)=84$$

$$|Ay|=\begin{vmatrix} 2 & 10 & 1 \\ 1 & 4 & 1 \\ 4 & -8 & -5 \end{vmatrix}$$

$$=2(-20+8)-10(-5-4)+1(-8-16)=42$$

$$A_z = \begin{vmatrix} 2 & 3 & 10 \\ 1 & -1 & 4 \\ 4 & -1 & -8 \end{vmatrix}$$

$$= 2(+8+4) - 3(-8-16) + 10(-1+4) = 126.$$

$$\therefore x = \frac{|A_x|}{|A|} = \frac{84}{42} = 2$$

$$y = \frac{|A_y|}{|A|} = \frac{42}{42} = 1$$

$$z = \frac{|A_z|}{|A|} = \frac{126}{42} = 3$$

Ans:-

Example: 3:- Given,

$$2x - 3y - 5z = 40$$
$$17x + 14y - 22z = 22$$
$$15x + 17y - 17z = -18$$

Solve the SLE using Cramers Rule:-

$\Rightarrow$ We know:-

In matrix from of $AX = B$

here, $A = \begin{bmatrix} 2 & -3 & -5 \\ 17 & 14 & -22 \\ 15 & 17 & -17 \end{bmatrix}$

$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$; $B = \begin{bmatrix} 40 \\ 22 \\ -18 \end{bmatrix}$

Now,

$$|A| = \begin{vmatrix} 2 & -3 & -5 \\ 17 & 14 & -22 \\ 15 & 17 & -17 \end{vmatrix}$$

$$= 2(-238+374) + 3(-289+330) - 5(289-210) = 0$$

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So, therefore, this's SLE is either inconsistent or more than one solutions.

Example: 4:- Given:

$2x - 5y + 6z = -27$

$10x - 11y - 9z = 0$

$-3x + 2z = 16$

Solve the SLE Using Cramer's Rule.

$\Rightarrow$ We know in matrix from,

$AX = B$

here,

$$A = \begin{bmatrix} 2 & -5 & 6 \\ 10 & -11 & -9 \\ -3 & 0 & 2 \end{bmatrix} ; \; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} ; \; B = \begin{bmatrix} -27 \\ 0 \\ 16 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & -5 & 6 \\ 10 & -11 & -9 \\ -3 & 0 & 2 \end{vmatrix}$$

$$= 2(-22-0) + 5(20-27) + 6(0-33)$$

$$= -277$$

$$|A_x| = \begin{vmatrix} -27 & -5 & 6 \\ 0 & -11 & -9 \\ 16 & 0 & 2 \end{vmatrix}$$

$$= -27(-22-0) + 5(0+144) + 6(0+176) = 2370$$

$$|A_y| = \begin{vmatrix} 2 & -27 & 6 \\ 10 & 0 & -9 \\ -3 & 16 & 2 \end{vmatrix} = 2(144) + 27(20-27) + 6(160) = 1059$$

$$|A_z| = \begin{vmatrix} 2 & -5 & -27 \\ 10 & -11 & 0 \\ -3 & 0 & 16 \end{vmatrix} = 2(-176) + 5(160) - 27(-33) = 1339$$

$\therefore x = \dfrac{|A_x|}{|A|} = \dfrac{2370}{-277}$; $y = \dfrac{|A_y|}{|A|} = \dfrac{1059}{-277}$

$z = \dfrac{|A_z|}{|A|} = \dfrac{1339}{-277}$.

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Example of Inversion Method:-

Examples:- Given,

$2x+y=1$

$x-2y=3$

Solve the following SOE using matrix Inversion method

Is given:-

$2x+y=1$

$x-2y=3$

We know:-

$X=A^{-1}B$

$A=\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$; $B=\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$A^{-1}=\frac{1}{\det(A)} \operatorname{adj}(A)$

$\operatorname{adj}(A)=(\text{Cofactor } A)^T$

Cofactor of A:-

$\operatorname{adj}(A)=\begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}^T$

$=\begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}$

$|A|=(-4-1)=-5$

$\therefore A^{-1}=\frac{1}{-5}\begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}$

Now:- $X=A^{-1}B=-\frac{1}{5}\begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$=-\frac{1}{5}\begin{bmatrix} -2-3 \\ -1+6 \end{bmatrix}=-\frac{1}{5}\begin{bmatrix} -5 \\ 5 \end{bmatrix}=\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\therefore x=1,\ y=-1$

Example! 3:- Given:-

$2x - y + 3z = 53$

$4x - z = -53$

$3x + 3y + 2z = 106$

Solve the following SLE using matrix Inversion and Justify your Answer.

We know:-

In matrix from

$$AX = B$$

$$\therefore A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 3 & 3 & 2 \end{bmatrix} ; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} ; B = \begin{bmatrix} 53 \\ -53 \\ 106 \end{bmatrix}$$

For Inversion methods:-

We know:-

$$X = A^{-1}B$$

Now, $|A| = \begin{vmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 3 & 3 & 2 \end{vmatrix}$

$= 2(+3) + 1(8+3) + 3(12-0) = 53$

$$adj(A) = \begin{bmatrix} 3 & -11 & 12 \\ 11 & -5 & -9 \\ 1 & 14 & 4 \end{bmatrix}^T$$

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$$\therefore A^{-1} = \frac{1}{53}\begin{bmatrix} 3 & 11 & 1 \\ 11 & -5 & 14 \\ 12 & -9 & 4 \end{bmatrix}$$

Now,

$$A^{-1}B = \frac{1}{53}\begin{bmatrix} 3 & 11 & 1 \\ 11 & -5 & 14 \\ 12 & -9 & 4 \end{bmatrix}\begin{bmatrix} 53 \\ -53 \\ 106 \end{bmatrix}$$

$$= \begin{bmatrix} -6 \\ 22 \\ 29 \end{bmatrix}$$

$\therefore x = -6$
$y = 22$
$z = 29$

Verification:-1

$AX = B$

$$\begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 3 & 3 & 2 \end{bmatrix}\begin{bmatrix} -6 \\ 22 \\ 29 \end{bmatrix} = \begin{bmatrix} 53 \\ -53 \\ 106 \end{bmatrix}$$

$$\begin{bmatrix} -12-22+87 \\ -24+0+(-29) \\ -18+66+58 \end{bmatrix} = \begin{bmatrix} 53 \\ -53 \\ 106 \end{bmatrix}$$

$$\begin{bmatrix} 53 \\ -53 \\ 106 \end{bmatrix} = \begin{bmatrix} 53 \\ -53 \\ 106 \end{bmatrix}$$

L.H.S = R.H.S

Verification:-2

R1:- $2x - y + 3z = 53$
$2\times -6 - 22 + 3\times 29 = 53$
$53 = 53$
L.H.S = R.H.S

R2:- $4x - z = -53$
$-24 - 29 = -53$
L.H.S = R.H.S

R3:- $3x + 3y + 2z = 106$
$3\times -6 + 3\times 22 + 2\times 29 = 106$
$106 = 106$
L.H.S = R.H.S

Ans:-

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#The Network Problem!

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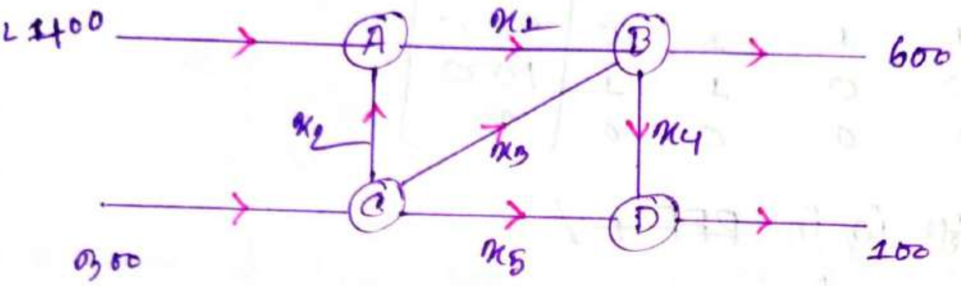
Example 4 :-

⇒ The network in the figure shows the traffic flow (in vehicles per hour) over the several one-way streets. Determine the general flow pattern for the Network



⇒ Each of the Network Four junctions gives rise to a linear equation. as shown below.

(A) $110 + x_2 = x_1 \Rightarrow x_1 - x_2 = 1400$

(B) $x_1 + x_3 = 600 + x_4 \Rightarrow x_1 + x_3 - x_4 = 600$

(C) $x_2 + x_3 + x_5 = 300$

(d) $x_4 + x_5 = 100$

⇒ Now the Augmented matrix is:-

$$\Rightarrow [A|b] = \left[\begin{array}{ccccc|c} 1 & -1 & 0 & 0 & 0 & 400 \\ 1 & 0 & 1 & -1 & 0 & 600 \\ 0 & 1 & 1 & 0 & 1 & 300 \\ 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right] \quad \tilde{R}_2 = R_2 - R_1$$

$$= \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 400 \\ 0 & +1 & 1 & -1 & 0 & 200 \\ 0 & 1 & 1 & 0 & 1 & 300 \\ 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right] \quad \tilde{R}_3 = R_3 - R_2$$

$$= \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 400 \\ 0 & 1 & 1 & -1 & 0 & 200 \\ 0 & 0 & 0 & 1 & 1 & 100 \\ 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right] \quad \tilde{R}_4 = R_4 - R_3$$

$$= \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 400 \\ 0 & 1 & 1 & -1 & 0 & 200 \\ 0 & 0 & 0 & 1 & 1 & 100 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Now, it in in REF /

Ans:- rank of $(A) = 3$

Rank of $[A|b] = 3$

The free variable are $= x_3, x_5$

Let,

$\boxed{x_3 = a}$

$\boxed{x_5 = b}$

Now, $x_1 + x_2 = 400$

$x_4 + x_5 = 100$

$\boxed{x_4 = 100 - b}$

So, $x_4 \geq 0$, then $x_5 \leq 100$

Again:- $x_1 - x_2 = 400$

$x_2 = x_1 - 400$

$\boxed{x_1 = 400 + x_2}$

$x_2 + x_3 - x_4 = 200$

$x_2 = 200 + x_4 - x_3$

$x_2 = 200 + 100 - b - x_3$

$= 300 - b - x_3$

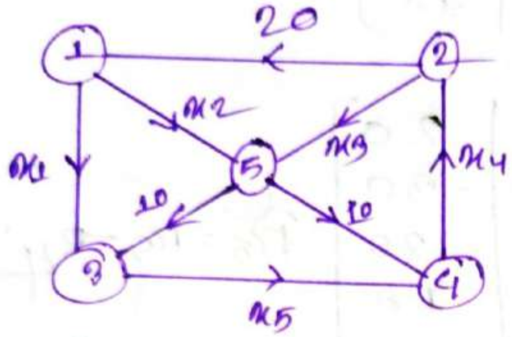
$\boxed{x_2 = 300 - (x_3 + x_5)}$

So, $x_2 \geq 0$, then $x_3 + x_5 \leq 300$

$x_3 \leq 300 - x_5$

Example: 2 :-

⇒ Set up a system of linear equations to represent the Network shown in the following figure. And solve the system.



⇒ Each of the Network's gives five junctions.

(1) $x_1 + x_2 = 20$

(2) $x_3 - x_4 = -20$

(3) $x_1 - x_5 = -10$

(4) $x_4 - x_5 = 10$

(5) $x_2 + x_3 = 10 + 10$

Now:-

$$[A|b] = \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 1 & 0 & 0 & 0 & -1 & -10 \\ 0 & 0 & 0 & 1 & -1 & 10 \\ 0 & 1 & 1 & 0 & 0 & 20 \end{array}\right] \quad R_3 = R_3 - R_1$$

$$= \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & -1 & 0 & 0 & -1 & -30 \\ 0 & 0 & 0 & 1 & -1 & 10 \\ 0 & 1 & 1 & 0 & 0 & 20 \end{array}\right] \quad \begin{array}{l} R_2 \leftrightarrow R_3 \\ R_4 \leftrightarrow R_5 \end{array}$$

$$= \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & -1 & 0 & 0 & -1 & -30 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 1 & 1 & 0 & 0 & 20 \\ 0 & 0 & 0 & -1 & -1 & 10 \end{array}\right] \quad R_4 = R_4 + R_2$$

$$= \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & -1 & 0 & 0 & -1 & -30 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 0 & 1 & 0 & -1 & -10 \\ 0 & 0 & 0 & 1 & -1 & 10 \end{array}\right] \quad \tilde{R_4} = R_4 - R_3$$

$$= \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & -1 & 0 & 0 & -1 & -30 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 0 & 0 & 1 & -1 & 10 \\ 0 & 0 & 0 & 1 & -1 & 10 \end{array}\right] \quad \tilde{R_5} = R_5 - R_4$$

$$= \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & -1 & 0 & 0 & -1 & -30 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 0 & 0 & 1 & -1 & 10 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

∴ Now it's in REF.

& Rank of $(A) = 4$

Rank of $[A|b] = 4$

So, the free variable is $= x_5$

Let, $\boxed{x_5 = a}$

So, $x_4 - x_5 = 10$

$\boxed{x_4 = 10 + x_5}$

$x_3 - x_4 = -20$

$x_3 = -20 + x_4$

$= -20 + 10 + x_5$

$\boxed{x_3 = -10 + x_5}$

So, $x_3 \geq 0$, then $x_5 \geq 10$

$-x_2 - x_5 = -30$

$\boxed{x_2 = 30 - x_5}$

$x_2 \geq 0$ then, $x_5 \leq 30$

$x_1 + x_2 = 20$

$x_1 = 20 - x_2$

$= 20 - (30 - x_5)$

$\boxed{x_1 = -10 + x_5}$

$x_1 \geq 0$ then $x_5 \geq 10$

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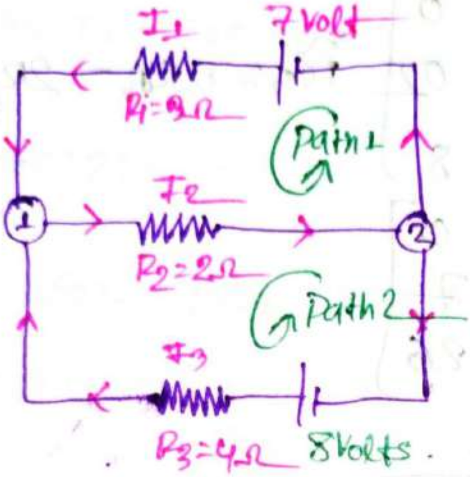
Example: 3:-

Determine the Current I_1, I_2, I_3 For electrical Network shown in the figure.



$\Rightarrow$ Appling KCL to each Junction :-

(1) $I_1 + I_3 = I_2$

(2) $I_2 = I_1 + I_3$

For Path 1

$P1 \rightarrow I_1R_1 + I_2R_2 = 7$

$3I_1 + 2I_2 = 7 \quad \ldots\ldots$ (i)

For Path 2 :-

$P2 \rightarrow I_2R_2 + I_3R_3 = 8$

$2I_2 + 4I_3 = 8 \quad \ldots\ldots$ (ii)

Now following SLE For I_1, I_2, I_3:-

$I_1 + I_3 - I_2 = 0$

$3I_1 + 2I_2 = 7$

$2I_2 + 4I_3 = 8$

$$[A|b] = \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 3 & 2 & 0 & 7 \\ 0 & 2 & 4 & 8 \end{array}\right] \quad \tilde{R}_2 = R_2 - 3R_1$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 5 & -3 & 7 \\ 0 & 2 & 4 & 8 \end{array}\right] \quad \tilde{R}_3 = 5R_3 - 2R_2$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 5 & -3 & 7 \\ 0 & 0 & 26 & 26 \end{array}\right]$$

Now, it's in REF

So, Rank of $A = 3$

Rank of $[A|b] = 3$

$26I_3 = 26$

$\therefore I_3 = 1A$

$5I_2 - 3I_3 = 7$

$I_2 = \frac{7+3}{5}$

$= 2A$

$I_1 - I_2 + I_3 = 0$

$I_1 = I_2 - I_3$

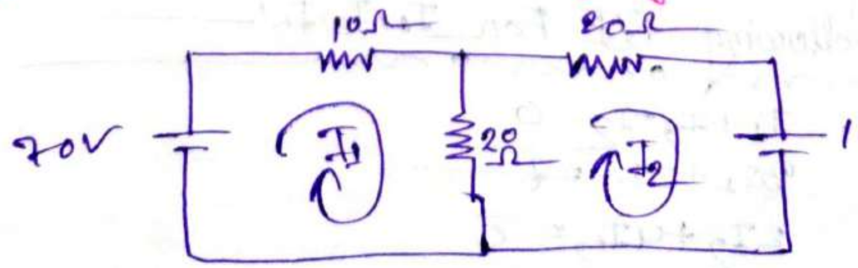
$= 2 - 1$

$= 1A$

Ans.

Examples 4:

⇒ Determine the loop currents I_1 and I_2 of the following circuit using mesh analysis.



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$\Rightarrow$ For Loop 1:—

$$10I_1 + 20I_1 - 20I_2 = 70$$

$$30I_1 - 20I_2 = 70$$

For Loop : 2:—

$$20I_2 + 20I_2 - 20I_1 = -100$$

$$-20I_1 + 40I_2 = -100$$

Now, SLE is:—

$$30I_1 - 20I_2 = 70$$
$$-20I_1 + 40I_2 = -100$$

So, $(A|b) = \left[\begin{array}{cc|c} 30 & -20 & 70 \\ -20 & 40 & -100 \end{array}\right]$ $\tilde{R}_2 = R_2 \times 3 + 2R_1$

$$= \left[\begin{array}{cc|c} 30 & -20 & 70 \\ 0 & 80 & -160 \end{array}\right]$$

$\therefore I_2 = -2A$

$$30I_1 - 20I_2 = 70$$

$\therefore I_1 = 1A$

Ans.

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Linear Programing Problem

Example:1:-

Example: A company manufactures and sells two models of lamps, L1 and L2. To manufacture each lamp, the manual work involved in model L1 is 20 minutes and for L2, 30 minutes. The mechanical (machine) work involved for L1 is 20 minutes and for L2, 10 minutes. The manual work available per month is 100 hours and the machine is limited to only 80 hours per month. Knowing that the profit per unit is 15 and 10 for L1 and L2, respectively, determine the quantities of each lamp that should be manufactured to obtain the maximum benefit.

⇒ Let:-

No. of lamps $L_1 = x$.

No. of lamps $L_2 = y$.

and objective function $f(x,y) = 15x + 10y$

Now, Convert the time into hours:-

$20 \text{ min} = \frac{1}{3} h$

$30 \text{ min} = \frac{1}{2} h$

$10 \text{ min} = \frac{1}{6} h$

Now:- Time taken in manual and machine:-

	L_1	L_2	Time
Manual	$\frac{1}{3} h$	$\frac{1}{2} h$	$100 h$
Machine	$\frac{1}{3} h$	$\frac{1}{6} h$	$80 h$

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So, the Eq: is:-

$$\frac{1}{3}x+\frac{1}{2}y\le 100 \quad \text{......} \quad (\mathrm{i})$$

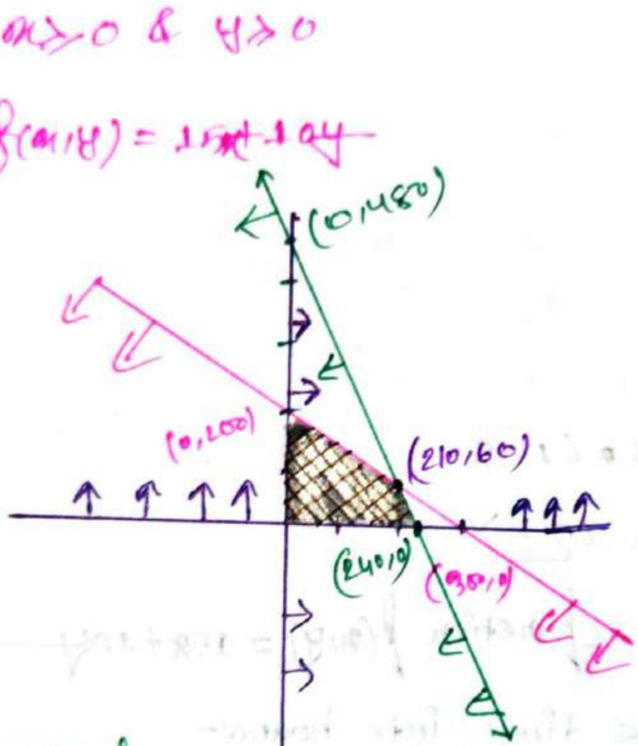
$$\frac{1}{3}x+\frac{1}{6}y\le 80 \quad \text{......} \quad (\mathrm{ii})$$

Since the lamps Number can't be <0.

So, $x\geq 0$ & $y\geq 0$

Let, $z=f(x,y)=15x+10y$

Now,



$$\frac{x}{300}+\frac{y}{200}\le 1$$

$$\frac{x}{240}+\frac{y}{480}\le 1$$

So, the soln of eq (i) and (ii)

$x=210$
$y=60$

$\therefore z=15x+10y$

$(0,200)\Rightarrow z=15\times 0+10\times 200=2000$ units
$(240,0)\Rightarrow z=15\times 240+10\times 0=3600$ "
$(210,60)\Rightarrow z=15\times 210+10\times 60=3750$ "

So, if $x=210$, $y=60$
then $f(x,y)$ is maximum.

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Example: A calculator company manufactures two types of calculator: a handheld calculator and a scientific calculator. Statistical data projects that there is an expected demand of at least 100 scientific and 80 handheld calculators each day. Since the company has certain limitations on the production capacity, the company can only manufacture 200 scientific and 170 handheld calculators per day. The company has received a contract to deliver a minimum of 200 calculators per day. If there is a loss of 2 taka on each scientific calculator that you sold and a profit of 5 taka on each handheld calculator, then how many calculators of each type the company should manufacture daily to maximize the net profit?

=> Let :-

No. of Scientific Calculator $= x$

No. of handheld " $= y$.

and the objective $P = f(x,y) = (x \times -2 + y \times 5)$

$= -2x + 5y$

Now :-

The lower bound : $100 \le x$, $80 \le y$

The uper " $200 \ge x$, $170 \ge y$

Contract bound . $x + y \ge 200$

So, the Condition is :-

$100 \le x \le 200$

$80 \le y \le 170$

$x + y \ge 200$

Since the calculator Number can't be negative.

So. $x \ge 0$, $y \ge 0$

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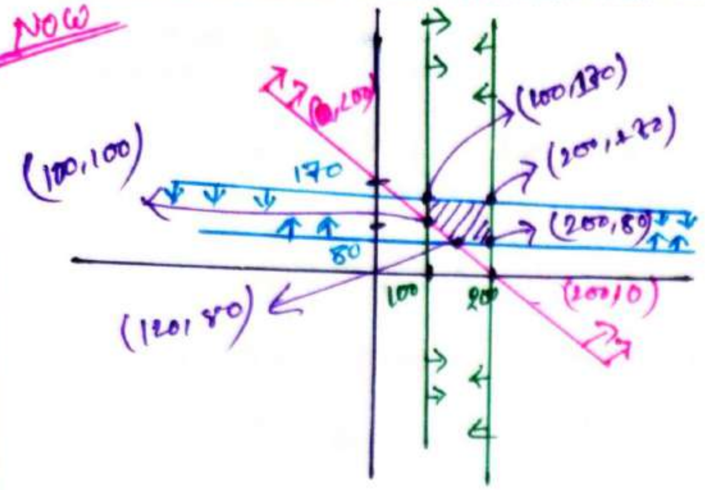
NOW



$x+y \geq 200$

$\frac{x}{200}+\frac{y}{200} \geq \frac{200}{200}$

$\frac{x}{200}+\frac{y}{200} \geq 1$

So, $P = f(x,y) = -2x + 5y$

$\Rightarrow (100,170) = -2\times100 + 5\times170 = 650$

$\Rightarrow (200,170) = -2\times200 + 170\times5 = 450$

$\Rightarrow (200,80) = -2\times200 + 80\times5 = 0$

$\Rightarrow (120,80) = -2\times120 + 80\times5 = 160$

$\Rightarrow (100,100) = -2\times100 + 5\times100 = 300$

So, if $x = 100$, $y = 170$

then profit $f(x,y)$ is maximum.

Ans:-

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Example : 3:-

. A company produces two products, *A* and *B*. The sales volume for *A* is at least 80% of the total sales of both *A* and *B*. However, the company cannot sell more than 110 units of *A* per day. Both products use one raw material, of which the maximum daily availability is 300 lb. The usage rates of the raw material are 2 lb per unit of *A*, and 4 lb per unit of *B*. The profit units for *A* and *B* are \$40 and \$90, respectively. Determine the optimal product mix for the company.

⇒ Let:-

No. of units of the products of $A = x$

No. of " " " " of $B = y$

Now the conditions:-

1. $x \geq 80\%(x+y)$

$\Rightarrow x \geq 0.8x + 0.8y$

$\Rightarrow 0.2x - 0.8y \geq 0 \quad \cdots \text{(i)}$

2. $x \leq 110 \quad \cdots \text{(ii)}$

3. $2x + 4y \leq 300 \quad \cdots \text{(iii)}$

And:- the objective function:-

$Z = 4x + 90y$

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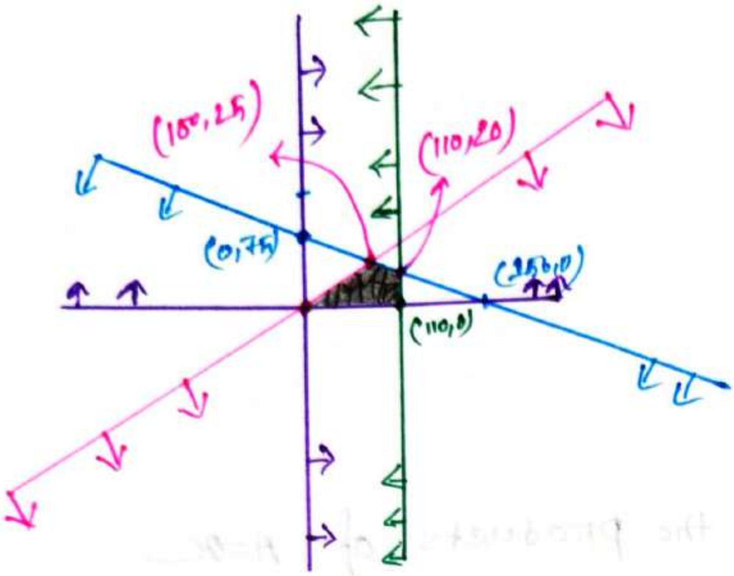
NOW



(i)

$2x+4y \le 300$

$\Rightarrow \frac{x}{150}+\frac{y}{75} \le 1$

(ii)

$0.2x-0.8y \ge 0$

$\Rightarrow 0.8y \le 0.2x$

$y \le \frac{0.2x}{0.8}$

$\therefore y \le \frac{1}{4}x$

(iii)

$x \le 110$

So, $P=f(x,y)= 4x+10y$

$\Rightarrow (100,25)=f(100,25)=6250$

$\Rightarrow (110,20)=f(110,20)=6200$

$\Rightarrow (110,0)=f(110,0)=4400$

So, if $x=100, y=25$

then profit $f(x,y)$ is maximum

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#Fourier Analysis

Ch:03

TOPIC NAME :

DAY :

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#Even & Odd Function:

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Even and Odd Function:-

Formula:-

1. A function $y=f(x)$ is called even if $f(-x)=f(x)$

2. A function $y=f(x)$ is called odd if $f(-x)=-f(x)$.

Example:-

① $f(x)=x$

$\Rightarrow f(-x)=-x$
$=-f(x)$

So, it's odd

② $f(x)=x^2$

$\Rightarrow f(-x)=(-x)^2$
$=x^2$
$=f(x)$

It's even.

③ $f(x)=e^x$

It's Not odd or not even.
that mean's Neither.

④ $f(x)=\sin x$

$f(-x)=\sin(-x)$
$=-\sin x$
$=-f(x)$

So' It's odd

⑤ $f(x)=\cos x$

$f(-x)=\cos(-x)$
$=\cos x$
$=f(x)$

So, It's even.

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Arithmatic Combination of Even and Odd function:—

Properties:—

1. odd × odd = even
2. even × even = even
3. odd × even = odd
4. even + even = even
5. odd + odd = odd
6. even − even = even
7. odd − odd = odd
8. even ± odd = neither
9. even/even = even
10. odd/odd = even
11. even/odd = odd

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Calculus Properties of Even and odd Function:-

Formula:-

① If $f(x)$ is an even function, that is $f(-x)=f(x)$ for all x then

$$\int_{-a}^{a} f(x)\cdot dx = 2\int_{0}^{a} f(x)\cdot dx$$

Ex:- $\int_{-3}^{3} x^2\cdot dx$

$= 2\int_{0}^{3} x^2\cdot dx$

$= 2\times[\frac{1}{3}x^3]_0^3$

$= 18$

Let

$f(x) = x^2$

$\therefore f(-x) = (-x)^2$

$= x^2$

$= f(x)$

So, It's a even function.

② If $f(x)$ is a odd function that is $f(x)=-f(x)$ for all x than.

$$\int_{-a}^{a} f(x)\cdot dx = 0$$

Ex:- $\int_{-4}^{4} x^3\cdot dx = 0$

Let

$f(x) = x^3$

$f(-x) = (-x)^3$

$= -x^3$

$= -f(x)$

So, It's a odd function.

Periodic Function:-

Formula:-

① Let $T > 0$,
A function $f(x)$ is said to be a periodic function if $f(x+nT) = f(x)$, $n \in Z$
where 'T' is called period of $f(x)$.

Exercise: 3.1

① Determine the period of the following functions.

ⓐ $f(x) = \sin x$
$= \sin(2\pi + x)$
$= \sin x$
So, Period $= 2\pi$

ⓑ $f(x) = \cos x$
$= \cos(2\pi + x)$
$= \cos x$
So, Period $= 2\pi$

ⓒ $f(x) = \tan x$
$= \tan(\pi + x)$
$= \tan x$
So, Period $= \pi$

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Short-Cut-Formula :-

(1.) if $f(x) = \sin(bx \pm c) + d$

so, Period $= \frac{2\pi}{b}$ (Also For $\sin^2 x$)

(2.) if $f(x) = \cos(bx \pm c) + d$

So, Period $= \frac{2\pi}{b}$

(3) if $f(x) = \tan(bx \pm c) + d$

So, Period $= \frac{\pi}{b}$

(4) if $f(x) = \cos^2(x)$

So, Period $= \frac{\cos(x)}{2}$

$= \frac{\frac{2\pi}{1}}{2}$

$= \pi$

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Exercise 3.1

(1) Find the period of the function:—

(a) $f(x) = \sin 5x$

$\Rightarrow$ we know:—

Period of $\sin ax = \frac{2\pi}{a}$

$\therefore \sin 5x = \frac{2\pi}{5}$

(b) $f(x) = \cos 7x$

$\Rightarrow$ we know:—

Period of $\cos ax = \frac{2\pi}{a}$

$\therefore \cos 7x = \frac{2\pi}{7}$

(c) $f(x) = \tan 7x$

we know:—

Period of $\tan ax = \frac{\pi}{a}$

$\therefore \tan 7x = \frac{\pi}{7}$

(d) $f(x) = \cos^2 x$.

$$f(\pi+x) = \cos(\pi+x)^2$$
$$= (-\cos x)^2$$
$$= \cos x^2$$
$$= f(x)$$

$\therefore$ Period $= \pi$

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(d) $f(x) = \sin^2 x - \cos^2 x$

we know:—

Period of $\sin ax / \sin^2 ax = \frac{2\pi}{a}$

$\therefore \sin^2 x = \frac{2\pi}{1} = 2\pi$

Period of $\cos x = \pi$

$\therefore \sin^2 x - \cos^2 x = 2\pi - \pi$
$= \pi$

Ans:—

(e) $f(x) = \sin 2x + \cos 3x$

we know:—

Period of $\sin ax = \frac{2\pi}{a}$

$\therefore \sin 2x = \pi$

Period of $\cos ax = \frac{2\pi}{a}$

$\therefore \cos 3x = \frac{2\pi}{3}$

$\therefore \sin 2x + \cos 3x = \left(\pi + \frac{2\pi}{3}\right)$
$= \frac{LCM(3\pi, 2\pi)}{3}$
$= \frac{6\pi}{3}$
$= 2\pi$

Ans:—

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2. Find the following function is even or odd.

(b) $f(x) = x^3 - x$

$\therefore f(-x) = (-x)^3 - (-x)$

$= -x^3 + x$

$= -(x^3 - x)$

$= -f(x)$

$\therefore f(x)$ is a odd function.

(c) $f(x) = \tan x$

$\therefore f(-x) = \tan(-x)$

$= -\tan x$

$= -f(x)$

$\therefore f(x)$ is a odd function.

(d) $f(x) = e^{x^2}$

$\therefore f(-x) = e^{(-x)^2}$

$= e^{x^2}$

$= f(x)$

$\therefore f(x)$ is a even function

(f) $f(x) = \dfrac{e^{3x} + e^{-3x}}{2}$

$\therefore f(-x) = \dfrac{e^{-3x} + e^{3x}}{2}$

$= \dfrac{e^{3x} + e^{-3x}}{2}$

$= f(x)$

$\therefore f(x)$ is a even function

(g) $f(x) = \dfrac{e^{x} - e^{-x}}{2}$

$\therefore f(-x) = \dfrac{e^{-x} - e^{x}}{2}$

$= \dfrac{-e^{x} + e^{-x}}{2}$

$= -\dfrac{(e^{x} - e^{-x})}{2}$

$= -f(x)$

$\therefore f(x)$ is a odd function

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#Fourier Series:—

Formula:—

⇒ $f(x)$ is definied in $-L<x<L$

(1) $$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi}{L} x\right) + b_n \sin\left(\frac{n\pi}{L} x\right)\right]$$

where:—

(i) $$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\cdot dx$$

(ii) $$a_n = \frac{1}{L}\int_{-L}^{L} \cos\left(\frac{n\pi}{L}x\right)\cdot f(x)\cdot dx \qquad [n=1,2,3\ldots]$$

(iii) $$b_n = \frac{1}{L}\int_{-L}^{L} \sin\left(\frac{n\pi}{L}x\right)\cdot f(x)\cdot dx \qquad [n=1,2,3,\ldots]$$

#Some properties:—

(1) If $f(x)$ is even then $b_n=0$ for all n

(2) If $f(x)$ is odd then $a_n=0$, $a_0=0$ for all n.

(3) $\sin n\pi = 0$, $n=1,2,3,4\ldots$

(4) $\cos n\pi = (-1)^n$, $n=1,2,3,4,\ldots$

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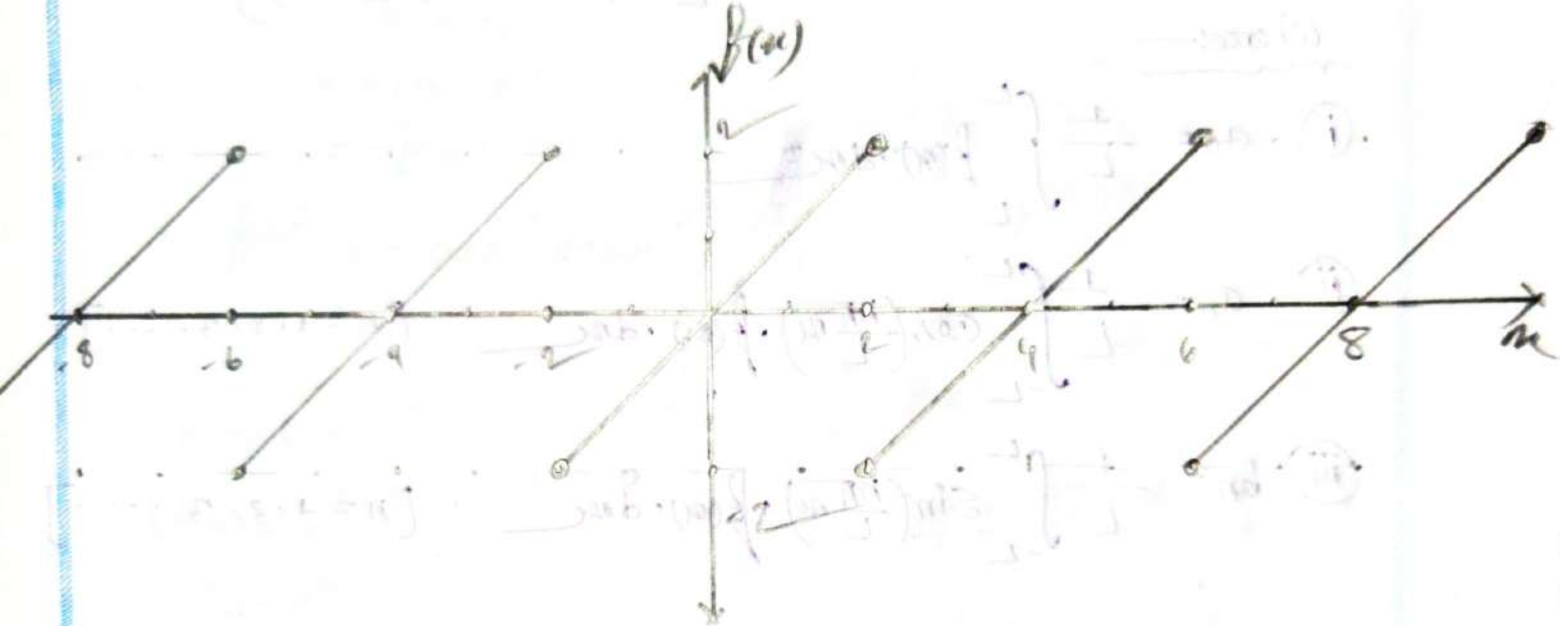
Example 1:

⇒ Find the FS (Fourier Series) of $f(x) = x$. defined in $-2<x<2$.
Also sketch the function $f(x+4) = f(x)$.

⇒



⇒ We know:

The FS:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi}{2}x\right) + b_n \sin\left(\frac{n\pi}{2}x\right)\right]$$

$\therefore$ $a_0 = \frac{1}{2}\int_{-2}^{2} f(x)\cdot dx$

$= \frac{1}{2}\int_{-2}^{2} x\cdot dx = 0$

Let,
$f(x) = x$
$\therefore f(-x) = -x$
$= -f(x)$
= odd function

$a_n = \frac{1}{2}\int_{-2}^{2} x\cdot \cos\left(\frac{n\pi}{2}x\right)\cdot dx$

$= 0$

cz:
We know:
odd × even = odd
$f(x) = x$ = odd
$\cos(x)$ = even

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$b_n = \frac{1}{2}\int_{-2}^{2} x\cdot \sin\left(\frac{n\pi}{2}x\right)\cdot dx$

we know;—
odd × odd = even

$= \frac{1}{2}\times 2\int_{0}^{2} x\cdot \sin\left(\frac{n\pi}{2}x\right)\cdot dx$

$= \left[(x)\left(-\frac{2}{n\pi}\right)\cos\left(\frac{n\pi}{2}x\right) - (1)\left(-\frac{2}{n\pi}\right)\left(\frac{2}{n\pi}\right)\cdot \sin\left(\frac{n\pi}{2}x\right)\right]_0^2$

$= \left\{2\times -\frac{2}{n\pi}\cos(n\pi)\right\} + \left\{\frac{4}{n^2\pi^2}\times \sin(n\pi)\right\}$

$= -\frac{4}{n\pi}(-1)^n + 0$

$= -\frac{4}{n\pi}(-1)^n$

$= \frac{4}{n\pi}(-1)^{n+1}$

Ans:—

So, the F.S:—

$f(x) = 0 + \sum_{n=1}^{\infty}\left[0\times \cos\left(\frac{n\pi}{2}x\right) + \frac{4}{n\pi}(-1)^{n+1}\cdot \sin\left(\frac{n\pi}{2}x\right)\right]$

$= \sum_{n=1}^{\infty}\left[\frac{4}{n\pi}(-1)^{n+1} \sin\left(\frac{n\pi}{2}x\right)\right]$

Ans:—

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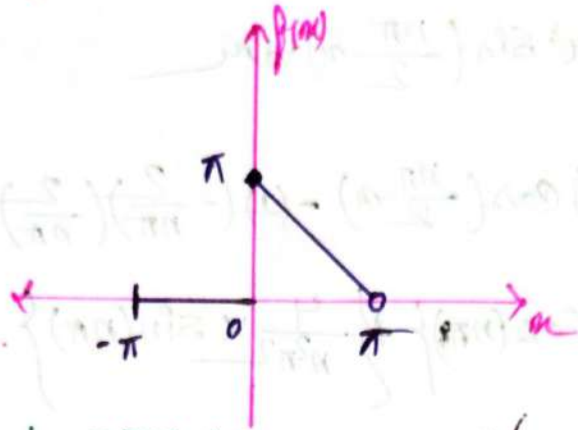
Example: 5:-

⇒ Find the F.S for the wave from below.



$$\therefore f(x) = \begin{cases} 0 & ; -\pi < x < 0 \\ \pi - x & ; 0 < x < \pi \end{cases}$$

$\therefore f(x)$ is define in $[-\pi, \pi]$

$$\frac{x}{a} + \frac{y}{b} = 1$$
$$\frac{x}{\pi} + \frac{y}{\pi} = 1$$
$$\frac{x+y}{\pi} = 1$$
$$\therefore \boxed{y = \pi - x}$$

we know:

The F.S:-

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi}{\pi}x\right) + b_n \sin\left(\frac{n\pi}{\pi}x\right)\right] \cdots (i)$$

$$\therefore a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cdot dx$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\cdot dx + \int_{0}^{\pi} (\pi - x)\cdot dx\right]$$

$$= \frac{1}{\pi}\left[\pi x - \frac{x^2}{2}\right]_0^{\pi}$$

$$= \pi - \frac{\pi}{2}$$

$$= \frac{\pi}{2}$$

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$$a_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\cdot \cos\left(\frac{n\pi}{\pi}x\right) + \int_{0}^{\pi} f(x)\cdot \cos\left(\frac{n\pi}{\pi}x\right)\right]$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\cdot \cos\left(\frac{n\pi}{\pi}\cdot x\right) + \int_{0}^{\pi} (\pi - x)\cdot \cos\left(\frac{n\pi}{\pi}x\right)\right]$$

$$= \frac{1}{\pi}\left[0 + \int_{0}^{\pi} (\pi - x)\cdot \cos\left(\frac{n\pi}{\pi}x\right)\right]$$

$$= \frac{1}{\pi}\left[\int_{0}^{\pi} (\pi - x)\cdot \cos(nx)\right]$$

$$= \frac{1}{\pi}\left[(\pi - x)\left(\frac{1}{n}\right)\sin(nx) - (-1)\left(\frac{1}{n}\right)\left(-\frac{1}{n}\right)\cos(nx)\right]_{0}^{\pi}$$

$$= \frac{1}{\pi}\left[0 - \frac{1}{n^2}\cos(nx)\right]_{0}^{\pi}$$

$$= \frac{1}{\pi}\left[0 - \left(\frac{1}{n^2}\cos(n\pi) - \frac{1}{n^2}\cos(0)\right)\right]$$

$$= \frac{1}{\pi}\left[\frac{1}{n^2} - \frac{1}{n^2}(-1)^{n}\right]$$

$$= \frac{1}{n^2\pi}\left(1 - (-1)^{n}\right]$$

$$b_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\cdot\left(\sin\frac{n\pi}{\pi}\cdot x\right) + \int_{0}^{\pi} (\pi - x)\cdot \sin(nx)\right]$$

$$= \frac{1}{\pi}\left[(\pi - x)\left(-\frac{1}{n}\cos nx\right) - (-1)\left(-\frac{1}{n^2}\sin nx\right)\right]_{0}^{\pi}$$

$$= \frac{1}{\pi}\left[-\left(\pi \times -\frac{1}{n}(-0)^{n}\right)\right]$$

$$= \frac{1}{\pi}\left[+\pi \times \frac{1}{n}\right]$$

$$= \frac{\pi}{\pi n} = \frac{1}{n}$$

so, The F.S:-

$$f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty}\left[\frac{1}{n^2\pi}\left[1-(-1)^{n}\right]\cos nx + \frac{1}{n}\sin nx\right]$$ Ans:-

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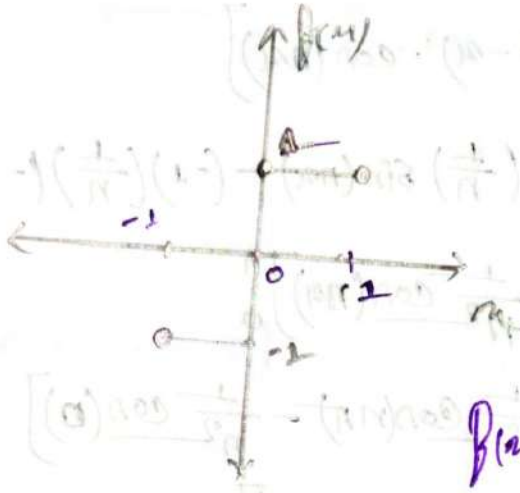
Example: 06:

⇒ Find the Fourier Series Expansion for the Standard Square wave. Also sketch the function.

$$f(x) = \begin{cases} -1 & ; \ (-1 < x < 0) \\ +1 & ; \ (0 \le x < +1) \end{cases}$$

⇒



f(x) define in $(-1, +1)$

We know:

The F.S:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(n\pi x) + b_n \sin(n\pi x) \right] \quad \text{----} \ (1)$$

Now,

$$a_0 = \frac{1}{1} \int_{-1}^{1} f(x) \cdot dx$$

$$= \int_{-1}^{0} -1 \cdot dx + \int_{0}^{1} 1 \cdot dx$$

$$= [-x]_{-1}^{0} + [x]_{0}^{1}$$

$$= -(+1) + 1$$

$$= 0$$

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$$a_n = \frac{1}{1}\left[\int_{-1}^{0} -1\times \cos(n\pi x)\,dx + \int_{0}^{1} 1\times \cos(n\pi x)\cdot dx\right]$$

$$= -\left[\frac{1}{n\pi}\sin(n\pi x)\right]_{-1}^{0} + \left[\frac{1}{n\pi}\sin(\pi n x)\right]_{0}^{1}$$

$$= \frac{-1}{n\pi}\sin n\pi + \frac{1}{n\pi}\sin n\pi$$

$$= 0$$

$$b_n = \frac{1}{1}\left[\int_{-1}^{0} -1\times \sin(n\pi x)\cdot dx + \int_{0}^{1} 1\times \sin(n\pi x)\cdot dx\right]$$

$$= -\left[-\frac{1}{n\pi}\cos(n\pi x)\right]_{-1}^{0} + \left(-\frac{1}{n\pi}\cos(n\pi x)\right]_{0}^{1}$$

$$= \frac{1}{n\pi}\left[1-\cos(n\pi)\right] - \frac{1}{n\pi}\left[\cos(n\pi) - 1\right]$$

$$= \frac{1}{n\pi} - \frac{1}{n\pi}\cos(n\pi) - \frac{1}{n\pi}\cos(n\pi) + \frac{1}{n\pi}$$

$$= \frac{2}{n\pi} - \frac{2}{n\pi}(-1)^n$$

$$= \frac{2}{n\pi}\left[1-(-1)^n\right]$$

$\therefore$ The F.S

$$F(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi}\left[1-(-1)^n\right]\sin(n\pi x).$$

Ans:

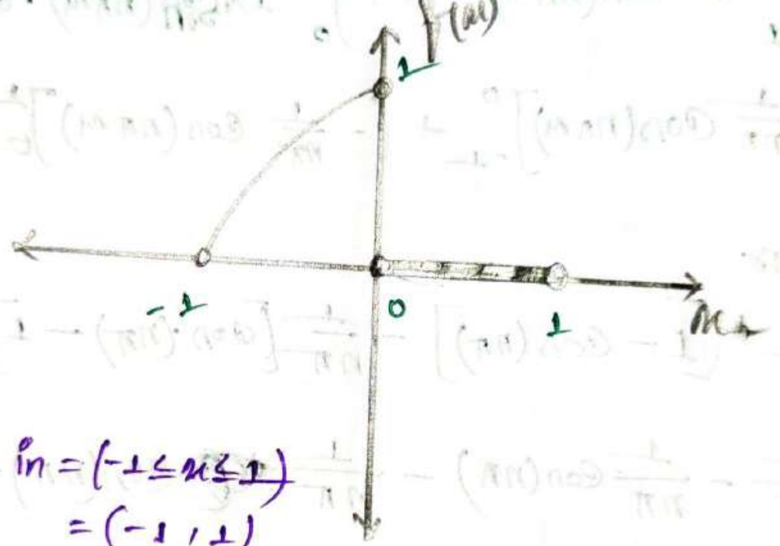
Example: 07:-

⇒ Find the Fourier Series for the functions $f(x)$ defined by.

$$f(x)=\begin{cases} 1-x^2 ; & (-1\le x<0) \\ 0 ; & (0\le x<1) \end{cases}$$

Also sketch the graph.

⇒



$f(x)$ is define in $=(-1\le x\le 1)$
$=(-1,1)$

⇒ We know:-

The F.S:-

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}\left[a_n\cos(n\pi x)+b_n\sin(n\pi x)\right] \quad \text{------ (1)}$$

$$\therefore \quad a_0=\frac{1}{1}\left[\int_{-1}^{0}\{(1)-(x)^2\}\cdot dx+\int_{0}^{1}0\cdot dx\right]$$

$$=\left[x-\frac{1}{3}x^3\right]_{-1}^{0}$$

$$=-\left(\frac{1}{3}-1\right)$$

$$=\frac{2}{3}$$

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$$a_n = \frac{1}{1}\left[\int_{-1}^{0}(1-x^2)\cdot\cos(n\pi x)\,dx + \int_{0}^{1} 0\cdot\cos(n\pi x)\cdot dx\right]$$

$$= \left[(1-x^2)\cdot\left(\frac{1}{n\pi}\right)\sin n\pi x - (-2x)\left(-\frac{1}{n^2\pi^2}\right)\cos(n\pi x) + (-2)\left(-\frac{1}{n^3\pi^3}\right)\sin(n\pi x)\right]_{-1}^{0}$$

$$= \left[\frac{(1-x^2)}{n\pi}\sin(n\pi x) - \frac{2x}{n^2\pi^2}\cos(n\pi x) + \frac{2}{n^3\pi^3}\sin(n\pi x)\right]_{-1}^{0}$$

$$= \left[\left(\frac{1-x^2}{n\pi} + \frac{2}{(n\pi)^3}\right)\sin(n\pi x) - \frac{2x}{(n\pi)^2}\cos(n\pi x)\right]_{-1}^{0}$$

$$= -\left[0 - \frac{2\times -1}{(n\pi)^2}\cos(n\pi)\right]$$

$$= \frac{-2}{(n\pi)^2}(-1)^n$$

$$b_n = \frac{1}{1}\left[\int_{-1}^{0}(1-x^2)\cdot\sin(n\pi x)\cdot dx + \int_{0}^{1} 0\cdot\sin(n\pi x)\cdot dx\right]$$

$$= \left[(+1-x^2)\left(-\frac{1}{n\pi}\right)\cos(n\pi x) - (-2x)\left(-\frac{1}{(n\pi)^2}\right)\sin(n\pi x) + (-2)\left(\frac{1}{(n\pi)^3}\right)\cos(n\pi x)\right]_{-1}^{0}$$

$$= \left[-\frac{1-x^2}{n\pi}\cos(n\pi x) - \frac{2x}{(n\pi)^2}\sin(n\pi x) - \frac{2}{(n\pi)^3}\cos(n\pi x)\right]_{-1}^{0}$$

$$= \left[\left(-\frac{1-x^2}{n\pi} - \frac{2}{(n\pi)^3}\right)\cos(n\pi x) - \frac{2x}{(n\pi)^2}\sin(n\pi x)\right]_{-1}^{0}$$

$$= \left[-\left\{\frac{1-x^2}{n\pi} + \frac{2}{(n\pi)^3}\cos(n\pi x)\right\} - \frac{2x}{(n\pi)^2}\sin(n\pi x)\right]_{-1}^{0}$$

$$= \left[-\left[\left\{\frac{1}{n\pi} + \frac{2}{(n\pi)^3}\right\} - \left\{\frac{2}{(n\pi)^3}(-1)^n\right\}\right]\right]$$

$$= -\frac{1}{n\pi} - \frac{2}{(n\pi)^3} + \frac{2}{(n\pi)^3}\times(-1)^n$$

$$= \frac{2}{(n\pi)^3}\left\{(-1)^n - 1\right\} - \frac{1}{n\pi}$$

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So, the F.S:—

$$f(x) = \frac{1}{3} + \sum_{n=1}^{\infty}\left[\frac{-2}{(n\pi)^2}(-1)^n \cos(n\pi x) + \left\{\frac{2}{(n\pi)^3}\{(-1)^n - 1\} - \frac{1}{n\pi}\right\}\sin(n\pi x)\right]$$

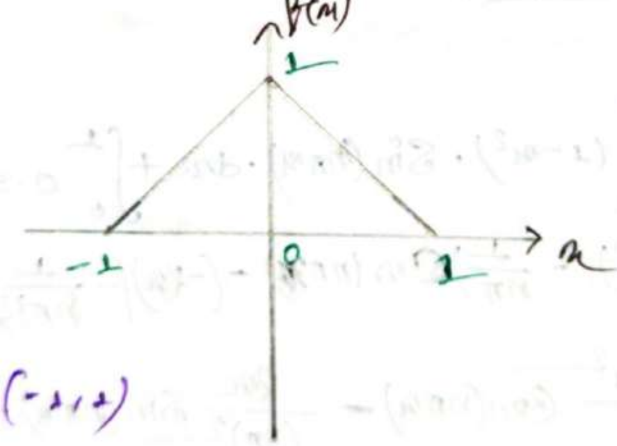
Ans.

Examples 08:—

$\Rightarrow$ Find the Fourier Series for the function $f(x)$ defined on the interval $[-1, 1]$ by

$$f(x) = 1 - |x|$$

$\Rightarrow$



$f(x)$ define in $(-1, 1)$

We know:—

The F.S:—

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi}{1}x\right) + b_n \sin\left(\frac{n\pi}{1}x\right)\right] \quad \cdots \quad (1)$$

$$\therefore a_0 = \frac{1}{1}\left[\int_{-1}^{1}(1-x)\cdot dx\right]$$

$$= 2\int_0^1 (1-x)\cdot dx$$

$$= 2\left[x - \frac{1}{2}x^2\right]_0^1$$

$$= 1$$

$f(x) = 1 - |x|$

$f(-x) = 1 - |-x|$

$= 1 - x$

$= f(x)$

$=$ even

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$$a_n = \frac{1}{1}\left[\int_{-1}^{1}\left[(1-x)\cdot \cos(n\pi x)\right]dx\right]$$

$$= 2\left[\int_0^1 \{(1-x)\cdot \cos(n\pi x)\}dx\right]$$

$$= 2\left[(1-x)\times\left(\frac{1}{n\pi}\right)\sin(n\pi x) - \left\{(-1)\left(-\frac{1}{(n\pi)^2}\right)\cos(n\pi x)\right\}\right]_0^1$$

$$= -2\times\left[\frac{1}{(n\pi)^2}\cos(n\pi x)\right]_0^1$$

$$= -\frac{2}{(n\pi)^2}(-1)^n + \frac{2}{(n\pi)^2}$$

$$= \frac{2}{(n\pi)^2}\left[1-(-1)^n\right]$$

$$b_n = \frac{1}{1}\left[\int_{-1}^{1}\underbrace{(1-x)}_{\text{Even}}\cdot \underbrace{\sin(n\pi x)}_{\text{odd}}\cdot dx\right]$$

$\therefore$ $b_n = 0$.

we know,

Even $\times$ odd = odd

So, the FS:-

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\left[\frac{2}{(n\pi)^2}\left[1-(-1)^n\right]\cos(n\pi x)\right]$$

Ans:-

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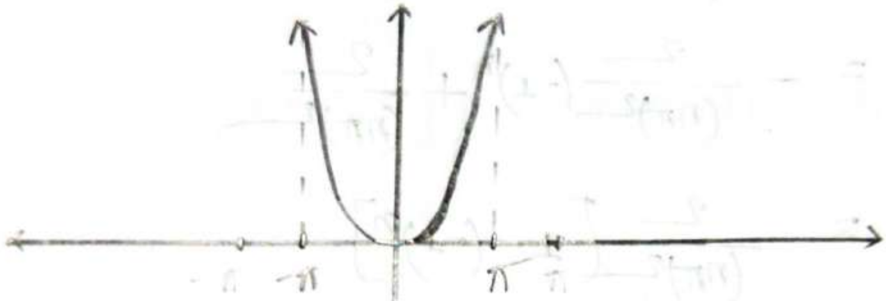
Exercise 3.2

(1.) Sketch the graph and find the Fourier coefficients and the fourier series of the the function of $f(x)=x^2$ in the interval $-\pi \le x \le \pi$

=>



$f(x)$ define in $=(-\pi,\pi)$

We know:-

The F.S =

$$f(x)= \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi}{\pi}x\right) + b_n \sin\left(\frac{n\pi}{\pi}x\right)\right] \cdots (1)$$

$$\therefore a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2 \, dx$$

$$= \frac{1}{\pi}\left[\frac{1}{3}x^3\right]_{-\pi}^{\pi}$$

$$= \frac{\pi^3}{3\pi} + \frac{\pi^3}{3\pi}$$

$$= \frac{2\pi^2}{3}$$

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$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \cos nx \cdot f(x) \cdot dx$$

$$= \frac{1}{\pi} \times 2\int_0^{\pi} \cos nx \cdot x^2 \cdot dx$$

$$= \frac{2}{\pi}\left[x^2 \times \left(\frac{1}{n}\right)\sin(nx) - 2x \times \frac{-1}{(n)^2} \times \cos(nx) + 2 \times \left(-\frac{1}{n}\right)^3 \cdot \sin(nx)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{x^2}{n}\sin nx + \frac{2x}{n^2}\cos(nx) - \frac{2}{(n)^3}\sin(nx)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\left\{\frac{x^2}{n} - \frac{2}{(n)^3}\right\}\sin(nx) + \frac{2x}{n^2}\cos(nx)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[0 + \frac{2\pi}{n^2}\cos(n\pi) - \frac{2 \times 0}{n^2}\cos(n \times 0)\right]$$

$$= \frac{2}{\pi} \times \left[\frac{2\pi}{n^2 \pi}(-1)^n\right]$$

$$= \frac{4(-1)^n}{n^2}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \sin nx \cdot f(x) \cdot dx$$

$$= \frac{1}{\pi} \times 0$$

$$= 0$$

cz.
$\sin nx$ = odd
x^2 = even
we know:-
odd × even = odd
$\therefore f(x)$ = odd = 0

so, the fourier series is:-

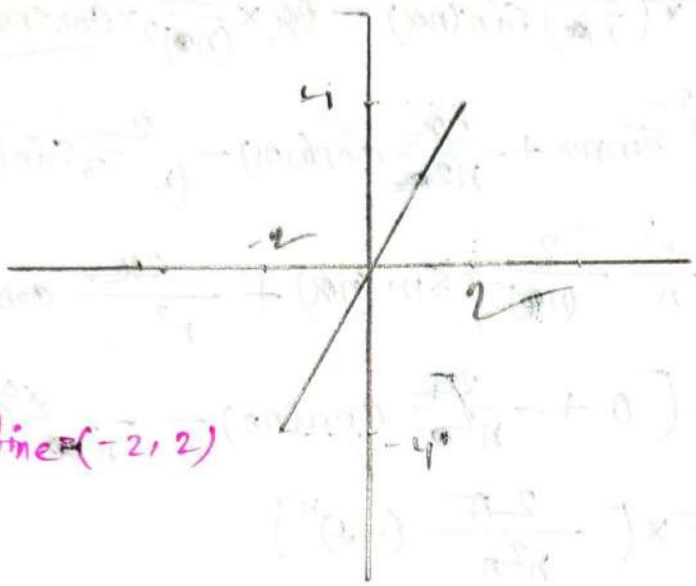
$$f(x) = \frac{2\pi^2}{3} + \sum_{n=2}^{\infty}\left[a_n \cos(nx) + b_n \sin(nx)\right]$$

$$= \frac{2\pi^2}{3} + \sum_{n=2}^{\infty}\left[\frac{4(-1)^n}{n^2}\cos(nx)\right]$$

Ans:-

2. Sketch the graph and find the fourier coefficients and the fourier series of the function $f(x)=2x$ in the interval $-2 \le x \le 2$

$\Rightarrow$



f(x) is define (-2, 2)

we know:-

The F.S:-

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi}{2}x\right) + b_n \sin\left(\frac{n\pi}{2}x\right) \right]$$

$$\therefore a_0 = \frac{1}{2}\int_{-2}^{2} f(x) \cdot dx$$

$$= \frac{1}{2}\int_{-2}^{2} 2x \cdot dx$$

$$= 0$$

CZ

$f(x) = 2x$

$\therefore f(-x) = 2 \times -x$

$= -2x$

$= -f(x)$

So, It's a odd function

$$a_n = \frac{1}{2}\int_{-2}^{2} \cos\left(\frac{n\pi}{2}x\right) \cdot 2x \cdot dx$$

$$= 0$$

CZ

$f(x) = 2x = \text{odd}$

$\cos x = \text{even}$

$\therefore \text{even} \times \text{odd} = \text{odd}$

So, the total function is a odd

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$$b_n = \frac{1}{2}\int_{-2}^{2} \sin\left(\frac{n\pi}{2}x\right) \cdot 2x \cdot dx$$

$$= \frac{2}{2}\left[2x \times \left(\frac{-2}{n\pi}\right)\cos\left(\frac{n\pi}{2}x\right) - 2 \times \left(-\frac{4}{(n\pi)^2}\right)\sin\left(\frac{n\pi}{2}x\right)\right]_0^2$$

$$= \left[\frac{-4x}{n\pi}\cos\left(\frac{n\pi}{2}x\right) + \frac{8}{(n\pi)^2}\sin\left(\frac{n\pi}{2}x\right)\right]_0^2$$

$$= \frac{-8}{n\pi}\cos(n\pi) + \frac{4\times 0}{n\pi} + \frac{8}{(n\pi)^2}\sin(n\pi) - 0$$

$$= \frac{-8(-1)^n}{n\pi} + 0 + 0 - 0$$

$\therefore$

So, the FS:-

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi}{2}x\right) + b_n \sin\left(\frac{n\pi}{2}x\right)\right]$$

$$= 0 + \sum_{n=1}^{\infty}\left[0 + \frac{-8(-1)^n}{n\pi}\sin\left(\frac{n\pi}{2}x\right)\right]$$

$$= \sum_{n=1}^{\infty}\left[-\frac{8(-1)^n}{n\pi}\sin\left(\frac{n\pi}{2}x\right)\right]$$

Ans:-

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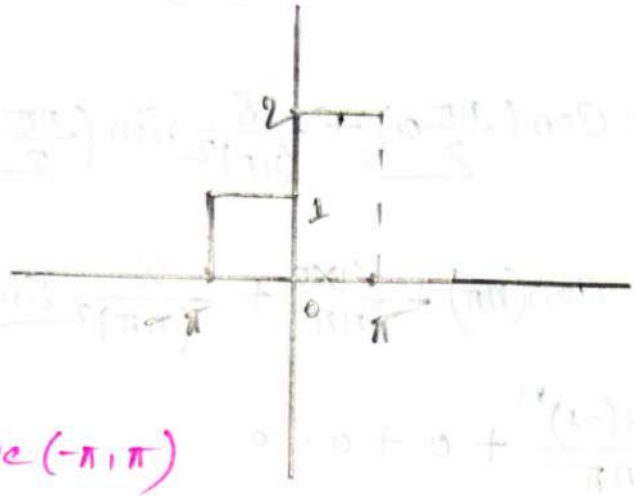
(3) Sketch the graph and obtain the Fourier Series of the function $f(x) = \begin{cases} 1 & ; -\pi < x < 0 \\ 2 & ; 0 < x < \pi \end{cases}$

$\Rightarrow$



$f(x)$ is define $(-\pi, \pi)$

We know:—

The F.S:—

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi}{\pi}x\right) + b_n \sin\left(\frac{n\pi}{\pi}x\right)\right]$$

$$\therefore a_0 = \frac{1}{\pi}\left[\int_{-\pi}^{0} 1 \cdot dx + \int_{0}^{\pi} 2 \cdot dx\right]$$

$$= \frac{1}{\pi}\left[[x]_{-\pi}^{0} + [2x]_{0}^{\pi}\right]$$

$$= \frac{1}{\pi} \times (\pi + 2\pi)$$

$$= 3$$

$$a_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} 1 \cdot \cos(nx) + \int_{0}^{\pi} 2\cos(nx)\right] dx$$

$$= \frac{1}{\pi}\left\{\left[\frac{1}{n}\sin(nx)\right]_{-\pi}^{0} + \frac{2}{n}\left[\sin(nx)\right]_{0}^{\pi}\right.$$

$$= \frac{1}{\pi}(0 + 0) = 0$$

$\sin n\pi = 0$

$$b_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} 1 \cdot \sin(nx)\cdot dx + \int_{0}^{\pi} 2\cdot \sin(nx)\cdot dx\right]$$

$$= \frac{1}{\pi}\left[-\frac{1}{n}\times\left[\cos(nx)\right]_{-\pi}^{0} - \frac{2}{n}\left[\cos(nx)\right]_{0}^{\pi}\right]$$

$$= \frac{1}{\pi}\left\{-\frac{1}{n}\times(1-(-1)^n) - \frac{2}{n}((-1)^n-1)\right\}$$

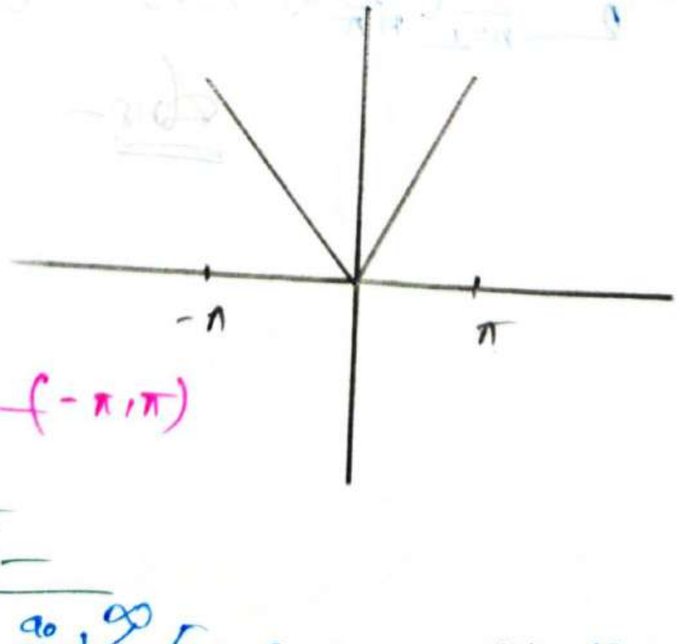
$$= -\frac{1}{n\pi}\left[((-1)^n-1)+2(1-(-1)^n)\right]$$

Ans:-

$$\therefore f(x) = \frac{3}{2} + \sum_{n=1}^{\infty}\left[-\frac{1}{n\pi}\left[\{(-1)^n-1\}+2(1-(-1)^n)\right]\sin(nx)\right].$$

4. Sketch the graph and obtain the Fourier Series of the function $f(x) = |x|$ in the interval $-\pi \le x \le \pi$

$\Rightarrow$



$f(x)$ is define $(-\pi, \pi)$

We know:-

The F.S:-

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos(nx)\cdot dx + b_n \sin(nx)\cdot dx\right].$$

$$\therefore a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} |x|\cdot dx$$

$$= \frac{2}{\pi}\left[\frac{1}{2}x^2\right]_0^{\pi} \Rightarrow \pi$$

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$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} |x|\cdot \cos(nx)\cdot dx$$

$$= \frac{2}{\pi}\int_{0}^{\pi} |x|\cdot \cos(nx)\cdot dx$$

$$= \frac{2}{\pi}\left[x \times \frac{1}{n}\sin(nx) - 1 \times \frac{-1}{n^2}\cos(nx)\right]_0^{\pi}$$

$$= \frac{2}{n^2\pi}\left[(-1)^n - 1\right]$$

$$b_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{|x|}_{\text{even}}\cdot \underbrace{\sin(nx)}_{\text{odd}}\cdot dx$$

$$= 0$$

at

even × odd = odd

So, the F.S:—

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty}\left[\frac{2}{n^2\pi}\left[(-1)^n - 1\right]\cos(nx)\right]$$

Ans:—

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#Complex Fourier Series

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Formulae:-

→ if $f(x)$ is define in $-L<x<L$

so, $f(x)=\sum_{n=-\infty}^{\infty} C_n e^{-i\frac{n\pi x}{L}}$, $n=0,1,2,3,\ldots$

Where:-

$$C_n=\frac{1}{2L}\int_{-L}^{L} f(x)\cdot e^{i\frac{n\pi x}{L}}\cdot dx$$

Examples:-

⇒ Find the complex form of fourier series function $f(x)=x^2$ in the region $-\pi\le x\le\pi$.

⇒ We know

The F.S:-

$$f(x)=\sum_{n=\infty}^{\infty} C_n e^{-inx}, \quad n=0,\pm1,\pm2,\ldots$$

$$\therefore C_n=\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2\cdot e^{inx}$$

$$=\frac{1}{2\pi}\int_{-\pi}^{\pi} x^2(\cos nx+i\sin nx)\cdot dx$$

$$=\frac{1}{2\pi}\int_{-\pi}^{\pi}(\underbrace{x^2\cos nx}_{\text{even}}+\underbrace{ix^2\sin nx}_{\text{odd}})\cdot dx$$

$$=\frac{2}{2\pi}\int_{0}^{\pi} x^2\cdot\cos(nx)\cdot dx$$

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$$= \frac{1}{\pi}\left[x^2 \times \frac{1}{n}\sin(nx) - 2x \times \frac{-1}{n^2}\cos(nx) + 2 \times \frac{-1}{n^3}\sin(nx)\right]_0^\pi$$

$$= \frac{1}{\pi}\left[2 \times \pi \times \frac{1}{n^2}\cos(n\pi) - 0\right]$$

$$= \frac{2\pi}{\pi n^2}(-1)^n$$

$$= \frac{2}{n^2}(-1)^n \; ; \; n \neq 0$$

If $n = 0$, then:—

$$C_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 \cdot dx$$

$$= \frac{2}{2\pi}\int_0^\pi x^2 \cdot dx$$

$$= \frac{1}{\pi}\left[\frac{1}{3}x^3\right]_0^\pi$$

$$= \frac{\pi^2}{3}$$

$\therefore$ The F.S:—

$$f(x) = \frac{\pi^2}{3} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{2}{n^2}(-1)^n e^{-inx}$$

Ans:—

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Exercise: 3.2

(i)

Find the complex form of F.S of the following function.

(1) $f(x) = 2x$; $-\pi \le x \le \pi$

we know—

The F.S:—

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{-i\frac{n\pi x}{\pi}}$$

$$\therefore c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} 2x(\cos nx + i\sin nx)\cdot dx$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} (\overset{odd}{2x}\cdot \overset{even}{\cos nx} + i\,\overset{odd}{2x}\cdot \overset{odd}{\sin (nx)})\,dx$$

$$= \frac{2}{2\pi}\left[i\times 2x\times -\frac{1}{n}\cos(nx) - i2\times -\frac{1}{n^2}\sin(nx)\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[-\frac{2\pi i}{n}(-1)^n - 0\right]$$

$$= -\frac{i2}{n}(-1)^n \;;\; n \ne 0$$

if $n = 0$ then

$$c_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} 2x\cdot dx$$

$$= 0$$

$\therefore$ The F.S:

$$f(x) = \sum_{\substack{n=-\infty \\ n\ne 0}}^{\infty} -\frac{i2}{n}(-1)^n e^{-inx}$$

Ans:—

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Half-Range Fourier-Series.

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Half-Range F.S.:-

$\Rightarrow f(x)$ is defined in $0<x<L$

So, the Half-range Fourier Sine series (odd extension)

$$\Rightarrow f(x) = \sum_{n=1}^{\infty} b_n \cdot \sin\left(\frac{n\pi}{L} x\right)$$

Where:-

$$b_n = \frac{2}{L}\int_0^L f(x)\cdot \sin\left(\frac{n\pi}{L} x\right)\cdot dx$$

And:- the Half-range Fourier Cosine Series (even extension)

$$\Rightarrow f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L} x\right)$$

Where:

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\cdot dx$$

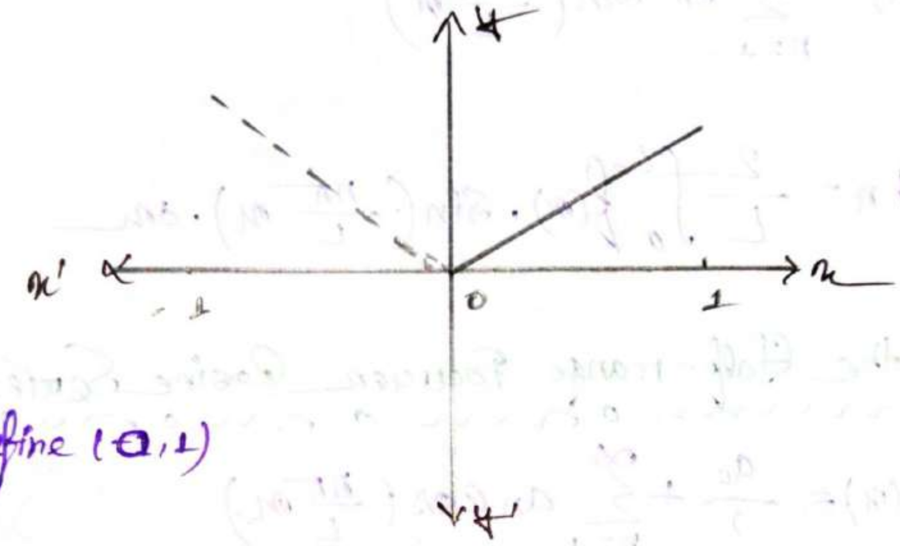
$$= \frac{2}{L}\int_0^L f(x)\cdot dx$$

$$a_n = \frac{2}{L}\int_0^L f(x)\cdot \cos\left(\frac{n\pi}{L} x\right)\cdot dx$$

Example: 1:-

⇒ Find the First three terms of half rang F. Cosine Series of the function $f(x) = x$; $0 \le x \le 1$. Also Sketch propee extension.

⇒



$f(x)$ is define $(0,1)$

even extension.

We know:-

The Half Fourier Cosine series:-

$$F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{1}\right).$$

$$\therefore a_0 = \frac{2}{1}\int_0^1 x \cdot dx$$

$$= 2\left[\frac{1}{2}x^2\right]_0^1$$

$$= 1$$

$$a_n = \frac{2}{1}\int_0^1 x \cdot \cos(n\pi x) \cdot dx$$

$$= 2\left[x \times \frac{1}{n\pi} \times \sin(n\pi x) + \frac{1}{(n\pi)^2}\cos(n\pi x)\right]_0^1$$

$$= 2 \times \frac{1}{(n\pi)^2}\left((-1)^n - 1\right)$$

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So, the H.R.F. cosine Series:—

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\left[\frac{2}{(n\pi)^2}\{(-1)^n - 1\}\right]\cos(n\pi x)$$

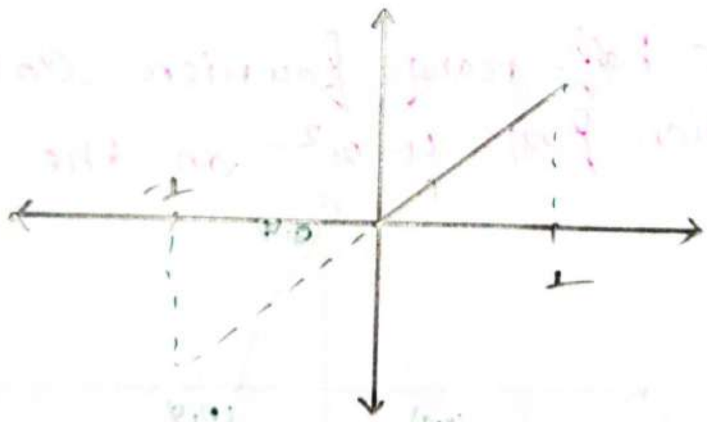
$$= \frac{1}{2} + \frac{2}{\pi^2}\left[-2\cos(\pi x) - \frac{2}{9}\cos(3\pi x) \ldots\right]$$

Ans:—

Ex:- 2:—

⇒ Same Question but Now Find the F. Sine Series also Sketch the proper extension.

⇒



Odd extension:—

We know:—

The F. Sine Series:—

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(n\pi x)$$

$$\therefore\ b_n = \frac{2}{1}\int_0^1 x \cdot \sin(n\pi x) \cdot dx$$

$$= 2\left[x \times -\frac{1}{n\pi} \times \cos(n\pi x) - 1 \times \frac{-1}{(n\pi)^2}\sin(n\pi x)\right]_0^1$$

$$= \frac{-2}{n\pi}(-1)^n$$

$\therefore$ the H.R. Sine Series:-

$$f(x) = \sum_{n=1}^{\infty} \frac{-2}{n\pi}(-1)^n \sin\left(\frac{n\pi x}{1}\right)$$

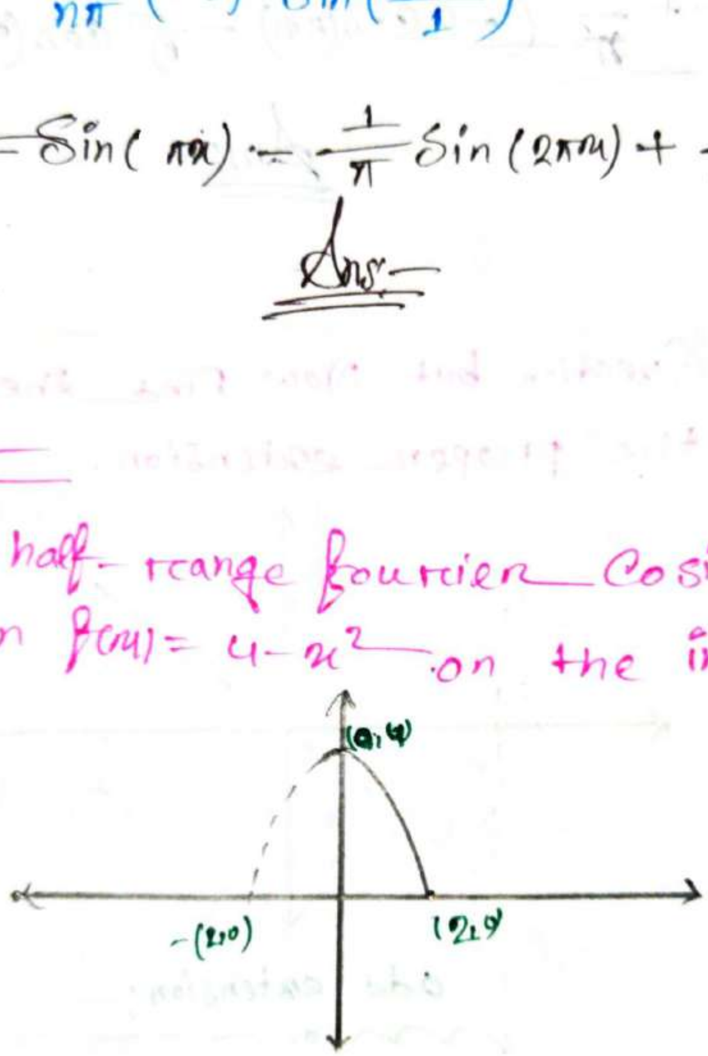
$$= \frac{2}{\pi}\sin(\pi x) - \frac{1}{\pi}\sin(2\pi x) + \frac{2}{3\pi}\sin(3\pi x) \ldots$$

Ans:-

Example: 03:-

$\Rightarrow$ Find the half-range fourier Cosine series for the function $f(x) = 4 - x^2$ on the interval $[0, 2]$

$\Rightarrow$



even extension.

We know:-

The H.R.F.C.S:-

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{2}\right)$$

$$\therefore a_0 = \frac{2}{2}\int_0^2 (4 - x^2) \cdot dx$$

$$= \left[4x - \frac{1}{3}x^3\right]_0^2 = \frac{16}{3}$$

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$$a_n = \frac{2}{2}\int_0^2 (4-x^2)\cdot \cos\left(\frac{n\pi}{2}x\right)\cdot dx$$

$$= \left[(4-x^2)\times\left(\frac{2}{n\pi}\right)\sin\left(\frac{n\pi}{2}x\right) - 2x\times -\frac{4}{(n\pi)^2}\cos\left(\frac{n\pi}{2}x\right) + 2 \times -\frac{4\times 2}{(n\pi)^3}\times \sin\left(\frac{n\pi}{2}x\right)\right]_0^2$$

$$= \left[-\frac{8x}{(n\pi)^2}\cos\left(\frac{n\pi}{2}x\right)\right]_0^2$$

$$= \frac{-16}{(n\pi)^2}(-1)^n$$

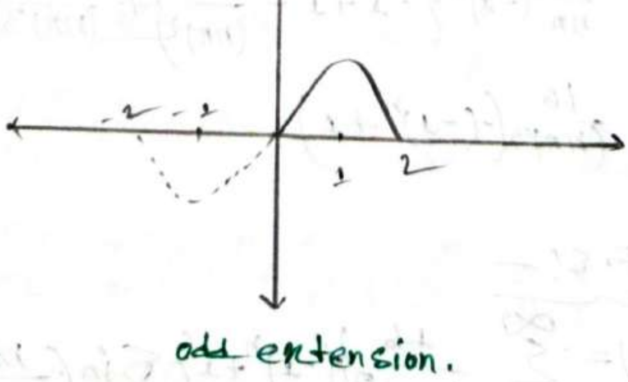
So, the Fourier Series is:—

$$f(x) = \frac{8}{3} + \sum_{n=1}^{\infty} \frac{16}{(n\pi)^2}(-1)^n \times \cos\left(\frac{n\pi x}{2}\right)$$

Example: 04:—

⇒ Find the half range Fourier Sine Series for the function $f(x) = 2x - x^2$ on the interval $[0, 2]$.

⇒



odd extension.

We know:—

The H.R.F.S.S:—

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$$

$$\therefore b_n = \frac{2}{2}\int_0^2 (2x-x^2)\cdot \sin\left(\frac{n\pi}{2}x\right)\cdot dx$$

$$= \left[(2x-x^2)\times\left(-\frac{2}{n\pi}\right)\cos\left(\frac{n\pi}{2}x\right) - (2-2x)\left(-\frac{4}{(n\pi)^2}\right)\sin\left(\frac{n\pi}{2}x\right) + (0-2)\left(\frac{8}{(n\pi)^3}\right)\cos\left(\frac{n\pi}{2}x\right)\right]_0^2$$

$$= \left[-\frac{4x}{n\pi}\cos\left(\frac{n\pi}{2}x\right) + \frac{2x^2}{n\pi}\cos\left(\frac{n\pi}{2}x\right) - \frac{16}{(n\pi)^3}\cos\left(\frac{n\pi}{2}x\right)\right]_0^2$$

$$= \left[\frac{2}{n\pi}\cos\left(\frac{n\pi}{2}x\right)\left\{-2x+x^2 - 8/(n\pi)^2\right\}\right]_0^2$$

$$= -\frac{8}{n\pi}(-1)^n + \frac{8}{n\pi}(-1)^n - \frac{16}{(n\pi)^3}(-1)^n + \frac{16}{(n\pi)^3}$$

$$= \frac{8}{n\pi}(-1)^n\left\{-1+1-\frac{2}{(n\pi)^2}\right\} + \frac{16}{(n\pi)^3}$$

$$= \frac{16}{(n\pi)^3}\left(-(-1)^n+1\right)$$

$\therefore$ The F.S:—

$$f(x) = \sum_{n=1}^{\infty} \frac{16}{(n\pi)^3}\left\{-(-1)^n+1\right\}\sin\left(\frac{n\pi x}{2}\right)$$

Ans:—

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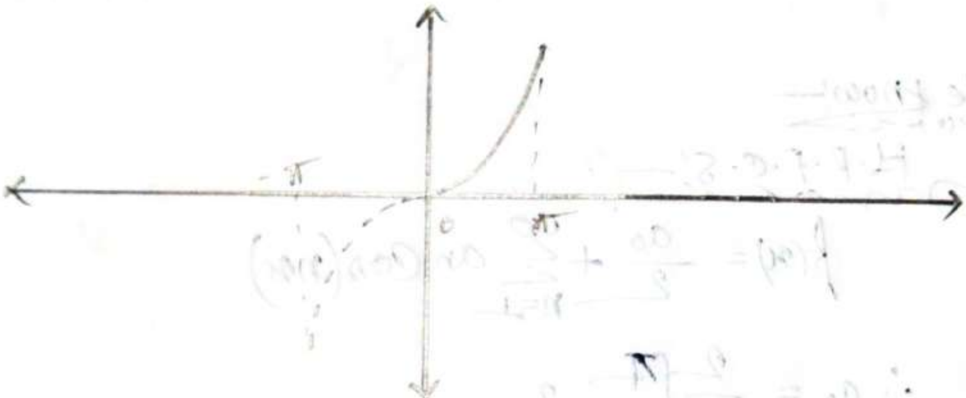
Exercise: 3.3

(2) Sketch the odd and even extension and express $f(x)=x^2$ as a half-range Fourier Sine and Cosine Series in the interval $0<x<\pi$

=> For half Range Fourier Sine series:-



odd extension.

We know:-

The F. Sine Series:-

$$f(x)=\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{\pi}\right)$$

$$\therefore b_n = \frac{2}{\pi}\int_0^{\pi} x^2 \cdot \sin(nx)\cdot dx$$

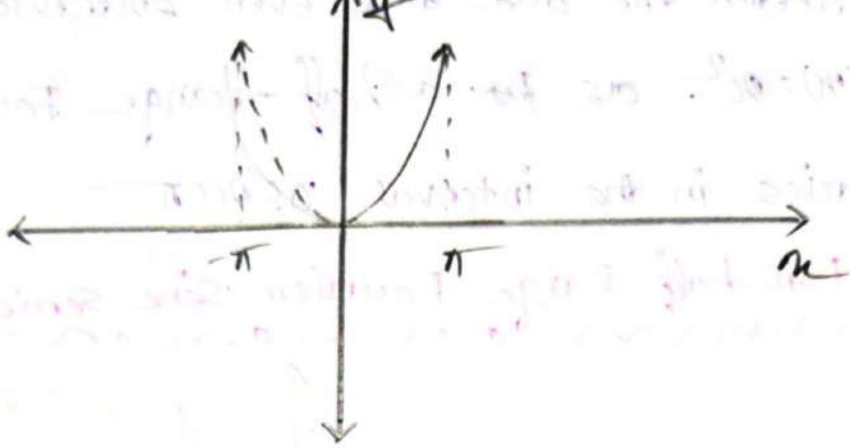
$$= \frac{2}{\pi}\left[x^2 \times \left(-\frac{1}{n}\right)\cos(nx) - 2x \times \left(-\frac{1}{n^2}\right)\sin(nx) + 2\times\frac{1}{n^3}\cos(nx)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[-\frac{\pi^2}{n}(-1)^n + \frac{2}{n^3}(-1)^n - \frac{2}{n^3}\right]$$

So the F. Sine Series:-

$$f(x)=\sum_{n=1}^{\infty}\left[\frac{2}{\pi}\left\{-\frac{\pi^2}{n}(-1)^n\right\} + \frac{2}{n^3}\left((-1)^n - 1\right)\right]\sin(nx)$$

For half-Range Fourier Cosine Series:-



We know:-

H.R.F.C.S:-

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx)$$

$$\therefore a_0 = \frac{2}{\pi}\int_0^{\pi} x^2 \cdot dx$$

$$= \frac{2}{\pi}\left[\frac{1}{3}x^3\right]_0^{\pi}$$

$$= \frac{2\pi^2}{3}$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} x^2 \cdot \cos(nx) \cdot dx$$

$$= \frac{2}{\pi}\left[x^2 * \left(\frac{1}{n}\right)\sin(nx) - 2x * \left(-\frac{1}{n^2}\right)\cos(nx) + 2 * \frac{-1}{n^3}\sin(nx)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{2\pi}{n^2}(-1)^n\right]$$

$$= \frac{4}{n^2}(-1)^n$$

So, the Series:-

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty}\left\{\left[\frac{4}{n^2}(-1)^n\right]\cos(nx)\right\}$$

Ans:-

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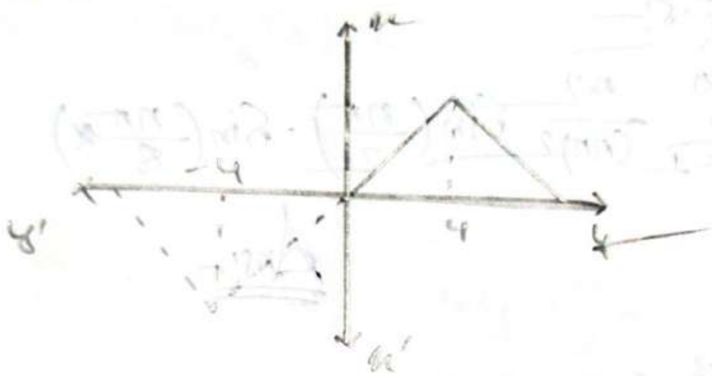
Q. Sketch the odd and even extension and find the half range fourier series for cosine and sine series of

$$f(x)=\begin{cases} x, & 0<x<4 \\ 8-x, & 4<x<8 \end{cases}$$

⇒ For half Range Fourier Sine Series:-



Odd extension.

We know:-

The H.R.F.S.S:-

$$f(x)=\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{8}x\right)$$

$$\therefore \quad b_n = \frac{2}{8}\int_0^8 f(x)\cdot \sin\frac{n\pi}{8}x \cdot dx$$

$$= \frac{2}{8}\left[\int_0^4 x\cdot \sin\left(\frac{n\pi}{8}x\right)\cdot dx + \int_4^8 (8-x)\cdot \sin\left(\frac{n\pi}{8}x\right)\cdot dx\right]$$

$$= \frac{1}{4}\left\{\left[x\times\left(-\frac{8}{n\pi}\right)\cos\left(\frac{n\pi}{8}x\right) - 1\times\left(-\frac{64}{(n\pi)^2}\right)\sin\left(\frac{n\pi}{8}x\right)\right]_0^4 + \left[(8-x)\times\left(-\frac{8}{n\pi}\right)\cos\left(\frac{n\pi}{8}x\right) - (-1)\left(\frac{-64}{(n\pi)^2}\right)\sin\left(\frac{n\pi}{8}x\right)\right]_4^8\right.$$

$$= \frac{1}{4}\left\{-\frac{32}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{64}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right) + \right.$$

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$$= \ldots -\left(\frac{-32}{n\pi} \times \cos\frac{n\pi}{2}\right) + \frac{64}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right)\Big\}$$

$$= \frac{1}{4}\Big\{ -\frac{32}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{32}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{128}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right)\Big\}$$

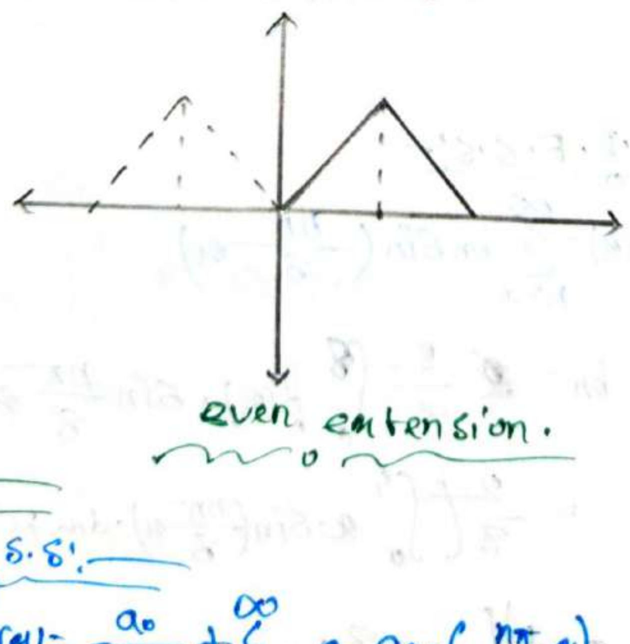
$$= \frac{32}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right)$$

∴ The F.S:—

$$bn = \sum_{n=1}^{\infty} \frac{32}{(n\pi)^2}\sin\left(\frac{n\pi}{2}\right)\cdot\sin\left(\frac{n\pi}{8}x\right)$$

Ans:-

For half-Range F. Cosine Series:—



even extension.

We know:—

The F.S.S:—

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{8}x\right)$$

$$\therefore a_0 = \frac{2}{8}\left[\int_0^4 x\cdot dx + \int_4^8 (8-x)\cdot dx\right.$$

$$= \frac{1}{4}\left[\left[\frac{1}{2}x^2\right]_0^4 + \left[8x - \frac{1}{2}x^2\right]_4^8\right]$$

$$= \frac{1}{4}\left\{8 + (64-32) - (32-8)\right\}$$

$$= 4$$

$$a_n = \frac{2}{8}\left[\int_0^4 x \cdot \cos\left(\frac{n\pi}{8}x\right)dx + \int_4^8 (8-x)\cos\left(\frac{n\pi}{8}\cdot x\right)dx\right]$$

$$= \frac{1}{4}\left[\left[x\cdot\left(\frac{8}{n\pi}\right)\sin\left(\frac{n\pi}{8}x\right) - \left(\frac{-64}{(n\pi)^2}\right)\cos\left(\frac{n\pi}{8}x\right)\right]_0^4 + \left[(8-x)\times\frac{8}{n\pi}\sin\left(\frac{n\pi}{8}x\right) - (-1)\left(-\frac{64}{(n\pi)^2}\right)\cdot\cos\left(\frac{n\pi}{8}x\right)\right]_4^8\right]$$

$$= \frac{1}{4}\left[\left\{\frac{32}{n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{64}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right) - \frac{64}{(n\pi)^2} + \left(-\frac{32}{n\pi}\sin\left(\frac{n\pi}{2}\right)\right. - \frac{64}{(n\pi)^2}\times(-1)^n + \frac{64}{(n\pi)^2}\cos\frac{n\pi}{2}\right\}\right]$$

$$= \frac{1}{4}\left[\frac{128}{(n\pi)^2}\times\cos\left(\frac{n\pi}{2}\right) - \frac{64}{(n\pi)^2}(-1)^n\right]$$

$$= \frac{32}{(n\pi)^2}\cos\left(\frac{n\pi}{2}\right) - \frac{16}{(n\pi)^2}(-1)^n$$

$$= \frac{16}{(n\pi)^2}\left\{2\cos\left(\frac{n\pi}{2}\right) - (-1)^n\right\}$$

So, the Series:-

$$f(x) = \frac{4}{2} + \sum_{n=1}^{\infty}\frac{16}{(n\pi)^2}\left\{2\cos\left(\frac{n\pi}{2}\right) - (-1)^n\right\}\cos\left(\frac{n\pi}{8}x\right).$$

Ans:-

Fourier Integral

ch:3

Fourier Integral:-

F.S → $f(x)$ is defined in $-L<x<L$

$f(x)$ is defined in $-\infty<x<\infty$

∴ Fourier Integral Series:-

$$f(x)=\int_{0}^{\infty}\left[A(\omega)\cos(\omega x)+B(\omega)\sin(\omega x)\right]d\omega$$

Where:-

1. $A(\omega)=\frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cdot\cos(\omega x)\cdot dx$

2. $B(\omega)=\frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cdot\sin(\omega x)\cdot dx$

Formula:-

1. $\int e^{ax}\cdot\sin bx\cdot dx=\frac{e^{ax}(a\sin bx-b\cos bx)}{a^2+b^2}$

2. $\int e^{ax}\cdot\cos bx\cdot dx=\frac{e^{ax}(a\cos bx+b\sin bx)}{a^2+b^2}$

3. $e^{-\infty}=0$

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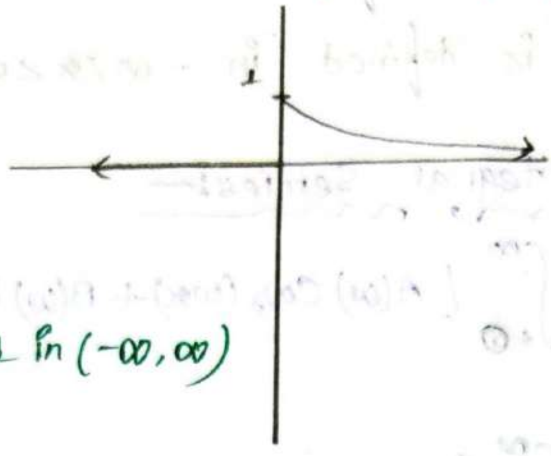
Example:1

⇒ Find the Fourier Integral of $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

⇒



$f(x)$ defined in $(-\infty, \infty)$

We know:

The FIS:

$$f(x) = \int_{0}^{\infty} [A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x)] \cdot d\omega$$

$$\therefore A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x) \cdot \cos(\omega x) \cdot dx$$

$$= \frac{1}{\pi}\left[\int_{-\infty}^{0} 0 \cdot \cos(\omega x) dx + \int_{0}^{\infty} e^{-x} \cdot \cos(\omega x) \cdot dx\right]$$

$$= 0 + \frac{1}{\pi} \times \left[\frac{e^{-x}(-\cos\omega x + \omega\sin\omega x)}{1^2 + \omega^2}\right]_{0}^{\infty}$$

$$= \frac{1}{\pi(1+\omega^2)}\left[e^{-x}(-\cos\omega x + \omega\sin\omega x)\right]_{0}^{\infty}$$

$$= \frac{1}{\pi(1+\omega^2)}\left[-e^{-x}\cos\omega x + e^{-x} \cdot \omega \cdot \sin\omega x\right]_{0}^{\infty}$$

$$= \frac{1}{\pi(1+\omega^2)}\left[-(0-1) + (0-0)\right]$$

$$= \frac{1}{\pi(1+\omega^2)} \quad \text{Ans:}$$

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cdot \sin(\omega x)\cdot dx$$

$$= \frac{1}{\pi}\left[\int_{-\infty}^{0} 0\cdot \sin(\omega x)\cdot dx + \int_{0}^{\infty} e^{-x}\cdot \sin\omega x \, dx\right]$$

$$= \frac{1}{\pi}\left[0 + \frac{e^{-x}(-\sin\omega x - \omega\cos\omega x)}{1^2+\omega^2}\right]_0^{\infty}$$

$$= \frac{1}{\pi(1+\omega^2)}\left[-e^{-x}\sin\omega x - e^{-x}\omega\cdot\cos\omega x\right]_0^{\infty}$$

$$= \frac{1}{\pi(1+\omega^2)}\left[-(0-0)-(0-\omega)\right]$$

$$= \frac{\omega}{\pi(1+\omega^2)}$$

So, the F.I.S:-

$$f(x) = \frac{1}{\pi}\int_0^{\infty}\left[\left(\frac{1}{1+\omega^2}\right)\cos(\omega x) + \frac{\omega}{1+\omega^2}\sin(\omega x)\right]d\omega$$

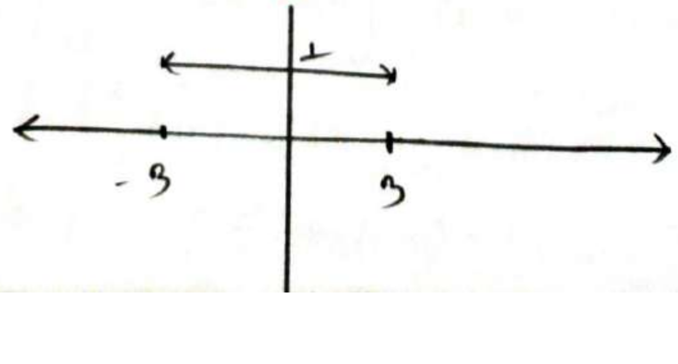
Ans

Example: 02

Find the Fourier Integral of the function:-

$$f(x) = \begin{cases} 0 & ; x<-3 \\ 1 & ; -3<x<3 \\ 0 & ; x>3 \end{cases}$$

$\Rightarrow$



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We know:—

The FIS:—

$$f(x) = \int_{0}^{\infty} [A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x)]\, d\omega$$

$$\therefore A(\omega) = \frac{1}{\pi}\left[\int_{-\infty}^{-3} 0\cdot\cos(\omega x)\cdot dx + \int_{-3}^{3} \overbrace{\cos(\omega x)}^{\text{even}}\cdot dx + \int_{3}^{\infty} 0\cdot\cos(\omega x)\cdot dx\right]$$

$$= \frac{1}{\pi}\left[0 + 2\times\frac{1}{\omega}\sin(\omega x)\right]_0^{3}$$

$$= \frac{1}{\pi}\left[\frac{2}{\omega}\sin(3\omega)\right]$$

$$= \frac{2\sin(3\omega)}{\pi\omega}$$

$$B(\omega) = \frac{1}{\pi}\left[\int_{-\infty}^{-3} 0\cdot\sin(\omega x)dx + \int_{-3}^{3} \overbrace{\sin(\omega x)}^{\text{odd}}\cdot dx + \int_{3}^{\infty} 0\cdot\sin(\omega x)dx\right]$$

$$\therefore = \frac{1}{\pi}[0+0+0]$$

$$= 0$$

So the FIS:—

$$f(x) = \int_{0}^{\infty}\left[\frac{2\sin(3\omega)}{\pi\omega}\times\cos(\omega x)\right] d\omega$$

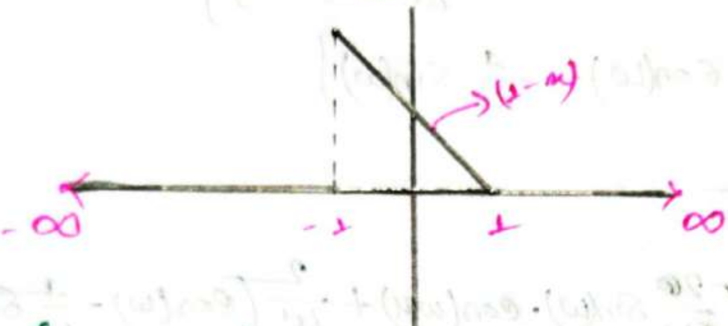
Ans:—

Ex: 3·4

1. Find the Fourier integral of the function

$$f(x)=\begin{cases}0 & ; x<-1\\ 1-x & ; -1\le x\le 1\\ 0 & ; x\ge 1\end{cases}$$

$\Rightarrow$



$f(x)$ is define $(-\infty,\infty)$

We know:-

The FIS:-

$$f(x)=\int_0^{\infty}\left[A(\omega)\cos(\omega x)+B(\omega)\sin(\omega x)\right]\cdot d\omega$$

$$A(\omega)=\frac{1}{\pi}\int_{-\infty}^{-1}0\cdot\cos(\omega x)\cdot dx+\int_{-1}^{1}(1-x)\cdot\cos(\omega x)dx+\int_{1}^{\infty}0\cdot\cos(\omega x)\cdot dx$$

$$=\frac{1}{\pi}\left[0+\int_{-1}^{1}\cos(\omega x)dx-\int_{-1}^{1}\overset{\text{odd}}{x}\cdot\overset{\text{even}}{\cos(\omega x)}\cdot dx+0\right]$$

$$=\frac{1}{\pi}\left[0+2\times\frac{1}{\omega}\sin(\omega x)+0\right]_0^1$$

$$=\frac{2}{\pi\omega}\sin\omega$$

$$B(\omega)=\frac{1}{\pi}\left(\int_{-\infty}^{-1}0\sin(\omega x)\cdot dx+\int_{-1}^{1}(1-x)\cdot\sin(\omega x)dx+\int_{1}^{\infty}0.\sin(\omega x)\cdot dx\right]$$

$$=\frac{1}{\pi}\left[\int_{-1}^{1}\overset{\text{odd}}{\sin(\omega x)}dx-\int_{-1}^{1}\overset{\text{odd}}{x}\cdot\overset{\text{even}}{\sin(\omega x)}\cdot dx\right]$$

$$=\frac{1}{\pi}\left[0-2\left[x\times\left(-\frac{1}{\omega}\right)\cos(\omega x)-1\cdot\left(-\frac{1}{\omega^2}\right)\sin(\omega x)\right]_0^1\right]$$

$$= \frac{1}{\pi}\left[-2\left[-\frac{x}{\omega}\cos(\omega x) + \frac{1}{\omega^2}\sin(\omega x)\right]_0^1\right]$$

$$= \frac{1}{\pi}\left[\frac{2x}{\omega}\cos(\omega x) - \frac{2}{\omega^2}\sin(\omega x)\right]_0^1$$

$$= \frac{1}{\pi}\left[\frac{2}{\omega}\cos(\omega) - \frac{2}{\omega^2}\sin(\omega)\right]$$

$$= \frac{2}{\pi\omega}\left[\cos(\omega) - \frac{1}{\omega}\sin(\omega)\right]$$

So, the FIS;—

$$f(x) = \frac{1}{\pi}\int_0^{\infty}\left[\frac{2}{\omega}\sin(\omega)\cdot\cos(\omega x) + \frac{2}{\omega}\left[\cos(\omega) - \frac{1}{\omega}\sin(\omega)\right]\sin(\omega x)\right]d\omega$$

Ans;-

(3) $f(x) = \begin{cases} 0 & ; x < 0 \\ \frac{1}{2} & ; x = 0 \\ e^{-x} & ; x > 0 \end{cases}$

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4. Find the Fourier Integral of $f(x) = e^{-kx}$ when $x>0$ and $f(-x) = -f(x)$ for $k>0$ and hence prove that

$$\int_0^\infty \frac{\omega \sin(\omega x)}{\omega^2+k^2}\, d\omega = \frac{\pi}{2} e^{-kx},\ k>0$$

⇒ We know:—

The FIS:—

$$f(x) = \int_0^\infty [A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x)]\, d\omega$$

$$\therefore A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} \underbrace{e^{-kx}}_{\to \text{odd}} \cdot \underbrace{\cos(\omega x)}_{\to \text{even}} \cdot dx$$

$$= 0$$

e.g. $f(-x) = -f(x)$
$=$ odd function

odd × even = odd

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} \underbrace{e^{-kx}}_{\to \text{odd}} \cdot \underbrace{\sin(\omega x)}_{\to \text{odd}} \cdot dx$$

$$= \frac{2}{\pi}\int_0^\infty e^{-kx} \cdot \sin(\omega x) \cdot dx$$

$$= \frac{2}{\pi}\left[\frac{e^{-kx}(-k\sin(\omega x) - \omega\cos(\omega x))}{k^2+\omega^2}\right]_0^\infty$$

$$= \frac{2}{\pi(k^2+\omega^2)}\left[-e^{-kx} k\sin(\omega x) - e^{-kx}\cdot\omega\cdot\cos(\omega x)\right]_0^\infty$$

$$= \frac{2}{\pi(k^2+\omega^2)}\left[-(0-0) - (0-\omega)\right]$$

$$= \frac{2\omega}{\pi(k^2+\omega^2)}$$

∴ The F.I.S:-

$$f(x) = \int_0^\infty \left[0 + \frac{2w}{\pi(k^2+w^2)} \sin(wx)\right] dw$$

$$e^{-kx} = \int_0^\infty \left[\frac{2w}{\pi(k^2+w^2)} \sin(wx)\right] dw$$

$$\int_0^\infty \frac{w}{w^2+k^2} \sin(wx) \, dw = \frac{e^{-kx} \times \pi}{2}$$

(proved)

Ans:-

(5) Find the Fourier Integral of $f(x) = e^{-kx}$ when $x > 0$ and $f(-x) = f(x)$ for $k > 0$ and hence prove that $\int_0^\infty \frac{\cos(wx)}{w^2+k^2} dw = \frac{\pi}{2k} e^{-kx}$, $k > 0$

= we know:-

The FIS:-

$$f(x) = \int_0^\infty [A(w) \cdot \cos(wx) + B(w) \sin(wx)] \, dw$$

$$\therefore A(w) = \frac{1}{\pi} \int_{-\infty}^{\infty} \underbrace{e^{-kx}}_{\to \text{even}} \cdot \underbrace{\cos(wx)}_{\to \text{even}} \cdot dx$$

$$= \frac{2}{\pi} \int_0^\infty [e^{-kx} \cdot \cos(wx)] \, dx$$

C7
$f(-x) = f(x)$
even function

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$$= \frac{2}{\pi}\left[\frac{e^{-kx}(-k\cos(wx) + w\sin(wx))}{k^2+w^2} \right]_0^{\infty}$$

$$= \frac{2}{\pi(k^2+w^2)}\left[-e^{-kx} \times k \times \cos(wx) + e^{-kx} \times w \times \sin(wx) \right]_0^{\infty}$$

$$= \frac{2}{\pi(k^2+w^2)}\left[-(0-k) + (0+0) \right]$$

$$= \frac{2k}{\pi(k^2+w^2)}$$

$\therefore$ The F.I.S:-

$$f(x) = \int_0^{\infty} \frac{2k}{\pi(k^2+w^2)} \cdot \cos(wx)\, dw$$

$$e^{-kx} = \int_0^{\infty} \frac{2k}{\pi(k^2+w^2)} \cdot \cos(wx) \cdot dw$$

$$\therefore \int_0^{\infty} \frac{\cos(wx)}{w^2+k^2} \cdot dw = \frac{\pi}{2k} e^{-kx}$$

(Proved)

Here: $B(w) = 0$ ez $\underbrace{e^{-kx}}_{\text{even}} \times \underbrace{\sin(wx)}_{\text{odd}} \cdot dx$

even $\times$ odd

$= \text{odd}$

$= 0$

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#Fourier Transform

Ch: 03

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Fourier Transform:—

Finite Fourier Transform:—

$\rightarrow f(x)$ is defined in $0<x<L$

So, Finite Fourier Sine Transform:—

$$F_s(n) = \int_0^L f(x) \cdot \sin\left(\frac{n\pi}{L} x\right) dx$$

Now:- Finite Fourier Cosine Transform:—

$$F_c(n) = \int_0^L f(x) \cdot \cos\left(\frac{n\pi}{L} \cdot x\right) \cdot dx$$

And:- Infinite Fourier Sine Transform:—

$$F_s(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cdot \sin(wx) \cdot dx$$

And:- Infinite Fourier Cosine Transform:—

$$F_c(f(x)) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cdot \cos(wx) \cdot dx$$

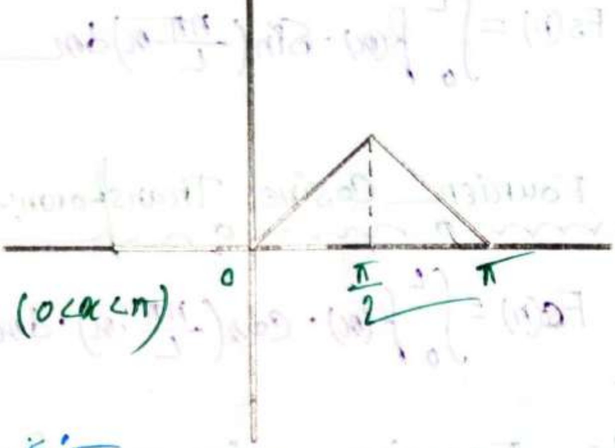
GOOD LUCK™

Example: 1:-

⇒ Sketch the function and find Fourcier Transforman of Fourcier Finite (F·F) Sine transform of the function

$$f(x)=\begin{cases} x \ ; & 0<x<\frac{\pi}{2} \\ \pi-x \ ; & \frac{\pi}{2}<x<\pi \end{cases}$$

We know:-



$f(x)$ is define in $(0<x<\pi)$.

The F.F. Sine S:-

$$F_S(n)=\int_0^{\pi} f(x)\cdot \sin(nx)\cdot dx$$

$$=\int_0^{\pi/2} \overset{\text{odd}}{x}\cdot \overset{\text{odd}}{\sin(nx)}\cdot dx+\int_{\pi/2}^{\pi}(\pi-x)\cdot \sin(nx)\cdot dx$$

$$=\left[x\times\left(-\frac{1}{n}\right)\cos(nx)-(1)\times\left(-\frac{1}{n^2}\right)\sin(nx)\right]_0^{\pi/2}+\left[(\pi-x)\times -\frac{1}{n}\cos(nx)-(-1)\left(-\frac{1}{n^2}\right)\sin(nx)\right]_{\pi/2}^{\pi}$$

$$=\left[-\frac{x}{n}\cos(nx)+\frac{1}{n^2}\sin(nx)\right]_0^{\pi/2}+\left[-\frac{\pi-x}{n}\cos(nx)-\frac{1}{n^2}\sin(nx)\right]_{\pi/2}^{\pi}$$

$$=-\frac{\pi}{2n}\cos\left(\frac{n\pi}{2}\right)+\frac{1}{n^2}\sin\left(\frac{n\pi}{2}\right)+\frac{\pi}{2n}\cos\left(\frac{n\pi}{2}\right)+\frac{1}{n^2}\sin\left(\frac{n\pi}{2}\right)$$

$$=\frac{2}{n^2}\sin\left(\frac{n\pi}{2}\right)$$

Ans

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Q:- Now For the previous Question Fine the F.F. Cosine transform.

we know:-

The F.F. Cosine S:-

$$F_c(n) = \int_0^{\pi} f(x) \cdot \cos(nx) \cdot dx$$

$$= \int_0^{\pi/2} x \cdot \cos(nx) \cdot dx + \int_{\pi/2}^{\pi} (\pi - x) \cdot \cos(nx) \cdot dx$$

$$= \left[x \cdot \frac{1}{n} \sin(nx) - 1 \cdot \frac{-1}{n^2} \cos(nx) \right]_0^{\pi/2} + \left[(\pi - x) \times \frac{1}{n} \sin(nx) - (-1) \times \left(\frac{-1}{n^2}\right) \cos(nx) \right]_{\frac{\pi}{2}}^{\pi}$$

$$= \left[\frac{x}{n} \sin(nx) + \frac{1}{n^2} \cos(nx) \right]_0^{\pi/2} + \left[\frac{\pi - x}{n} \sin(nx) - \frac{1}{n^2} \cos(nx) \right]_{\pi/2}^{\pi}$$

$$= \left[\frac{\pi}{2n} \sin\left(\frac{n\pi}{2}\right) + \frac{1}{n^2} \cos\left(\frac{n\pi}{2}\right) - \frac{1}{n^2} \right] + \left[\frac{-\pi}{2n} \sin\left(\frac{n\pi}{2}\right) - \frac{1}{n^2} (-1)^n + \frac{1}{n^2} \cos\left(\frac{n\pi}{2}\right) \right]$$

$$= \frac{2}{n^2} \cos\left(\frac{n\pi}{2}\right) - \frac{1}{n^2} \left(1 + (-1)^n\right)$$

Ans:-

Example 2:-

⇒ Sketch the functions and find the Infinite Fourier (I.F) Sine and Cosine transform of the function

$$f(x) = \begin{cases} x \; ; & 0 < x < 1 \\ 2 \; ; & 1 < x < 2 \\ 0 \; ; & x > 2 \end{cases}$$

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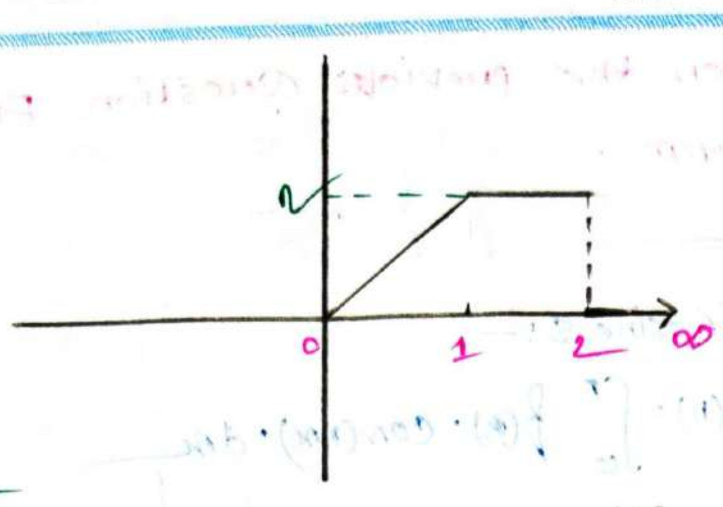
$\Rightarrow$



We know:—

The I.F. F. Sine. S:—

$$F_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x)\cdot \sin(\omega x)\cdot dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\int_0^1 x\cdot \sin(\omega x)\cdot dx + \int_1^2 2\sin(\omega x)\cdot dx + \int_2^{\infty} 0\cdot \sin(\omega x) dx \right]$$

$$= \sqrt{\frac{2}{\pi}} \left\{ \left[x \times \left(-\frac{1}{\omega}\right) \cos(\omega x) + \frac{1}{\omega^2} \sin(\omega x) \right]_0^1 + 2\left[-\frac{1}{\omega}\cos(\omega x) \right]_1^2 \right\}$$

$$= \sqrt{\frac{2}{\pi}} \left\{ \left[-\frac{x}{\omega} \cos(\omega x) + \frac{1}{\omega^2} \sin(\omega x) \right]_0^1 + 2\left[-\frac{1}{\omega}\cos(\omega x) \right]_1^2 \right\}$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}} \left[-\frac{1}{\omega}\cos(\omega) + \frac{1}{\omega^2}\sin(\omega) - \frac{2}{\omega}\cos(2\omega) + \frac{2}{\omega}\cos(\omega) \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{1}{\omega}\cos(\omega) + \frac{1}{\omega^2}\sin(\omega) - \frac{2}{\omega}\cos(2\omega) \right]$$

Ans:—

The I.F. F. Cosine transform:—

$$F_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x)\cdot \cos(\omega x)\cdot dx$$

$$= \sqrt{\frac{2}{\pi}} \left\{ \int_0^1 \overset{\text{odd}\to\text{even}}{x\cdot \cos(\omega x)}\cdot dx + \int_1^2 2\cos(\omega x) dx + \int_2^{\infty} 0\cdot \cos(\omega x)\cdot dx \right\}$$

$$= \sqrt{\frac{2}{\pi}}\left\{0 + 2\times\left[\frac{1}{w}\sin(wx)\right]_1^2 + 0\right\}$$

$$= \sqrt{\frac{2}{\pi}} \times 2\left[\frac{1}{w}\sin(2w) - \frac{1}{w}\sin(w)\right]$$

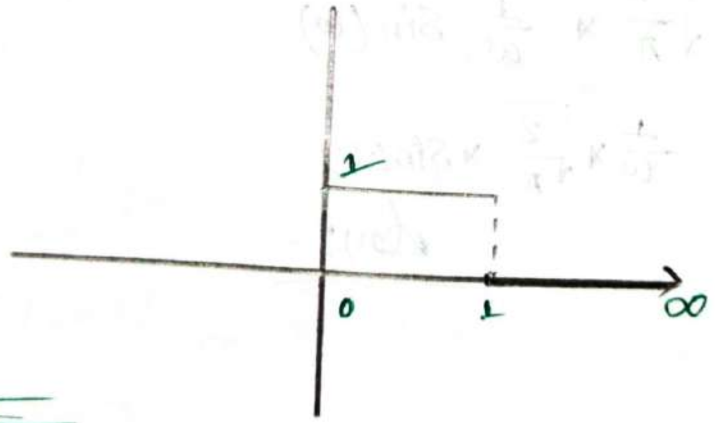
$$= \sqrt{\frac{2}{\pi}} \times \left\{\frac{2}{w}\sin(2w) - \frac{2}{w}\sin(w)\right\}$$

Ans:-

Example:3:-

⇒ Sketch the graph and find the infinite Fourier Sine and Cosine transform $f(x) = \begin{cases} 1 ; & 0 \le x < 1 \\ 0 ; & x \ge 1 \end{cases}$

⇒



we know:-

The IF Sine Transform:-

$$F_s(w) = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cdot\sin(wx)\cdot dx$$

$$= \sqrt{\frac{2}{\pi}}\left[\int_0^1 \sin(wx)\cdot dx + \int_1^\infty 0\cdot\sin(wx)\cdot dx\right]$$

$$= \sqrt{\frac{2}{\pi}}\left\{\left[\left(-\frac{1}{w}\right)\cos(wx)\right]_0^1\right.$$

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$$= \sqrt{\frac{2}{\pi}} \left\{ \left(-\frac{1}{\omega}\right) \cos(\omega) + \frac{1}{\omega} \right\}$$

$$= \sqrt{\frac{2}{\pi}} \times \frac{1}{\omega} \left\{ 1 - \cos(\omega) \right\}$$

Ans:-

The IF Cosine Transform:-

$$F_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cdot \cos(\omega x) \cdot dx$$

$$= \sqrt{\frac{2}{\pi}} \times \left[\int_0^1 \cos(\omega x) + \int_1^\infty 0 \cdot \cos(\omega x) \right] dx$$

$$= \sqrt{\frac{2}{\pi}} \times \left\{ \left[\frac{1}{\omega} \sin(\omega x) \right]_0^1 \right.$$

$$= \sqrt{\frac{2}{\pi}} \times \frac{1}{\omega} \sin(\omega)$$

$$= \frac{1}{\omega} \times \sqrt{\frac{2}{\pi}} \times \sin(\omega)$$

Ans:-

Exercise : 3.5

Sketch the graph and then find the Finite Fourier Sine transform and Finite Fourier Cosine transform For the functions:-

1. $f(x) = 2x \; ; \; 0 < x < 4$

$\Rightarrow$

We know:-

The Fourier F. Sine T:-

$$F_S(n) = \int_0^4 f(x) \cdot \sin\left(\frac{n\pi}{4} x\right) \cdot dx$$

$$= \int_0^4 2x \cdot \sin\left(\frac{n\pi}{4} x\right) \cdot dx$$

$$= \left[2x \times \left(-\frac{4}{n\pi}\right) \cdot \cos\left(\frac{n\pi}{4} x\right) - 2 \times \frac{-16}{(n\pi)^2} \sin\left(\frac{n\pi}{4} x\right) \right]_0^4$$

$$= \left[\frac{-8x}{n\pi} \cos\left(\frac{n\pi}{4} x\right) + \frac{32}{(n\pi)^2} \sin\left(\frac{n\pi}{4} x\right) \right]_0^4$$

$$= -\frac{32}{n\pi} (-1)^n$$

Ans:-

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The F.F.C.T:-

$$F_c(n) = \int_0^4 f(x)\cdot \cos\left(\frac{n\pi}{4}x\right)\cdot dx$$

$$= \int_0^4 2x\cdot \cos\left(\frac{n\pi}{4}\cdot x\right)\cdot dx$$

$$= \left[2x\cdot\left(\frac{4}{n\pi}\right)\sin\left(\frac{n\pi}{4}\cdot x\right) - 2\times\left(\frac{-16}{(n\pi)^2}\right)\cos\left(\frac{n\pi}{4}\cdot x\right)\right]_0^4$$

$$= \left[\frac{8x}{n\pi}\sin\left(\frac{n\pi}{4}x\right) + \frac{32}{(n\pi)^2}\cos\left(\frac{n\pi}{4}x\right)\right]_0^4$$

$$= \frac{32}{n\pi}\sin(n\pi) + \frac{32}{(n\pi)^2}(-1)^n - \frac{32}{(n\pi)^2}$$

$$= 0 + \frac{32}{(n\pi)^2}\left\{(-1)^n - 1\right\}$$

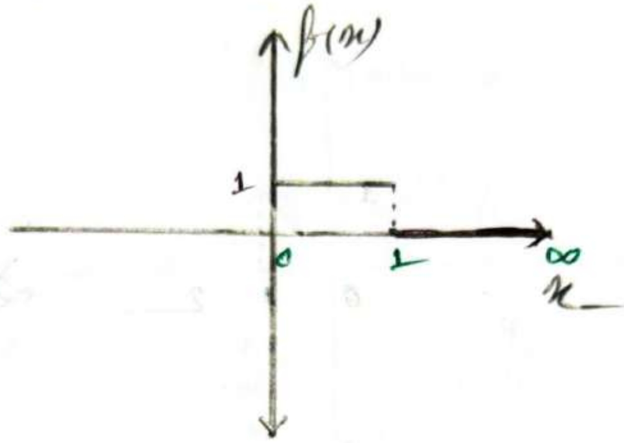
$$= \frac{32}{(n\pi)^2}\left((-1)^n - 1\right)$$

Ans:-

Q. Sketch the graph and then find the Infinite Fourier Sine transform, and infinite Fourier cosine Transform of

$$f(x)=\begin{cases} 1 & ; \ 0\le x\le 1 \\ 0 & ; \ x\ge 1 \end{cases}$$

⇒



We know:—

The IFST:—

$$F_s(\omega)=\sqrt{\frac{2}{\pi}}\left\{\int_0^\infty f(x)\cdot \sin(\omega x)\cdot dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left[\int_0^1 1\cdot \sin(\omega x)dx+\int_1^\infty 0\cdot \sin(\omega x)\cdot dx\right]$$

$$=\sqrt{\frac{2}{\pi}}\left[-\frac{1}{\omega}\cos(\omega x)\cdot dx\right]_0^1$$

$$=\sqrt{\frac{2}{\pi}}\left[-\frac{1}{\omega}\cos(\omega)+\frac{1}{\omega}\right]$$

$$=\sqrt{\frac{2}{\pi}}\times\frac{1}{\omega}\left[1-\cos(\omega)\right]$$

The IFCT:—

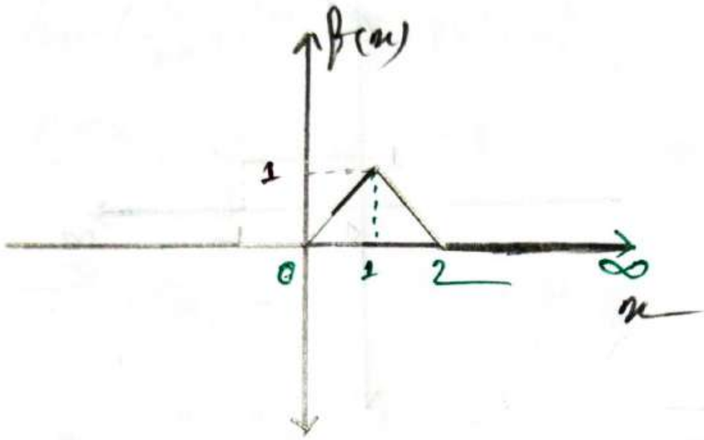
$$F_c(\omega)=\frac{\sqrt{2}}{\sqrt{\pi}}\left\{\int_0^\infty f(x)\cdot \cos(\omega x)\cdot dx\right\}$$

$$=\frac{\sqrt{2}}{\sqrt{\pi}}\left\{\int_0^1 1\cdot \cos(\omega x)\cdot dx+\int_1^\infty 0\cdot \cos(\omega x)\cdot dx\right\}$$

$$=\frac{\sqrt{2}}{\sqrt{\pi}}\left[\frac{1}{\omega}\sin(\omega x)\right]_0^1=\frac{\sqrt{2}}{\sqrt{\pi}}\times\frac{1}{\omega}\times\sin\omega$$ Ans:-

4. Sketch the graph and then find the Infinite Fourier Sine transform, and Infinite Fourier Cosine transform of $f(x)=\begin{cases} x & ; 0<x<1 \\ 2-x & ; 1<x<2 \\ 0 & ; x>2 \end{cases}$

$\Rightarrow$



We know:—

The IFST:—

$$F_S(\omega)=\sqrt{\frac{2}{\pi}}\left\{\int_0^{\infty} f(x)\cdot \sin(\omega x)\cdot dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\int_0^{1} x\cdot \sin(\omega x)\cdot dx+\int_1^{2}(2-x)\cdot \sin(\omega x)\cdot dx+\int_2^{\infty} 0\,\sin(\omega x)\,dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\left[x\times\left(-\frac{1}{\omega}\right)\cos(\omega x)-1\times\left(-\frac{1}{\omega^2}\right)\sin(\omega x)\right]_0^1+\left[(2-x)\times\left(-\frac{1}{\omega}\right)\cos(\omega x)-(-1)\left(-\frac{1}{\omega^2}\right)\sin(\omega x)\right]_1^2\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\left[-\frac{x}{\omega}\times\cos(\omega x)+\frac{1}{\omega^2}\sin(\omega x)\right]_0^1+\left[-\frac{2-x}{\omega}\cos(\omega x)-\frac{1}{\omega^2}\sin(\omega x)\right]_1^2\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\left[-\frac{1}{\omega}\cos(\omega)+\frac{1}{\omega^2}\sin(\omega)\right]+\left[\frac{1}{\omega}\cos(\omega)-\frac{1}{\omega^2}\sin 2\omega+\frac{1}{\omega^2}\sin\omega\right]\right\}$$

$$=\sqrt{\frac{2}{\pi}}\times\frac{1}{\omega^2}(2\sin\omega-\sin 2\omega)$$

Ans:—

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The I.F.C.T:-

$$F_c(\omega)=\sqrt{\frac{2}{\pi}}\left\{\int_0^\infty f(x)\cdot\cos(\omega x)\cdot dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\int_0^1 x\cdot\cos(\omega x)\cdot dx+\int_1^2(2-x)\cdot\cos(\omega x)\cdot dx+\int_2^\infty 0\cdot\cos(\omega x)\cdot dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\left[x\times\frac{1}{\omega}\times\sin(\omega x)-1\times\frac{-1}{\omega^2}\times\cos(\omega x)\right]_0^1+\left[(2-x)\cdot\frac{1}{\omega}\times\sin(\omega x)-(-1)\times\frac{-1}{\omega^2}\cos(\omega x)\right]_0^2\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\left[\frac{x}{\omega}\sin(\omega x)+\frac{1}{\omega^2}\cos(\omega x)\right]_0^1+\left[(2-x)\times\frac{1}{\omega}\sin(\omega x)-\frac{1}{\omega^2}\cos(\omega x)\right]_0^2\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\left[\frac{1}{\omega}\sin\omega+\frac{1}{\omega^2}\cos\omega-\frac{1}{\omega^2}\right]-\frac{1}{\omega^2}\cos(2\omega)+\frac{1}{\omega^2}\right\}$$

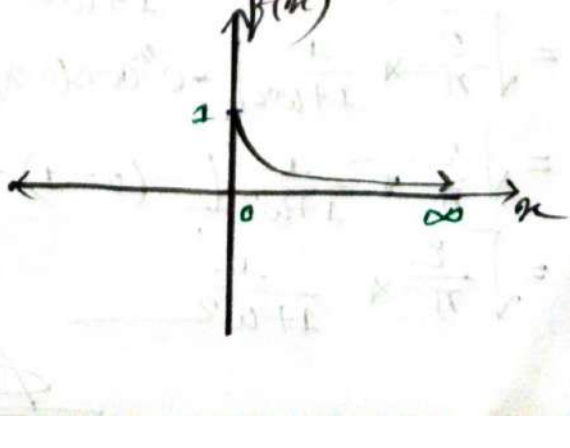
$$=\sqrt{\frac{2}{\pi}}\times\frac{1}{\omega}\left\{\sin\omega+\cos\omega\times\frac{1}{\omega}-\frac{1}{\omega}\cos 2\omega\right\}$$

$$=\sqrt{\frac{2}{\pi}}\times\frac{1}{\omega}\left\{\sin\omega+\frac{1}{\omega}\cos\omega-\frac{1}{\omega}\cos 2\omega\right\}$$

Ans:-

(5.) Sketch the graph and find the Infinite Fourier sine transform, and Infinite Fourier Cosine transfrom of $f(x)=e^{-x}$; $x\geq 0$

$\Rightarrow$



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we know:-

The IFST:-

$$F_s(\omega)=\sqrt{\frac{2}{\pi}}\left\{\int_0^{\infty} f(x)\quad \sin(\omega x)\cdot dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\int_0^{\infty} e^{-x}\cdot \sin(\omega x)\cdot dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left\{\left[\frac{e^{-x}(-\sin(\omega x)-\omega\cos(\omega x))}{1^2+\omega^2}\right]_0^{\infty}\right.$$

$$=\sqrt{\frac{2}{\pi}}\times\frac{1}{1+\omega^2}\left\{-e^{-x}\times\sin(\omega x)-e^{-x}\omega\cos(\omega x)\right\}\Big]_0^{\infty}$$

$$=\sqrt{\frac{2}{\pi}}\times\frac{1}{1+\omega^2}\times\left[\left(-(0-0)-(0-\omega)\right)\right]$$

$$=\sqrt{\frac{2}{\pi}}\times\frac{\omega}{1+\omega^2}$$

Ans:-

The IFC.T:-

$$F_c(\omega)=\sqrt{\frac{2}{\pi}}\left\{\int_0^{\infty} f(x)\cdot\cos(\omega x)\cdot dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\times\left\{\int_0^{\infty} e^{-x}\cdot\cos(\omega x)\cdot dx\right\}$$

$$=\sqrt{\frac{2}{\pi}}\left[\frac{e^{-x}(-\cos(\omega x)+\omega\sin(\omega x))}{1+\omega^2}\right]_0^{\infty}$$

$$=\sqrt{\frac{2}{\pi}}\times\frac{1}{1+\omega^2}\left[-e^{-x}\cos(\omega x)+e^{-x}\sin(\omega x)\right]_0^{\infty}$$

$$=\sqrt{\frac{2}{\pi}}\times\frac{1}{1+\omega^2}\left[-(0-1)+(0-0)\right]$$

$$=\sqrt{\frac{2}{\pi}}\times\frac{1}{1+\omega^2}$$

Ans:-